



KADUNA AGILE PROJECT

Adolescent Girls Initiative For Learning And Empowerment

ADOLESCENT GIRLS INITIATIVE FOR LEARNING AND
EMPOWERMENT (AGILE) KADUNA STATE



DIGITAL LITERACY

TRAINING MANUAL



**INSTRUCTORS
MANUAL**

**Adolescent Girls Initiative for Learning and
Empowerment (AGILE) Project, Kaduna State**

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Technology from the advent of the GPS to the days of Telephones, Satellite navigation and now emails and smartphones is a constant part of our lives. The high expectation of employers on prospective workers to have the basic understanding of technology makes it imperative to possess these skills. The absence of these skills becomes a disadvantage for individuals, both locally and globally.

This manual is developed by Lexington Technologies for the Adolescent Girls' Initiative for Learning and Empowerment (AGILE) Project, Kaduna State for the training of adolescent girls and boys on digital skills under the digital literacy subcomponent.

The development of this curriculum is based on the UNESCO Digital Literacy Framework and incorporates elements of Microsoft Standard Digital Literacy Curriculum standard 4, developed by Microsoft Inc.

The Digital Literacy Training Manual for Instructors

This manual includes learning courses and the assessments for instructors. The topics include demonstrations, simulations, hands-on activities and links to external resources to enhance learning.

The Topics

The training manual is subdivided into several sections and each section is clustered into lessons. Each lesson has sub-foundational topics divided into grades, (SS1, SS2 and SS3) covering basic, intermediate and advanced digital skills.

- Foundational Computer Skills (use of computers and digital mobile devices):
- Navigating the Internet and the World Wide Web (search engines, web browsing, evaluating information on the internet):
- Multimedia services (deducing information from multimedia):
- Productivity programs (productivity tools, e.g. spreadsheets, word processors, presentations software and graphics programs):
- Computer Security and Privacy (Firewalls and anti-virus software):
- Networking and collaboration (LinkedIn and other online communities):

Curriculum: Digital Literacy

Curriculum Type: Seminar

Location Training Room

Curriculum Description

In this curriculum, the students will be introduced to computers and the Internet, some productivity programs available, and some devices associated with digital audio, video, and photography. This curriculum also explains some of the common threats to your computer and how to safeguard your computer from these

threats. The students will also be introduced to different career opportunities available for them in the world of digital technology.

Curriculum Objectives

After completing this curriculum, the students will be able to:

1. Perform basic computer operations.
2. Describe basic Internet and World Wide Web terminologies, browse information, and perform transactions on the Internet.
3. Perform basic tasks in productivity programs, such as word processors, spreadsheets, presentation programs, and databases.
4. Identify the primary risks to computers and the steps to minimize these risks.
5. Explain the role of digital technology in everyday life and the basic concepts of digital audio, video, and photography.

Prerequisites

To attend a seminar course, the students should be able to read and write. Students who can perform basic computer operations can skip the “Perform basic computer operations” objective.

Instructional Methods

You can use a combination of instructional strategies to deliver this course. This combination would depend on factors such as the type of content and the skill level of the students. Some instructional strategies are as follows:

- Lecture
- Demonstration
- Case studies
- Problem solving
- Role playing
- Experiments and observations
- Computer assisted instruction

In this Programme, the students will learn to use computers effectively. They will accomplish this through a combination of self-paced learning and classroom activities that would add to the concepts taught in the e-learning courses. The e-Learning content will be used for lectures and demonstrations. You can use a projector to display the eLearning course or provide you the Print version of the e-Learning course. You can also give homework assignments and self-paced reading materials.

CLASS SCHEDULE AND OBJECTIVES

DAY ONE: HOURS ALLOCATED: 4 HOURS

HOUR ONE (1)

COURSE 1: FOUNDATIONAL COMPUTER SKILLS (USE OF COMPUTERS AND DIGITAL MOBILE DEVICES)

COURSE DESCRIPTION

Computing has become a very integral part of everyday life in the twenty-first century. From music and photos to banking and communicating, computers have changed the way we work and live. This course introduces you to the fundamentals of computing, explains the components of a computer, explores operating system basics, and shows you how to use a mouse and a keyboard. Also explains how computers can be used in different aspects of life.

COURSE BENEFITS

Computers are used in every field. They help organizations and individuals to conduct business transactions efficiently and quickly. Today, one of the basic skills necessary to succeed at a workplace is to know how to use the computer. To be able to get better jobs, you need to know how to use a computer.

After completing the Computer Basics course, successful students are able to describe the importance of computers in today's world; identify the main parts of a computer; describe basic computer performance factors; and define the operating systems, networks, and the uses for typical programs at work and at home.

LESSON 1: INTRODUCTION TO COMPUTERS

OBJECTIVES:

- 1.1. DESCRIBE THE IMPORTANCE OF COMPUTERS IN TODAY'S WORLD.
- 1.2. IDENTIFY THE MAIN PARTS OF A COMPUTER.
- 1.3. IDENTIFY THE STEPS FOR STARTING A COMPUTER.
- 1.4. IDENTIFY THE DIFFERENT GROUPS OF KEYS ON A KEYBOARD.
- 1.5. PERFORM DIFFERENT TASKS USING A MOUSE.

A computer is an electronic device that you can use to store and process information. A computer can be used to help you perform various tasks, such as:

1. Send e-mail
2. Make Presentations
3. Maintain Records
4. Write Text
5. Organize Files
6. And more

Roles of Computer

Computers impact our lives everyday. Consider a trip to the store and how many different computers are involved. Computers are inside your car, manage the traffic lights, control the temperature, and calculate the cost of your groceries. Let's look more at how computers play a regular role in your life.

Parts of computer

Computers contain many parts including keyboards, monitors, processing units, memory, and more. In this lesson we will look at all the parts of a computer. We will organize these parts by categories including input, output, microchips, storage, and ports.

Steps of Starting Up a computer system

To start a computer, you press the power button. This may cause a small light to turn on and then the monitor to turn on. The computer will check all of its components and if everything is running smoothly, it will display the welcome screen, and then to the user screen. Here you may login into your personal user, or as a guest.

All new computers now have an energy-saving mode called Sleep that allows the computer to temporarily shut down all nonessentials. Usually, there is a light flashing to indicate that it is asleep. This is a huge energy saver, using as much electricity as a Night light and taking seconds to start up. The computer may seem to be shut down, except all of your work will still open and will not be lost. It takes seconds to turn back on and continue to work on any pages that were open, sleep is a very useful tool on a computer. There are other options; such as Log off, Restart, and Shut down. Before doing any of those options, make sure to save all unsaved items because they can become lost.

Do not press the power button to shut down the computer unless the whole computer has stopped responding and you can no longer restart it

Logging Off

Logging off will exit you from the current user that you are using. Any open programs will remain open and once logged back in will open up, unless the computer is shut down. Any other users may have their information open as well.

Restarting

The Restart option is usually used when something is not corresponding. The Restart option will shut down the computer and automatically begin to turn on the computer again. Remember that any unsaved information will be lost if it is not saved if the computer is restarted.

Shutting Down

Shutting down the computer will completely turn off the whole system. Any unsaved information will be lost, so remember to save before shutting down the computer.

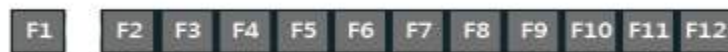
Overview of the Keyboard

The keyboard is an input device that goes into your computer which allows you to enter information such as letters from the alphabets, numerical data, symbols, and commands. There are many keyboards, although not all have the same layout style. The most common kind is referred to as a QWERT keyboard, which is named after the keys on the top rows of letters. The keyboard is divided into several groups based on the function.



Groups of keys on the Keyboard

Function keys: The function keys, labeled from F1 to F12, are used to perform specific tasks based on the operating system and software used. For example, in many programs, the key F1 is commonly used as the Help file. In some cases, it can control the monitor brightness or the sound volume. The functionality of these keys differentiates from program to program and operating system to operating system



Special keys: The special keys are used alone or in combination with other keys to perform certain actions. These include the Control (ctrl), Alternate (Alt), Escape (Esc), Windows, Tab, Caps Lock, and Shift keys. For an example, in many programs pressing “Ctrl+S” will save a file automatically



The Shift key is a combination key used with another letter to type an uppercase letter.

The Caps Lock key allows all letters to be typed in uppercase without having to press shift. This function can be turned on or off. An indicator light might be present to show that the function is on. Most keyboards include a special key called the Windows logo key. This key is used to open the Start menu, or is used in combination with a second key to perform common Windows tasks.

Alphanumeric keys: The alphanumeric keys cover most of the typical keyboard; this is for letters, numbers, punctuations, symbol keys, the space bar, tab, backspace, Caps Lock, shift, and enter key. The Space Bar allows a space to be indicated when pressed



The Enter key allows the cursor to be moved to the beginning of the next line



The Backspace key deletes the character before the cursor, or the selected text chosen



The Backspace key deletes the character before the cursor, or the selected text chosen

Navigation keys The navigation keys are used for moving around in documents or web pages and editing text. Navigation keys include Delete, Insert, Home and End, and Page Up and Page Down keys. The Home key is used to move the cursor to the beginning of a line or to move to the top of the webpage



The Left, Right, Up, and Down Arrow keys is used to move the cursor or selection one space or line in the direction of the arrow, or scroll a web page in the direction of the arrow

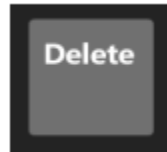


The End key moves the cursor to the end of a line or move to the bottom of a webpage.

The Page Up key moves the cursor or page up one screen and The Page Down key moves the cursor or page down one screen



The Delete key deletes the character after the cursor or the selected text. If used in Windows, the Delete Key can be used to move selected items to the recycle bin

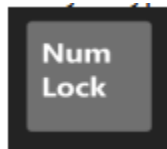


The Insert key can be turned on or off. When used, the text that you type is inserted at the cursor. When the Insert key is not used, the text that you type replaces existing characters



The Insert key can be turned on or off. When used, the text that you type is inserted at the cursor. When the Insert key is not used, the text that you type replaces existing characters

Numeric Key



Three keys that you may have seen on your keyboard known as PrtScn (Print Screen), ScrLk (Scroll Lock), and Pause/Break have different uses for different programs. PrtScn is used to capture an image of your entire screen known as a screen shot which copies into your computer's clipboard.



The keyboard is an input device that you can use to type commands or text into a computer. Different keyboards have different key layouts. In addition, the functions of some keys may vary from program to program.



Perform different tasks by using a mouse.

A mouse is a small device that you can use to move, select, and open items on your computer screen.



1. The mouse is usually kept on the desk next to the keyboard.
2. Most mouse devices have at least two buttons, left and right.
3. Most of the actions are performed by clicking the left button. The right button is used for specific functions.
4. Some advanced types of mouse devices provide additional buttons to speed up common tasks, such as scrolling.
5. As you move the mouse on your desk, a pointer moves correspondingly on your screen.
6. The mouse allows you to select an item on the screen.
7. As you move the pointer over different areas of the screen, the items or the pointer change.
8. These changes indicate that you can click an item to open it, or to see more of its options.
9. You can open an item or show more options by moving the pointer to it, and then clicking the left or right mouse button.
10. In a document, you can use the mouse to select a position to start typing.
11. To move an item, you need to click it, and then holding the mouse button down, move the item to a different location.
12. After you move the item to the new location, you release the mouse button.
13. The right button on the mouse is often used to display a menu.

14. Menu options can include common tasks, such as copy or paste as well as advanced options.
15. These are called context menus. These menus help you complete tasks quickly.
16. Most mouse devices also have a wheel that helps you to scroll through documents or pages.
17. To scroll, place your finger on the wheel and roll it back and forth.
18. This action moves the document up and down.
19. Some mouse devices have a "back" button on the side that works in browsers and Windows Explorer.
20. There are different types of mouse devices available in the market.

LESSON 2: CAREER OPPORTUNITIES

OBJECTIVES:

- 2.1. DESCRIBE HOW COMPUTERS HAVE BECOME A CENTRAL PART OF EVERYDAY LIFE.
- 2.2. IDENTIFY DIFFERENT CAREER OPPORTUNITIES AVAILABLE FOR A PERSON WHO IS COMPUTER-LITERATE.

Overview of career opportunities

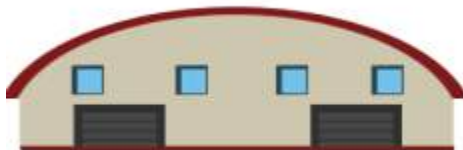
Remember that computers play a major role in daily life in many industries, schools, government offices, and shops. Computer literacy is a valuable skill whether you work in education, farming, business, health care, scientific research, government affairs, arts, entertainment, publishing and many other industries.



An information worker, known as a home and office user, does not particularly work with computers although they normally use a computer to enter and maintain records. An information worker can include a data entry operator, a warehouse manager, graphic designers, a travel agent, a human resource manager, or an entrepreneur.



A warehouse manager keeps an account of the warehouse stocks, manages receipts, storage, training employees, and such. It is vital that a warehouse manager uses a computer to keep track of their company. The use of computers, word processors, and data processing systems has made the task easier. A warehouse manager has to dispatch and send a wide range of products. They must organize their inventory, stock shelves, transfer and export inventory. Without the use of a computer and a data-entry program, companies can be short on inventory and therefore can cost the company money.



A data entry operator can use the computer to create documents, enter data such as names, addresses, statistical information, business information, account numbers, or other data, handle word processors, and maintain records. These records can be analyzed by the computer to generate reports. The words “data entry” is broad and could be used for a position that types information into a computer program such as Microsoft’s excel.



Travel agents arrange their clients with travel arrangements, look up quotes about plane tickets, hotels and restaurants, collect payments for transportation and accommodations, and provide information about mode of transportation, finances, and flight schedules. To provide their clients with this information, they must use a computer. A travel agent can search up their client’s accommodations, ticket time and price, information about a specific place, and compute the costs of their trip, transportation, restaurants, and hotels, with a computer.



Graphic designers create designs and art that are visually-appealing to the target audiences. Graphic designers can work with drawn, painted, photographed, typography, or computer generated-images. Most graphical designers work with computers to complete their task. Normally, they develop creative ideas and concepts to create. Graphic designers usually create their pieces of work using a computer with an advance photo editing program. They can create a poster, a billboard, a business logo, a magazine ad, an album cover, a website, and more. If you want to work closer with computers, you can seek various job opportunities in the field of computers, if you have relevant knowledge. Depending on your skills, you can work as an information worker, an IT professional, or a software developer.



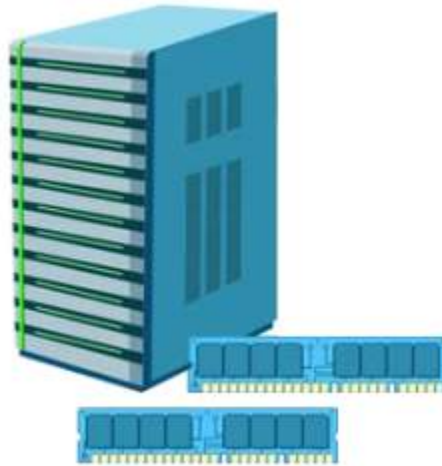
Information Technology professionals, known as I.T. professionals are specialists that apply their technical expertise to the implementation, monitoring, or maintenance of IT systems. IT professionals can specifically work in one area such as computer networking, database administration, system administration, security and information assurance, IT audit, database administration, web administration and more. Information technology careers vary depending on your education level. You can get a certification, teach yourself, and get a degree in information and technology. I.T. jobs are based specifically on the computer. Examples of IT professionals include network administrators, and database administrators.

Network administrators are responsible for managing a local area network (LAN) or a wide area network (WAN) for an organization. Network administrators can organize, install, and support an organization's computer system by maintaining network security, installing new hardware and applications, monitoring software upgrades, enforcing licensing agreements, developing a storage management program, and ensure routine backups. Network administrators make sure that the network is reliable, secure, and efficient.



Database Administrators

A database is an organized collection of information on a computer. Database administrators are in charge of storing, organizing, and managing data. Database administrators make sure the database is operating efficiently, maintaining a secure successful database, fine-tuning, upgrading, and testing modifications if needed.



Bank clerks use computers to process transactions for their clients. Using computers, they provide information such as creating a new account, looking up account balance and interest rates, for their customers.



Entrepreneurs In today's society, entrepreneurs use blogs and social network sites to promote their business and products. These networking tools give their businesses a wider reach and help them meet possible prospects.



Software developers create applications, software, and programs, that businesses and consumers use every day. They can range from game developers, mobile application developers, to making software for a company. They write programs, diagnose and fix problems, design, and organize projects. Every time you open an application on your phone or on your computer, this is the work of a software developer.



LESSON 3: COMMON COMPUTER TERMINOLOGY

OBJECTIVES:

- 3.1. IDENTIFY THE PRIMARY HARDWARE COMPONENTS OF A COMPUTER.
- 3.2. DEFINE WHAT AN OPERATING SYSTEM IS AND ITS ROLE.

Computer Hardware Components

An automobile, such as a car or a van, is available in different models and colours, but its essential components remain the same. All automobiles have an engine, a body, and wheels. Similarly, computers are available in various sizes and shapes, but they all have common components that work in the same manner.

The essential components of a computer are:

1. Hardware
2. Software

Introduction to hardware

Every time you press a key or click your mouse a signal is sent to the processor. The processor then decides what the best course of action is for the signal. The CPU handles millions of instructions per second which is why you can watch videos, listen to music and chat with friends all at the same time. Let's look at a simple example of how signals are processed.

- **Define what an operating system is and its role.**

Introduction to operating systems

Operating systems, OS, allow you to use files and devices. The OS controls all the programs you are running, by sharing the CPU's time. They also provide help when installing new devices. Most OS now have a graphical user interface which allows more people to use computers. Let's look at how the OS works. Every time you press a key or click your mouse is an example of how signals are processed.

Roles of an operating system

In any computer, the operating system performs the following roles:

1. Controls the backing store and peripherals such as scanners and printers.
2. Deals with the transfer of programs in and out of memory.
3. Organizes the use of memory between programs.
4. Organizes processing time between programs and users.
5. Maintains security and access rights of users.
6. Deals with errors and user instructions.
7. Allows the user to save files to a backing store.
8. Provides the interface between the user and the computer - for example, Windows Vista and Apple OSX. For more information, see the User Interfaces study guide.
9. Issues simple error messages.

HOUR TWO (2)

LESSON 3: COMMON COMPUTER TERMINOLOGY (CONTINUED)

OBJECTIVES:

3.3. DEFINE THE TERM PROGRAM.

3.4. EXPLAIN WHAT IS MEANT BY DATA.

3.5. DEFINE THE TERM NETWORK AND IDENTIFY THE BENEFITS OF NETWORKING.

3.6. DEFINE THE TERM INTERNET.

Computer Program

Programs run computers, without programs commanding the computer on what to do and the input data to the program, the computer would just be a screen and components. The programs that run computers are like recipes. These programs have a set of variables, which are like ingredients, and statements, which are like directions.

Computers may run other programs, which are also known as applications, which are meant to do other tasks. There are many types of programs, from programs that are used to create files like a letter or brochure to making a presentation. There are also programs created for entertainment, such as playing games, watching movies, or communicating with others.

In order for a program to work, there must be input of data. Much like a car, the car must have the key inserted and turned for the car to do anything. Once the car is on, the output is being able to drive or listen to the radio. In a program, information must be inserted for tasks to be done. When the information is inserted, it will display on the screen. You will soon click on the next objective, that inputs data that gives you the output of being able to move on to the next objective.

Once data is in, the file can be saved so that it can be opened or viewed later on. If a program has inserted information and is closed without saving, all that information will be lost.

Introduction to Network

Many employees need to print or share documents in their workspace in order to fulfill their tasks. Buying a printer and memory stick for every single employee would be very expensive and time consuming. Workspaces normally have a network, a network consists of two or more computers that are linked in order to share resources, exchange files, or allow electronic communications. Networks allow all employees to share information while sitting at different desk, print to the same printer, and more.

Most networks, if not all, have the following components:

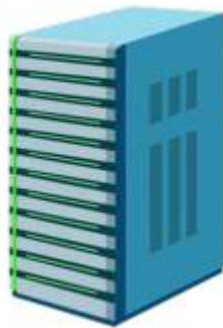
- A server
- A workstation
- Communication Channel

The size of the workplace matters because it will indicate whether you use a Local Area Network (LAN) or a Wide Area Network (WAN).

- LAN
- WAN

A Server

The main computer or drive that provides the services to all the other computers. The server is in charge of allowing devices or computers to access certain items.



Workstation

The actual device that is connected to the network.



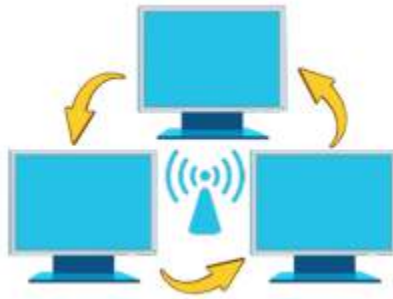
Communication channel

The path or link that actually connects computers or devices to transfer the information.



LAN

A small network that is confined to a small and limited area that is usually the size of a home or a small group of offices.



WAN

A very large network that can be connected throughout the whole world; via long wires, optical cables, or satellites. A great example of a WAN is the Internet.



Benefits of Networking

The main benefits of networking are:

1. File sharing - you can easily share data between different users, or access it remotely if you keep it on other connected devices.
2. Resource sharing - using network-connected peripheral devices like printers, scanners and copiers, or sharing software between multiple users, saves money.
3. Sharing a single internet connection - it is cost-efficient and can help protect your systems if you properly secure the network.
4. Increasing storage capacity - you can access files and multimedia, such as images and music, which you store remotely on other machines or network-attached storage devices.

Using the Internet

The Internet is a worldwide collection of public networks that are linked to each other for information exchange. The Internet started as a network to facilitate communication between government and educational departments in the United States. When other networks were connected to this network, it became a vast medium for exchanging information and ideas.

Today, the Internet connects many commercial, government, and educational networks, as well as individual computers, which share data.

The Internet offers a range of services to its users, such as file transfers between Internet users and electronic mail for sending messages. The World Wide Web (WWW), or the Web, is another service that the Internet provides. WWW includes specially formatted documents that are interlinked and stored on servers around the world.

You can use the Internet and its services to send messages to other Internet users, search and apply for jobs, watch movies, and buy and sell products. Many organizations use a special type of network to communicate and share information within the organization. Such a network is called an intranet. An intranet is similar to the WWW but is accessible only to authorized users of the organization. An intranet is much smaller than the Internet and can provide services such as document distribution, software distribution, access to databases, and training

LESSON 4: COMPUTER PERFORMANCE AND FEATURES

OBJECTIVES:

- 4.1. IDENTIFY AND COMPARE THE FEATURES OF DIFFERENT TYPES OF COMPUTERS.
- 4.2. EXPLAIN THE ROLE OF MEMORY.
- 4.3. EXPLAIN THE BASICS OF COMPUTER PERFORMANCE AND HOW IT RELATES TO PRODUCTIVITY.
- 4.4. DESCRIBE THE DIFFERENT TYPES OF PRODUCTIVITY PROGRAMS AND THEIR USES.
- 4.5. DESCRIBE THE DIFFERENT TYPES OF COMMUNICATIONS PROGRAMS AND THEIR USES.
- 4.6. DESCRIBE THE USES OF EDUCATIONAL AND ENTERTAINMENT PROGRAMS.

Consider a situation where you want to buy a television. There are many brands and models available in the market. You need to make a decision based on the features that you want and the price of the product. Similarly, there are different types of personal computers available in the market. They differ on the basis of features such as price, size, and speed. In addition, these factors affect the overall performance of the computer.

After you buy a television, you need to choose from different kinds of channels that are available. These channels may offer entertainment, sports, or news. You can choose to view a channel based on your preferences. In the same way, after you start using a computer, there are different types of programs available that help you perform different tasks. You can use a word processor to create documents or a spreadsheet to perform mathematical calculations. Communication programs can help you talk to people at distant locations. With entertainment programs, you can watch movies, listen to music, or play games.

The Role of Memory

When a computer performs a task, it stores data in its memory. Computer data is represented by 0's and 1's. Each 0 or 1 is a bit.

Computer Performance

In computing, computer performance is the amount of useful work accomplished by a computer system. Outside of specific contexts, computer performance is estimated in terms of accuracy, efficiency and speed of executing computer program instructions. Also the productivity(output) depend on the performance

Productivity Programs and Their Uses

There is software for just about every task or project you do in that requires writing, organizing, record keeping or creating images.

Word processing

Word processing programs are used to create and edit text documents. They come with built-in features that spell check and format your writings any way you choose. Some are extremely specialized such as software that aids with screenwriting. Others are made to create high-end professional looking documents that combine text and graphics to produce such things as brochures or magazines.

Note taking

Note taking programs allow an easy way to take, collect and organize notes, recordings and other data. You can then store that information for quick reference in one convenient location.

Presentations

With a presentation program you can create slideshows and business proposals that are professional looking and easily edited or modified. You can even add audio/visual elements to make your presentations even better.

Spreadsheets

A spreadsheet program is used to manage charts and graphs. You can organize budgets, streamline complex calculations and manage data tables. Spreadsheets use cells in vertical and horizontal rows to custom create any chart you need.

Databases

A database is used to organize and store data in a searchable and manageable format. You can create data reports that can track sales, customer information and inventory.

Graphics

Graphics programs help you create pictures and drawings. The software allows you to edit what you draw or enhance photographs. These programs often offer digital art tools such as custom colour options and various built-in pencils, paintbrushes and pen types.

Communication programs

The Internet allows us to quickly and easily communicate with people all around the world.

Sending e-mail: Sending e-mail is the exchange of messages from one computer user to another computer user.

Instant messaging: Instant messaging programs allow you to send and receive messages immediately.

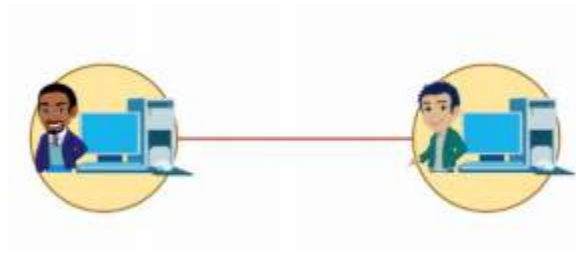
Social media: You can build or join social groups with common interests by using social media and social networking sites.



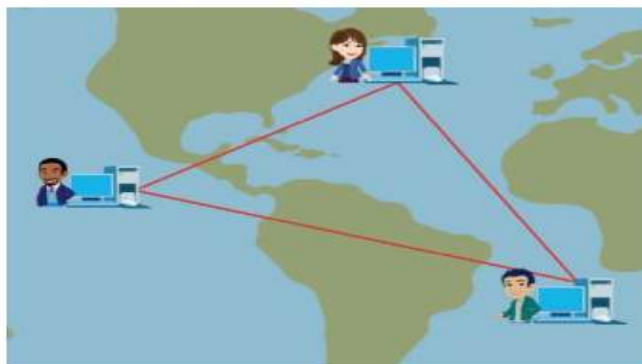
Sending an E-mail

Sending e-mail is the exchange of messages from one computer user to another computer user.

1. This exchange can be within a local area or from one part of the world to another.



2. You can send an e-mail message to, or receive an e-mail message from, one or several persons at any time of day.



3. To use e-mail, you must have an Internet connection and an email account.
4. You can obtain an Internet connection from an Internet service provider and use several services, such as Windows Live Mail, to create an e-mail account.
5. If you have an e-mail account, it will be similar to username@example.com, where the username is set by the account owner. The “at” symbol and then example.com is the domain name.
6. A domain name identifies the name and type of organization with whom you have an e-mail account. After you have an e-mail account, you need to know the e-mail address of any person to whom you want to send an email message.
7. You can send both text and pictures through e-mail; however, this can depend on various factors, such as the type of service you have or the kind of picture you are sending.



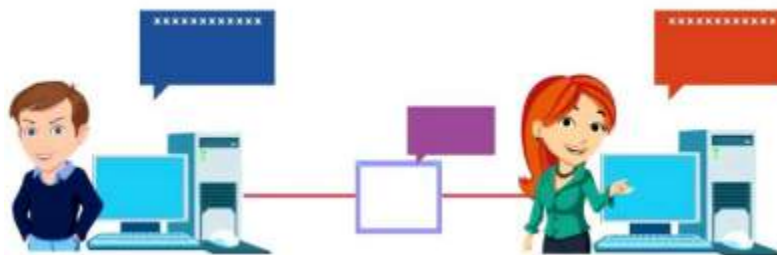
8. Sending and receiving e-mail messages is an instant way of communicating with anyone.
9. It only takes a few seconds to send and receive an e-mail message depending on the speed of your Internet connection.



Instant messaging

Instant messaging programs allow you to send and receive messages immediately.

1. When you are chatting with someone over an instant messaging program,



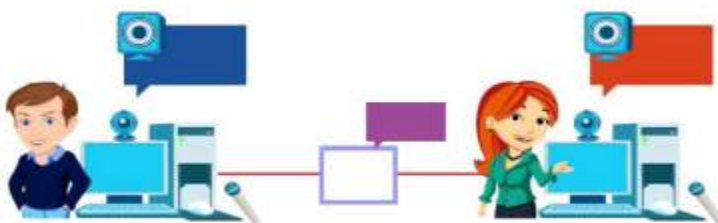
2. the person on the other end receives your messages immediately.
3. You can use these programs to communicate with several people at the same time.



4. A commonly used communication program is Windows Skype.
5. You can share pictures and other files through Windows Skype.



6. With some chat programs including Windows Skype you can talk to the person you are chatting with over a microphone.
7. This form of chat is called voice chat. You can also see the person you are talking to by using a device called a webcam. This form of chat is called video chat.



Social media

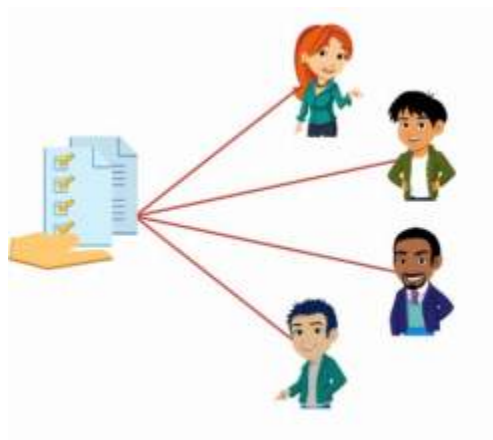
You can build or join social groups with common interests by using social networking sites.



1. To use a social networking site, you need to create an account and a profile.
2. This profile contains all the information that you want to share with your contacts.



3. With social networking sites you can share ideas, share files, view photos, and hold discussions.
4. Windows Live allows you to view highlights of updates from your contacts on Outlook, Skype and other social networks.



5. Some of the social networking sites also allow you to post job openings and promote products.



Educational and Entertainment programs

Computers can give you access to educational and entertainment programs that may not be available in your community.

Educational programs

You may want to learn a new language or work toward a college degree but you don't have a school that offers the courses you want in your hometown. With your computer you can find and download educational software that focuses on the lessons you want to study. Often these programs are designed to teach the course just like a school does, and offer audio/visual elements to make the learning experience more immersive or interactive. You can discover educational software that accommodates any age group or subject with a simple web search.

Entertainment programs

Entertainment options are nearly endless on your computer. Don't have a movie theater or arcade in your town? You can find the latest games and movies from your computer while sitting at home and download them to watch anytime. You can do the same with music and video games. With a PC or laptop, you can also watch or listen to your music or movies on a CD, DVD or Blu-ray disk, depending on what your computer is compatible with.

LESSON 5: COMPUTER OPERATING SYSTEMS

Objectives:

- 5.1. EXPLAIN THE COMMON FUNCTIONS OF AN OPERATING SYSTEM.
- 5.2. IDENTIFY THE DIFFERENT COMPONENTS OF THE WINDOWS 8 INTERFACE.
- 5.3. WORK WITH THE WINDOWS 8 INTERFACE WITHIN PROGRAMS.
- 5.4. MANAGE FILES AND FOLDERS IN MICROSOFT WINDOWS EXPLORER.
- 5.5. PERFORM BASIC FILE OPERATIONS.

An operating system performs four primary functions. It manages and controls the hardware connected to a computer. It helps other programs running on a computer to use the hardware. It helps you organize and manage files and folders on the computer. It provides a user interface that allows you to interact with the hardware, the operating system itself, and other programs.

Operating systems, OS, allow you to use files and devices. The OS controls all the programs you are running, by sharing the CPU's time. They also provide help when installing new devices. Most OS now have a graphical user interface which allows more people to use computers. Let's look at how the OS works. Every time you press a key or click your mouse is an example of how signals are processed.

Anytime you create a file or folder is as a result of the file management system which is one of the functions of an operating system.

Functions of an operating system

1. An operating system controls how programs work with each other
2. It also controls how they interact with the computer hardware.
3. It also creates the file system that determines how your data is stored within a storage device.
4. It serves as an interface between the user and the computer
5. It controls and coordinate all other software application

Windows 8 interface

The interface for Windows 8 includes performance improvements and security enhancements.

Windows 8 start screen

The Windows 8.1 start screen allows a new level of personalization. In the start screen, you will be able to customize to your own personal preferences. You can pin your favourite applications (apps), websites, friends, files, and folders, to your start screen. You can also change your background and colour the way you want it to appear.



Personalize your desktop background

The Start screen allows you to customize it the way you prefer it. You can change the start screen with themes, background colours, accent colours, and motion backgrounds. To personalize your computer, you must first access the charms. You can find the charms by moving your cursor to the upper-right corner or the lower-right corner of the screen. The charms will display five different settings: Search, Share, Start, Devices, and Settings. Click Settings, and then click Personalize. You will now be able to see the Personalize tab that allows you to choose from different themes, background colours, and accent colours.

Adding apps

When installing an application, Microsoft's Windows 8.1 operating system will usually create a tile on the Start Screen. In some cases, you will need to add the tile yourself. To add a tile, you must be on the Windows 8 Start Screen, and then left-click the All-Apps button located on the bottom right corner. The All-Apps button will reveal a list of all installed apps. Move your cursor to the desired application that you would like to appear on the Start screen, and then right-click the appropriate application. A panel will appear on the bottom giving you different options. These different options can include: Pin to Start, Pin to taskbar, Open new window, Run as administrator, and Open file location. Click on the Pin to Start button. Now the selected application will be added to the Start Screen. After you have added an application, you will automatically slide back to the Start Screen to view the location of your newly listed application.

Arranging tiles Now

You may want to arrange the tiles on your Start Screen. You can arrange them into specific categories depending on their purpose or however you want them to appear. To move a tile, you must click on a specific tile and hold it. Now, you will be able to drag it to any place on the screen. To arrange multiple tiles into a group, you need to drag one tile to an open space. When the gray bar appears, release the title. You can start drag more tiles to the selected group. Another way to create a group is to right-click all the applications you want to move. After right-clicking the tiles you would like to move, you need to left-click one of the selected tiles and hold it moving towards the desired location.

Changing tile sizes

Another way to personalize your Start screen is by changing tile sizes located on the Start screen. Windows 8.1 offers four different sizes: small, medium, wide, and large. You can set your favorite or most used apps, websites, friends, files, and folders on your start screen. This way you can have easy access to them. Once

you get all your desired applications in the right place on the Start Screen, you will be able to change the sizes of tiles. To resize a tile, you must right-click the application you want to resize. A panel will appear on the bottom of your screen giving you different options. These options may include Unpin from Start, Uninstall, Resize, and Turn live tile off. Click the Resize button. It will give you options of resizing your tiles to small, medium, wide, and large. Some of these options will not appear depending on the application.

Running multiple apps

You may want to run multiple applications at once. The Windows 8.1 allows you to use up to four applications, depending on your screen's resolution. While running these applications, you can snap them, which means that you can adjust the size of each application displayed on the screen. For example, you can now have your email open, while watching a video, and completing a work task. This makes multitasking easier.

Changing your account picture

In the upper-right corner, you might have noticed your account name and a default account picture. You can customize this to meet your personal information. To change your picture, right-click your name or default icon picture, it will give you options such as: Change account picture, Lock, Sign out, and Switch account. If there are multiple signed on registered accounts, you will get an option to sign on. You will be redirected to the Your account menu. In here it will give you the option of connecting to your Microsoft account, changing your account picture, or you can create an account picture. Underneath the "Account picture" section, there should be an option called, Browse. If you already have a picture in your PC, you can locate it by using Browse. Once you have clicked Browse, it will automatically display your Pictures folder. Click the desired picture, or locate the picture you want inside another folder. After finding the picture that you would like to use, click Choose image. You can also create an account picture underneath the section called "Create an account picture". It gives you the option of Camera and People. The Camera option is used when there is a webcam installed onto your computer. You can create an account picture easily like this or you can even create a video to use instead of a photo. To do this, press the Camera option, you will be directed to your installed camera application. This will allow you to take pictures or record videos which will then be your default picture. The People option allows you to sync your picture and all other information from your Microsoft account.

Using the search charm

The Search charm has been provided in Windows 8.1. This will provide global search results powered in Bing to find the best answer for your query. It searches from your local PC by applications, settings, and files, and then it will give you searches from the World Wide Web if you are connected to the Internet. Now, you will be able to search from the Internet without having to open a browser. Microsoft 8.1 Blue version has added this handy feature to provide this to make getting things done easier and quicker. For example, you can now search your favorite artist group, and the Search charm will provide information about them, discography, songs, album names, images, videos, and different websites that feature your search term. To access the Search charm, you can click the Windows key + F or move your cursor to the upper-right or lower-right corner. You will be able to view your charms, click on the Search charm.

Start button

One of the major features that Microsoft has brought back in Windows 8.1 was the Start button. You can quickly access the Start screen by moving your cursor to the bottom-left corner of the screen in Windows 8.1 or by clicking the Windows icon located on your keyboard. When right-clicking the Start button, you will be able to see the PC settings. Windows 8.1 offers the updated Settings feature meaning that you will not have to go into your Control Panel. You have access to Programs and Features, Power Options, System, Task Manager, Computer Management, Search, Shut Down, and such. When you have a program open, clicking the Start button will allow you to go back to the Start screen. Windows 8.1 gives a clean and easy way to access your applications from your Start screen. You may prefer to find these applications directly

from your desktop or from a different Start screen. You change the Windows 8.1 Start screen by going to your desktop. To go to your desktop, press the application that says Desktop or you can locate it through Apps. Once you are on the desktop screen, right-click on the Windows 8.1 taskbar, and then choose Properties. Click on the Navigation tab, then under the Start screen section, you are able to view what options you can choose from. These options should include: Go to the desktop instead of Start when I sign in, show my desktop background on Start, always show Start on my main display when I press the Windows logo key, show the Apps view automatically when I go to Start, Search everywhere instead of just my apps when I search from the Apps view, and List desktop apps first in the Apps view when it's sorted by category. If you want to load to the Start screen, click on the box next to Go to the desktop instead of Start when I sign in and click Apply. When applied, every time you start your computer, you will be directed straight to the desktop view.

Break: 1 Hour

HOURL THREE (3)

COURSE 2: NAVIGATING THE INTERNET AND THE WORLD WIDE WEB (SEARCH ENGINES, WEB

BROWSING, EVALUATING INFORMATION ON THE INTERNET)

COURSE DESCRIPTION

This course shows students how to connect to the Internet; browse Web sites; use search engines; exchange e-mail and instant messages with others; explore the features of online communities; and it explains how Web authoring software is used to create and publish Web pages.

LESSON 1: THE INTERNET

OBJECTIVES:

- 1.1. DEFINE WHAT THE INTERNET IS AND ELABORATE ON ITS USES.
- 1.2. IDENTIFY THE DIFFERENT COMPONENTS REQUIRED FOR AN INTERNET CONNECTION.
- 1.3. IDENTIFY THE FEATURES OF DIFFERENT TYPES OF INTERNET CONNECTIONS.
- 1.4. EXPLAIN THE MEANING OF THE TERM BANDWIDTH IN RELATION TO THE DIFFERENT TYPES OF INTERNET CONNECTIONS.

The Internet is an international computer network that allows you to go to websites for certain objectives, such as:

- Email
- Information
- Education
- Entertainment

- Communication
- And more

Imagine a scenario where you want to exchange information between a computer on your network and a computer on another network. For this exchange, you can use the Internet to connect both the networks. The Internet is a collection of networks that are connected together to exchange information. Today, the Internet is being used worldwide.

A new trend on the Internet is cloud services, which allows users to store data or run programs on remote machines and access that data on any computer. A cloud service is replicated to several distribution centres across the globe. This replication allows other machines to resume activity of a cloud service if the active centre experiences an interruption in service or loses connection.

Uses of the Internet

1. You can do the following in the Internet:
2. Perform banking transactions
3. Communicate with people all around the world
4. Obtain up-to-date news on world events
5. Search for information on any topic
6. Study any course of your choice and even take exams
7. Access entertainment including music, games, TV, movies, and pictures
8. Buy and sell products

The requirements for an Internet connection

Connecting to the Internet requires specific hardware and services. You need to have a computer and a connection device such as a modem. Service is provided by an Internet service provider (ISP). Let's look at what is required to connect to the Internet.

1. To connect to the Internet, you need a computing device, a connection device, and an Internet Service Provider, or ISP.



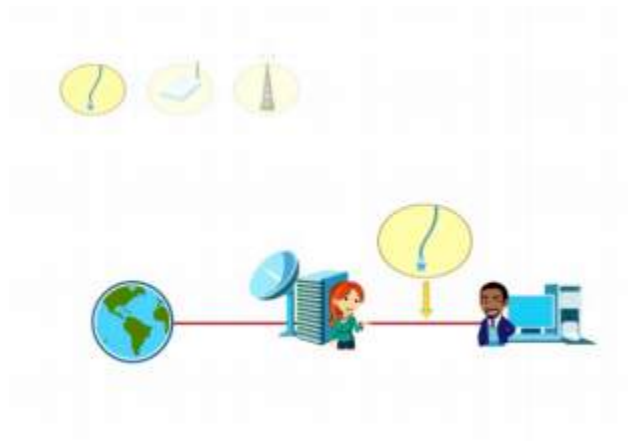
2. The computing device can be a personal computer, a portable computer, a tablet computer, or even a mobile device, such as a cell phone, a smart phone, or a handheld device



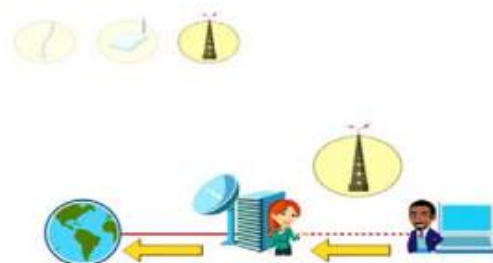
3. To connect your computer to the Internet, you need a connection device, such as a modem. A modem converts digital information into analog information and transmits it over a phone line.



4. You can use cables to connect to the Internet. Such a connection is called a physical connection.



5. Another device you need for wireless connectivity is an access point (AP). This is used to connect a wireless computing device to a wired network. This wired network may belong to an ISP. You can then connect to the Internet through the ISP.



6. The wireless device such as a USB modem provides access to the Internet via 4G, 3G, or general packet radio service (GPRS), or circuit switched data (CSD).



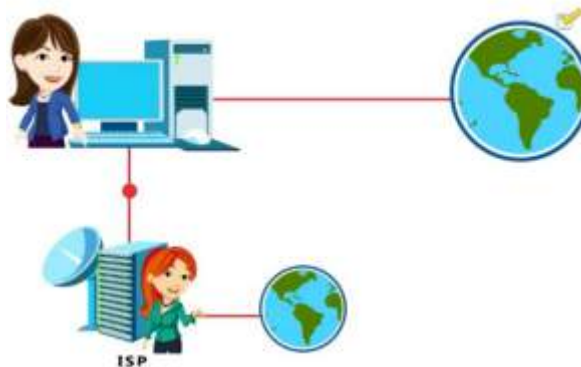
Features of two types of Internet connections

There are two common ways to connect to the Internet; dial up and dedicated connections. A dial up connection is less expensive but can be very slow. Dedicated connections are very fast but require a certain infrastructure that may not exist in rural areas. Please continue to view video demonstrations of the two types of Internet connections.

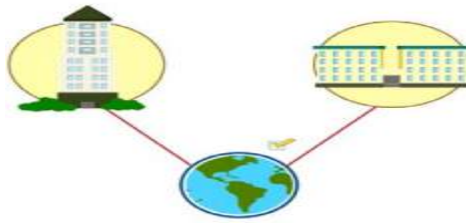
Dedicated connection video: A dedicated Internet connection keeps the computer connected to the Internet all the time.



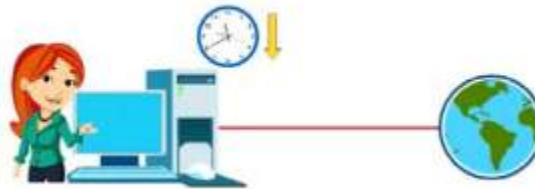
In a dedicated Internet connection, you do not need to request your ISP for a connection every time you want to connect to the Internet.



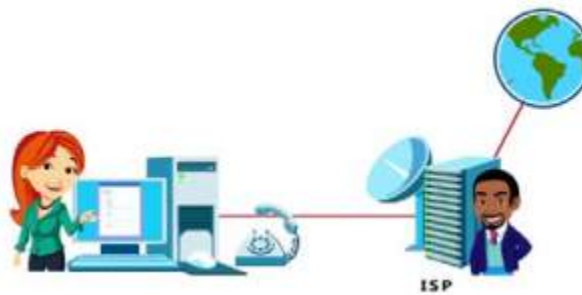
Many large organizations and universities use dedicated Internet connections because these organizations and universities need to use the Internet extensively.



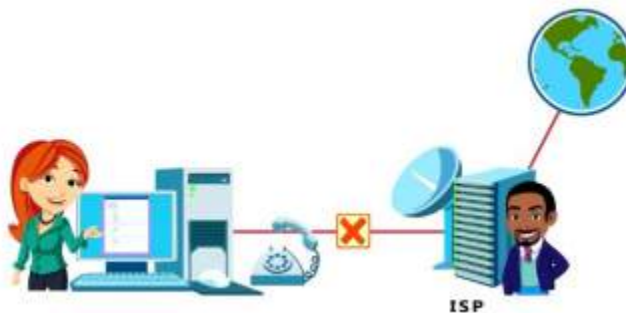
Dial-up connection video: This worker only wants to be connected to the Internet for only a short time, and therefore uses a dial-up Internet connection.



In a dial-up connection, you need to connect the ISP every time you want to connect to the Internet.



You can disconnect from the Internet after completing your work.



Bandwidth and Internet connections

Bandwidth is the amount of data that can be transmitted over a network in a certain amount of time and is measured in Mbit/s (megabits per second), Kbit/s (kilobits per second), or bit/s (bits per second). If the bandwidth of a network is 1 Mbit/s, then 1 megabit of data can be transmitted over that network in 1 second. The actual rate of transfer of data will however vary depending on a number of factors such as the equipment of the ISP, the type of Internet connection, and the number of people using the same connection at a single point in time.

LESSON 2: THE WORLD WIDE WEB

OBJECTIVES:

- 2.1. DEFINE THE WORLD WIDE WEB AND ELABORATE ON ITS USES.
- 2.2. EXPLAIN HOW WEB ADDRESSES WORK.
- 2.3. EXPLAIN HOW TO USE A BROWSER TO NAVIGATE THE WEB.

Web components

The Internet is a series of connected networks. To access information on the Internet you need to know how and where to access the specific information or network. We use Internet Protocol (IP) addresses to label and recognize websites and Internet users. Web browsers access these IP addresses to open web pages or networks. Continue to the next simulation to see how IP addresses work.

The basic features of Internet Explorer 11

You can use the Internet to avail a range of services, such as transferring files between Internet users, sending messages through electronic mail, and browsing for recent news and information in Web sites. You can also use the Internet and its services to search and apply for jobs, watch movies, and buy and sell products. In this simulation, you will learn about the various features of Internet Explorer 11.

1. Click the URL in the navigation bar at the bottom of the window.
2. To visit a website type in the URL of the webpage you want to view and press enter.



3. To add a webpage to your favourites list, click the “Favourites” button on the navigation bar to open the favourites list.



4. Click the “Add to favourites” button, Then click “Add” to finish adding the web page to your favourites.



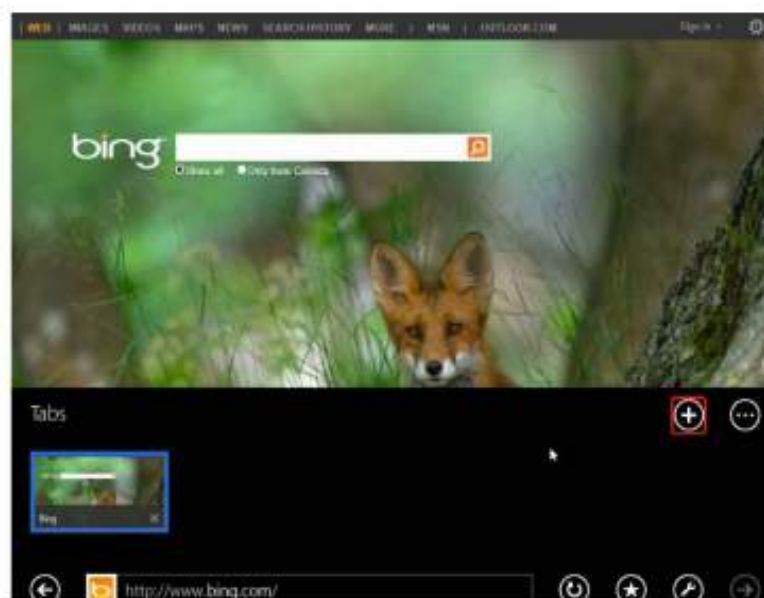
5. You can go back to the page you were previously on by clicking the “Back” button on the navigation bar



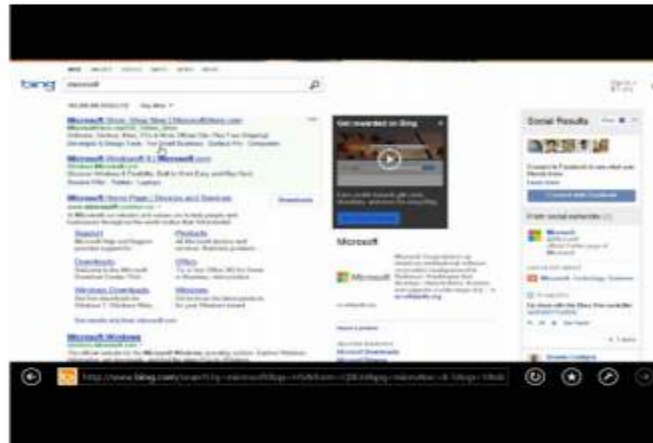
6. After going back to the previous page, you can go forward again by clicking the “Forward” button.



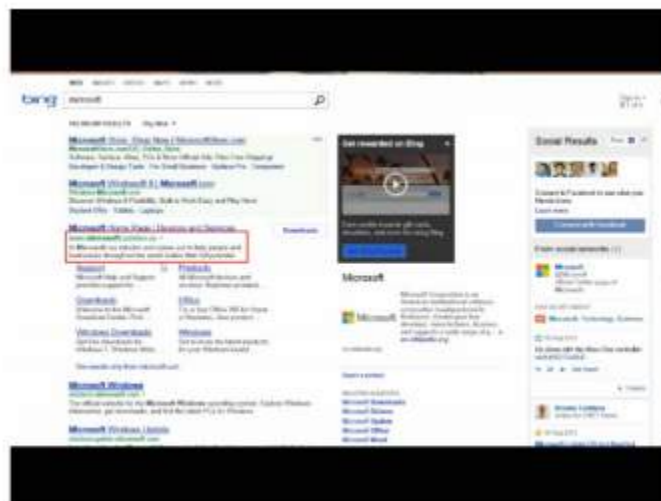
7. Right click over a blank area of the current page to open the tabs menu and then click the “New tab” button to open a new tab.
8. You can then type in the URL for the website you wish to view.
9. You can quickly switch between tabs by clicking on them in the tabs menu. You can close tabs by clicking the “X” button on the tab you wish to close.



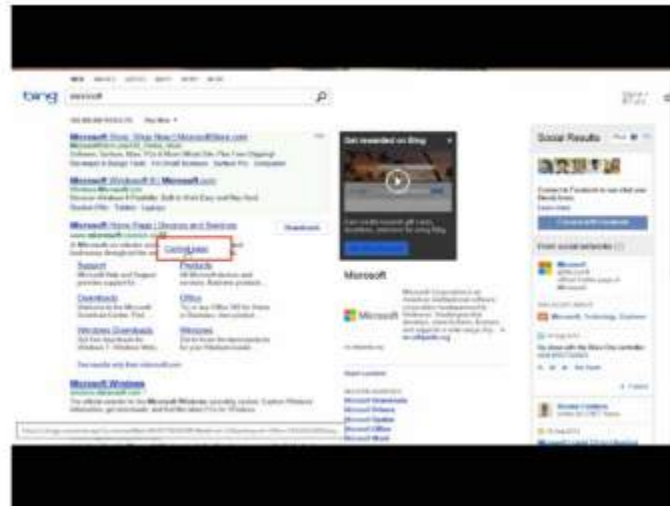
2. To search with Bing type what you are looking for in the search box.
3. Related suggestions will appear below the search box on a results page.
4. The results page will show you how many results match your keyword or keywords and will also show you related searches for your keyword or keywords.



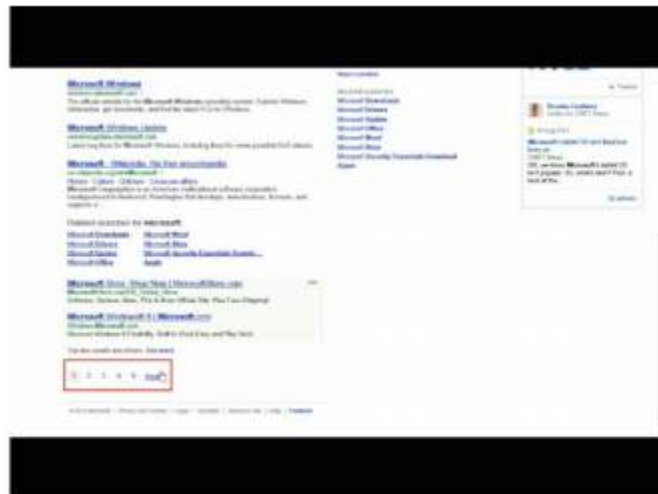
5. Each result contains a brief description and a link to the website.



6. If the website is not available, the search engine may have a saved version of that website. This 'out of date' version can be viewed by selecting the 'Cached Page' command.



7. Your search may result in thousands of web pages. Each result page only contains a few results, to see the rest you can select the links at the bottom of the page.



E-commerce

Businesses have websites on the Internet that allow you to perform online transactions. These types of transactions include buying or selling merchandise, ordering tickets to a location or event, and performing banking transactions. Let's look at what is involved with online transactions.

- An online transaction refers to a business transaction that is made over electronic systems, such as the Internet.
- You can buy and sell items and gifts online with a debit or credit card.



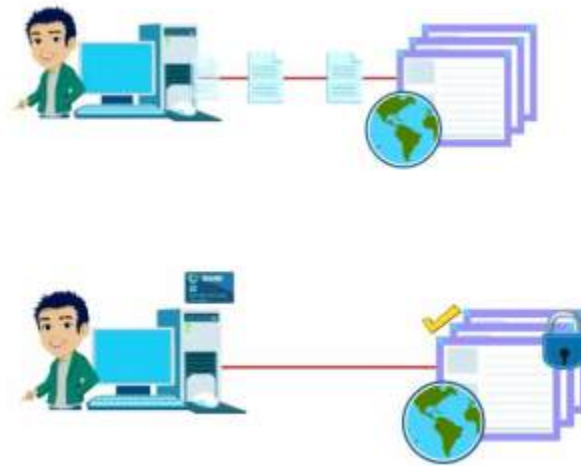
- ATM transactions such as deposits, withdrawals, inquiries, and transfers are some examples of online transactions.



- An online transaction usually requires you to provide personal information such as user name, password, address, and bank account or credit card details.



- Because there is a great deal of personal information exchanged in an online transaction, you must ensure that the transactions are done in a secure manner and only on credible Websites.



LESSON 3: COMMUNICATING ON THE INTERNET

Objectives:

- 3.1. EXPLAIN HOW E-MAIL WORKS.
- 3.2. WRITE AND SEND E-MAIL MESSAGES.
- 3.3. MANAGE EMAIL MESSAGES
- 3.4. IDENTIFY THE FEATURES OF ONLINE COMMUNITIES.
- 3.5. EXPLAIN HOW INSTANT MESSAGING WORKS.
- 3.6. EXPLAIN HOW WEB AUTHORING SOFTWARE IS USED TO CREATE AND PUBLISH WEB PAGES.

Using Email

Electronic mail or email is similar to traditional postal service mail. Instead of a pen and a paper, you use a software program to write your message, and a network server takes the place of the postal employee in delivering a message to its recipient. A message can travel halfway around the world in seconds, and can have the following content:

- Text
- Pictures
- Data Files
- Audio
- Video Clips
- And more

How email works

Electronic mail, commonly known as email, allows you to communicate with friends and family through the Internet. Email is a great way to communicate and share information quickly. In this lesson we will learn about email.

Create an email address

To use email, you need to register with an email service. In this lesson we will show you how to create a Hotmail account through Microsoft Live. Creating an email is as easy as 1, 2, 3. By using your email account, you can send emails to multiple email clients all around the world. Sending an email is a lot like sending a text message on a phone. You need the other person's phone number; where in an email, you need their email domain. Once the recipient has been chosen and the subject for the email has been written, we move from the header to the body. The body of the email is much like the actual text sent in a text message. A text message can hold a picture or video along with text, much like an email. Except, an email can hold much more; it can include files such as Word documents, PowerPoint documents, and more.

1. First go to www.live.com by typing it into Internet Explorer.



2. First time users need to select the “Sign up now” link



- Microsoft asks for your name, birthdate and gender.

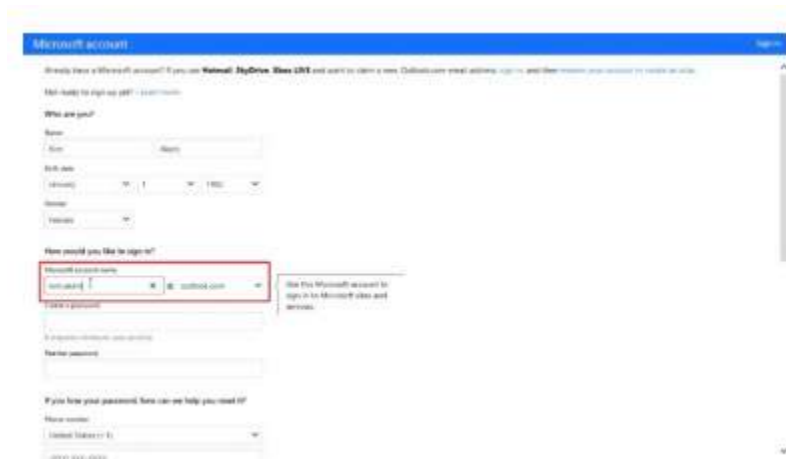


The screenshot shows the 'Microsoft account' creation page. The 'What are you?' section is highlighted with a red box. It contains the following fields:

- Name:** First name (text input) and Last name (text input).
- Birth date:** A date picker showing '1/1/1900'.
- Gender:** A dropdown menu with options: Male, Female, Not specified.

Below this section, there are fields for 'Microsoft account name' (text input), 'Create a password' (password input), and 'Verify your password' (password input). At the bottom, there is a section for 'Phone number' with a dropdown for 'Country/Region' and a text input for the number.

- Your Microsoft account name needs to be unique, try some variations if the first choice you want is not available.



This screenshot is similar to the previous one, but the 'Microsoft account name' field is highlighted with a red box. To the right of this field, there is a message: 'Use this Microsoft account to sign in to Microsoft sites and devices.' Below the 'Microsoft account name' field, there is a 'Create a password' field and a 'Verify your password' field. At the bottom, there is a section for 'Phone number' with a dropdown for 'Country/Region' and a text input for the number.

5. Passwords must have at least 8 characters and contain at least two of the following: uppercase letters, lowercase letters, numbers, and symbols.



The screenshot shows the Microsoft account creation page. The 'Create a password' section is highlighted with a red box. It includes a text input field for the password, a strength indicator showing 'A strong password', and a 'Remember password' checkbox. Below this, there is a section for 'Phone number' and 'Email address'.

6. Microsoft needs two ways to contact you in case you need to reset your account password. i.e phone number or another different email



The screenshot shows the Microsoft account creation page. The 'Phone number' and 'Email address' sections are highlighted with a red box. The 'Phone number' field is labeled 'Phone number' and 'Country/region'. The 'Email address' field is labeled 'Email address' and 'Country/region'. A tooltip on the right side of the page states: 'If you need to reset your password, we'll send a code to this phone number or email address to help you reset it.'

7. Microsoft will store a security question just in case you do need to reset your account password.



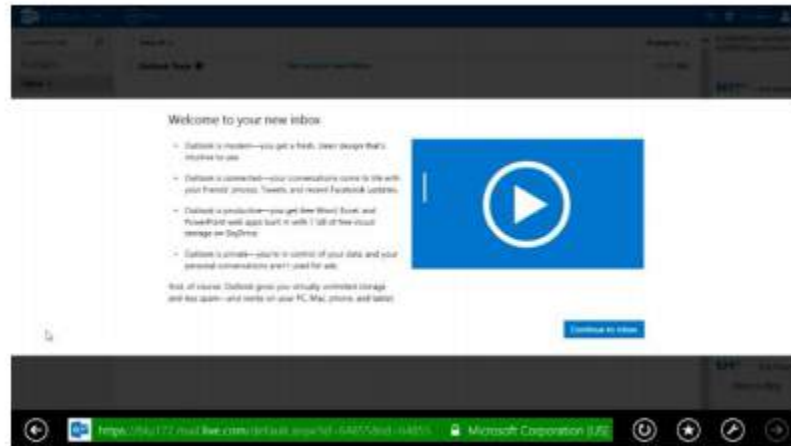
The screenshot shows the Microsoft account creation page. The 'Security question' section is highlighted with a red box. It includes a text input field for the security question, a 'Remember this question' checkbox, and a 'Show hint' button. Below this, there is a section for 'Where are you from?' with a 'Country/region' dropdown menu.

8. Make sure all the required fields are filled in depending on your country or region.

9. Finally, captchas are used to make sure accounts are created by a human and not another program.

10. All that is left is to accept the service agreement

11. You will be presented with a video when you login the first time. Take the tour and learn how to use your email account.



Identify email etiquette

It is always best to use proper etiquette when writing emails. How you write depends greatly on whether you are writing an email for business or friends and family just like writing formal and informal letter.

Features of online communities

The Internet is full of online communities. An online community is a gathering of people who share a similar interest. A community shared interest might be a favourite band, a common hobby, or a local sports team. In this lesson we will look at the many types of communities the Internet has to offer.

Instant messaging and web-cam video communication

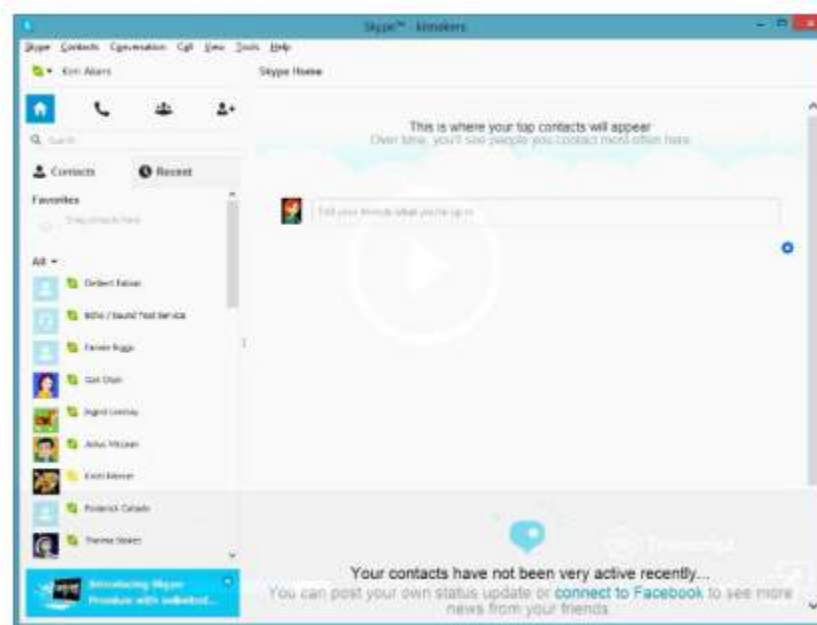
Instant messaging, IM, lets you communicate with friends and family instantly. IM software also lets you share files and images with contacts. Some IM software like Microsoft's Skype lets you also talk and see your contacts if microphones and cameras are supported. In this lesson we will learn how to use Microsoft's Skype.

Microsoft Skype

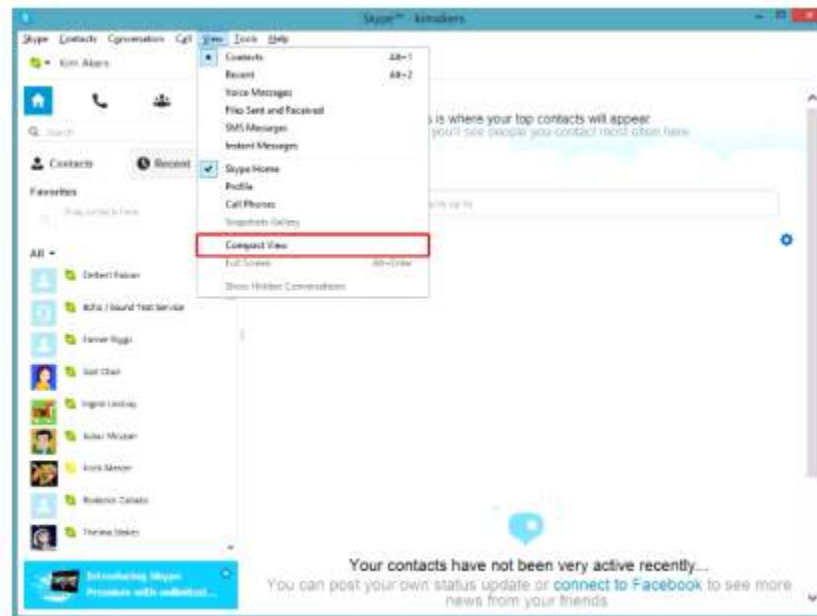
In this lesson we will use Microsoft's Skype. Once you install the software you will need to create an account and sign in



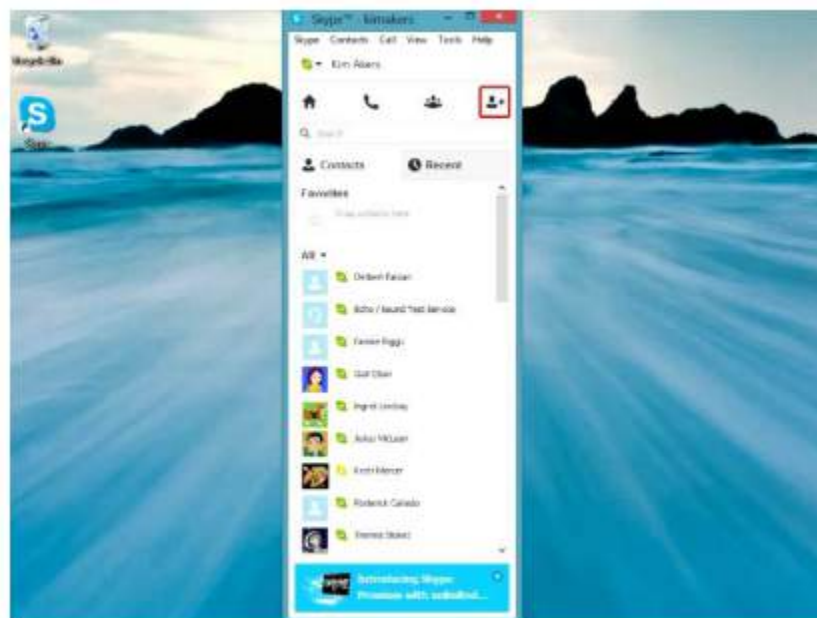
1. When you first open Skype, it will be in full view. This mode shows all your contacts and news feeds from social websites.



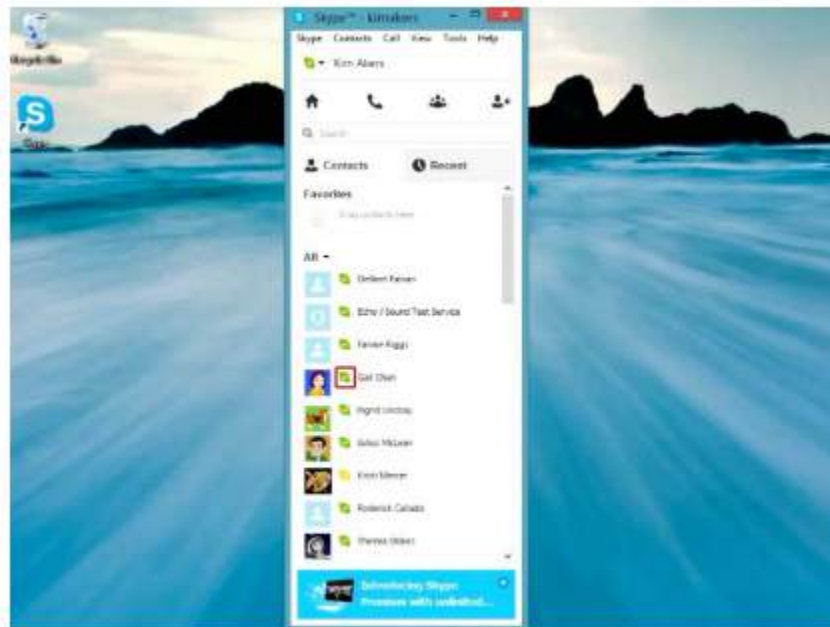
2. If you are only interested in seeing which contacts are online change to compact View from the View menu. Compact view will separate your contacts from the newsfeed so you can close the news window.



3. Contacts can be added by selecting the 'Add a Contact' command on the right and typing a name in the text box below. Once your contact confirms your invitation, you can communicate with them.



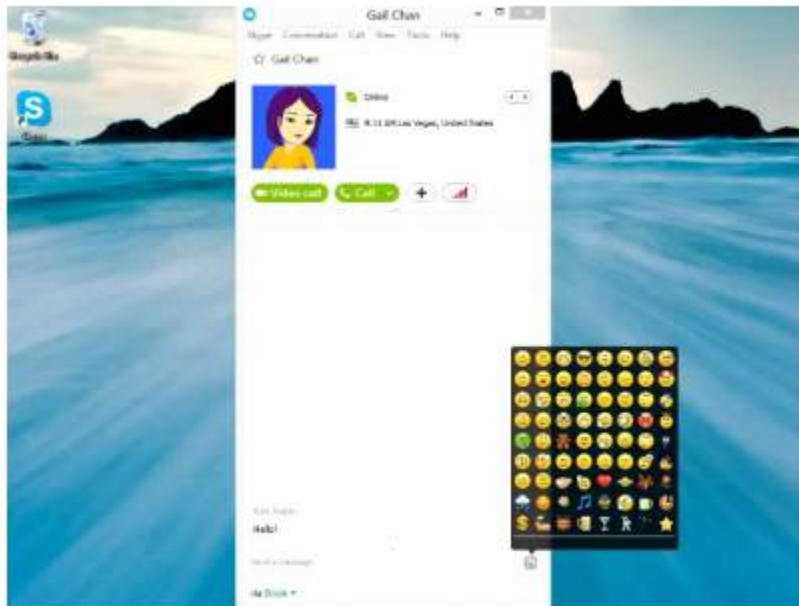
- 



5. To send a text message double click on the contact and start typing in the text area.



- Text messages can be enhanced by adding emoticons from the selection on the right-hand side.

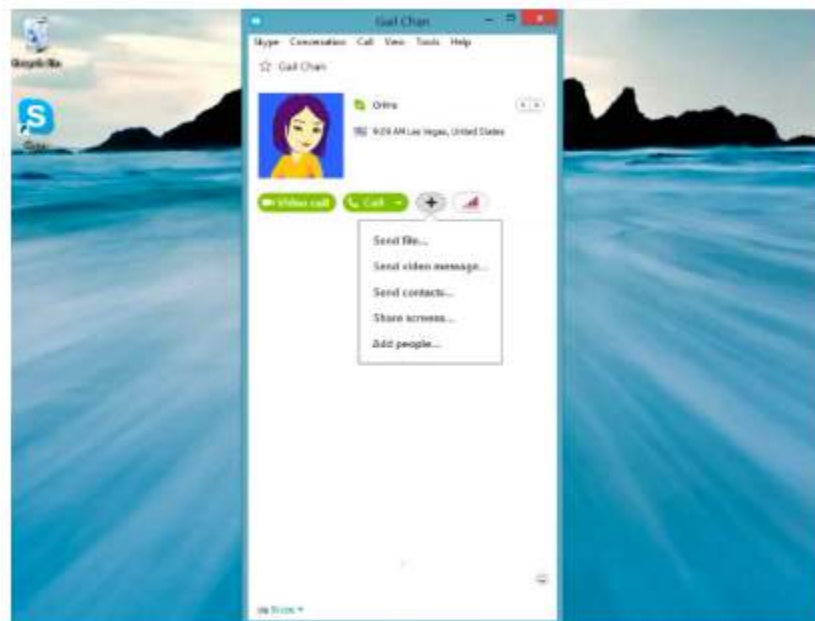


- If your computer has a microphone and a webcam you can select the 'Call' or 'Video Call' button to start a voice or video chat.





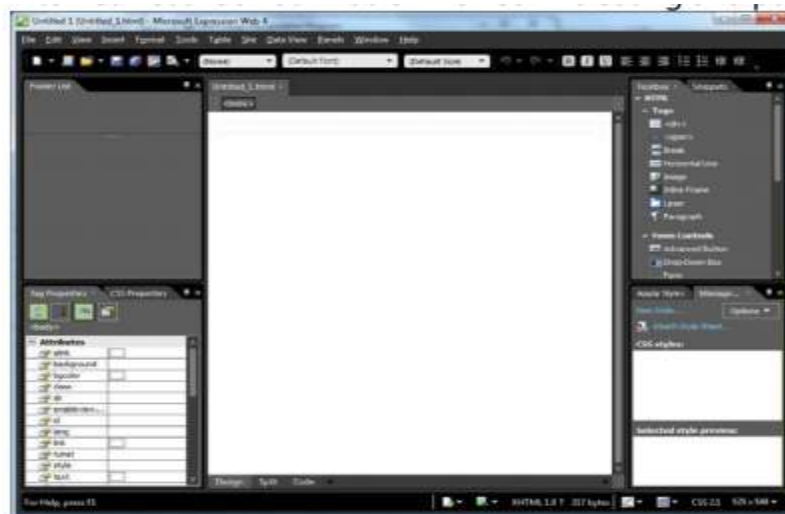
8. Finally, IM software lets you share files with contacts. Simply press the 'Plus sign' to see your options.



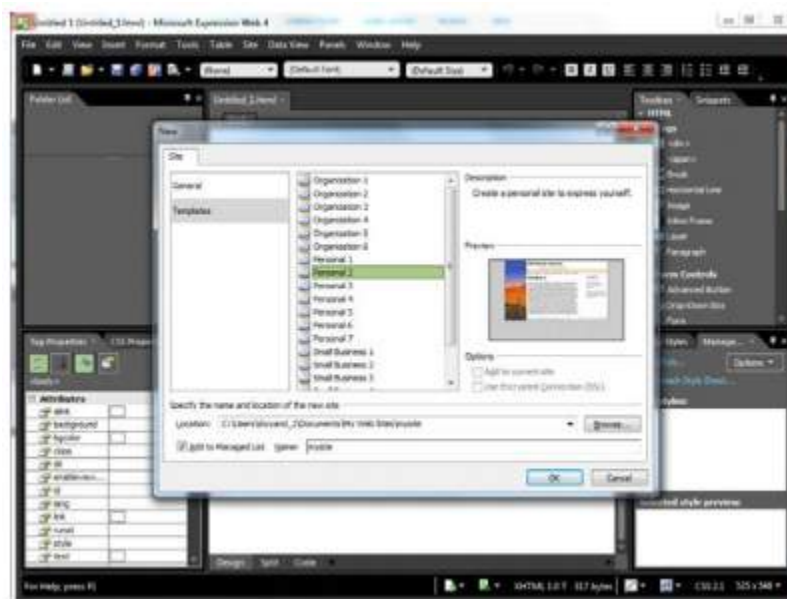
How to create and publish web pages

People and businesses have websites on the Internet that provide information about themselves or a service, allow for online transactions, or to perform searches of data. Using software such as Expression Web 4 it is easy to set up a web page and publish it to the Internet. Let's look at what is involved in creating and publishing a web site.

Expression Web 4 may be used to create an entire web site based on a template. We begin by clicking on the New Site button.



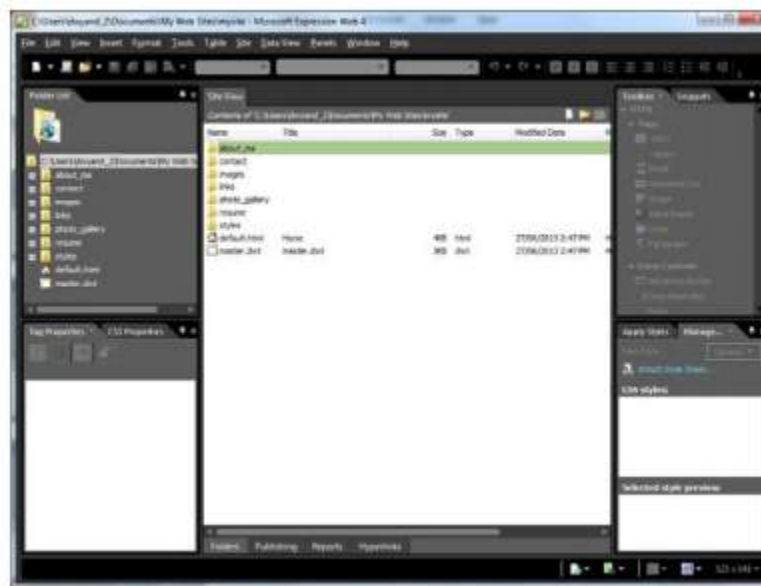
From the New Site dialog, we click on the Templates category and choose one of the personal website templates. We then give our website a name and click on the OK button.



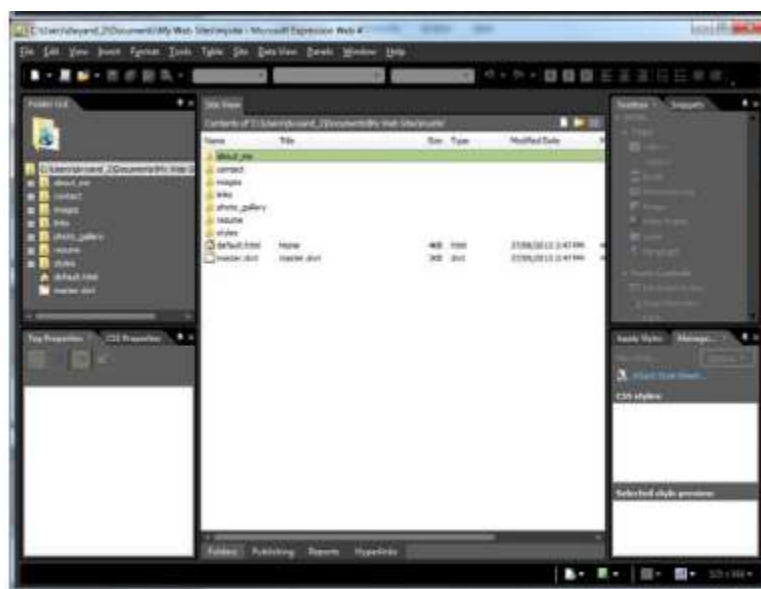
Expression Web 4 then builds all of the necessary files and folder structure for our web site.

On the left is the Folder List tab, this tab provides easy access to all of our web site's files.

In the middle is the Site View tab where we can edit our web site's pages.



We will place a horizontal line just below our heading. We will now edit our web site's home page, which is contained within the default.html file. Double click on the file and Expression Web 4 opens the page for us to make it ready for editing



While in the Design view we are able to make changes right on the web page itself. We will now change the text, with a content header.



And change the style to heading three.



The right side of Expression Web 4 is the Toolbox where we can drag out various HTML tags and controls onto our web page.



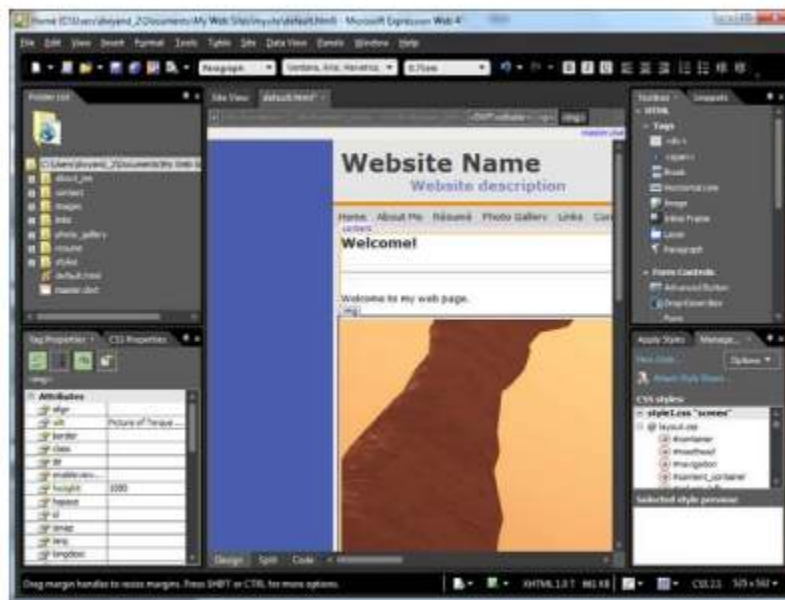
To change the body text of the web page we first select it and then enter in our text.



We may also add images to our web page by dragging and dropping the Image tag from the Toolbar. Next, double click on the image icon to open the picture properties dialog box.



Expression Web 4 allows us to resize the image by changing its properties. I will make the image smaller by changing its height and width attributes.





We may have the image take the viewer to another web page or even another web site by adding a link to it. To add a link using Expression Web 4 we right click on the image and choose the Hyperlink item.

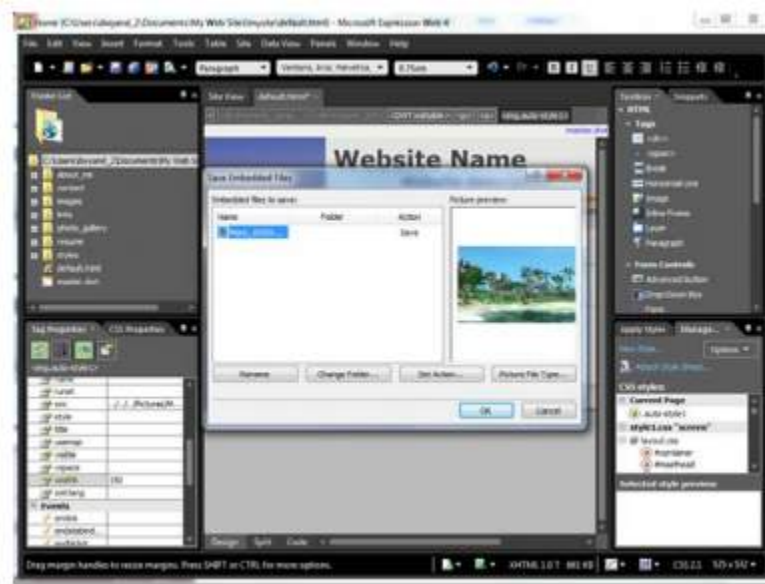


Next, type or paste in the link to the web page we wish the user to go to if they click the image and click on the OK button.

With our changes complete we click on the Save button to store our changes

When our web page contains images or other files that are embedded within it, Expression Web 4 makes a copy of that image and places it within our file list. We may then close the preview and continue to modify our web site in Expression Web 4.

We can confirm this by clicking on the OK button of the Save Embedded Files dialog



To see how our web page looks in a web browser we go to the File menu and select Preview In Browser.

By previewing our website, we ensure that the viewer's experience matches our design.

In this demonstration you saw how to create and publish your webpage by using Expression Web



DAY TWO: HOURS ALLOCATED: 3 HOURS

HOUR ONE (1)

COURSE 3: PRODUCTIVITY PROGRAMS (PRODUCTIVITY TOOLS, E.G. SPREADSHEETS, WORD PROCESSORS, PRESENTATIONS SOFTWARE AND GRAPHICS PROGRAMS)

COURSE DESCRIPTION

This course explores the most common productivity software applications used in business, in education, and at home. Students are taught to select the right software for a project, and how to perform basic tasks by using word processing, spreadsheets, presentation software, and database software.

LESSON 1: COMMON FEATURES AND COMMANDS

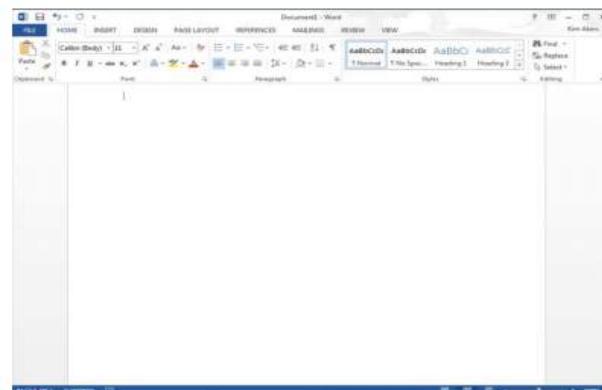
OBJECTIVES:

- 1.1. IDENTIFY THE MAIN COMPONENTS OF THE USER INTERFACE.
- 1.2. IDENTIFY THE PURPOSE OF THE COMMANDS ON THE MENU BAR.
- 1.3. WORK WITH THE BUTTONS ON THE TOOLBAR.
- 1.4. WORK WITH THE POINTER IN A PROGRAM.
- 1.5. WORK WITH TEXT AND CHARACTERS IN A PROGRAM.
- 1.6. EXPLAIN THE USE OF PRIMARY KEYBOARD SHORTCUTS AND KEY COMBINATIONS.

Introduction to User Interface

The user interface is the means by which you interact with the productivity programs in Office 2013. You will find many common user-interface features and commands across all the Office 2013 applications.

This lesson will introduce you to the common user interface you will see in office 2013 productivity programs.

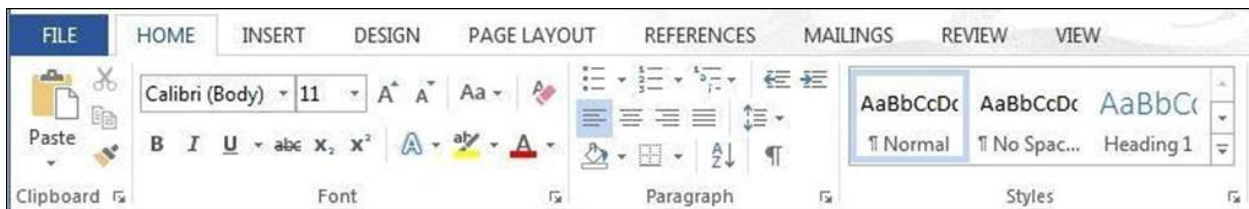


Quick Access Toolbar: This is a customisable toolbar placed by default above the Ribbon - you can add icons for frequently used commands to this toolbar. It can also be placed below the Ribbon.



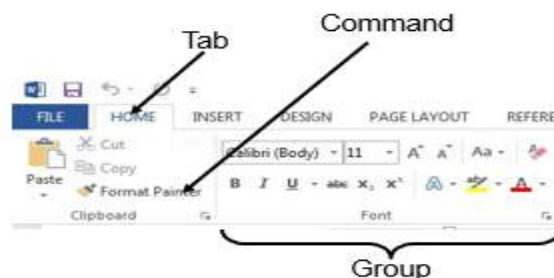
Status Bar: Located at the bottom of the screen, this provides information about the total number of pages, total number of words, and different layouts, including zoom options.

Ribbon: The Ribbon displays commands used to work with your document. Commands related to each other are arranged in named groups. Buttons next to the group names launch dialog boxes which allow you to access more commands and options.



Parts of the Ribbon: The Ribbon consists of three parts:

1. Tabs
2. Commands
3. Groups



In productivity applications, such as Microsoft Word 2013, Excel 2013, and PowerPoint 2013, the ribbon consists of tabs. Each tab is divided into groups. A group helps you perform a set of similar tasks, such as adding picture shapes in a document.

The ribbon also consists of contextual tabs, which appear only when necessary. For example, in a Word 2013 document, the contextual tab for editing a table appears when a table is inserted in the document.

Using the pointer

A pointer is the on-screen arrow or shape that moves when you move your mouse or other pointing device. Using the pointer, you can quickly move around a document and perform commands.

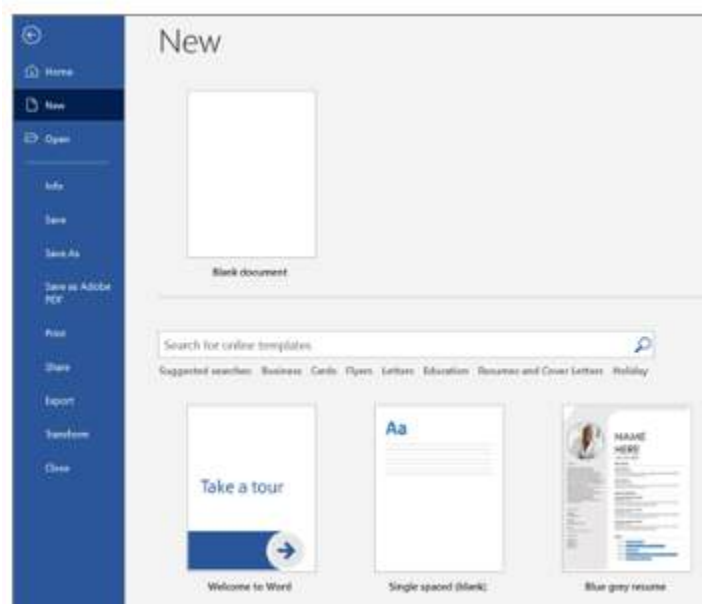
A **cursor** is a blinking vertical bar on the screen that indicates the location where you can insert text or graphics in a document. For example, in Word 2013, you can set the location of the cursor by moving the pointer to some text and then clicking the left mouse button.

Work with documents in Word for Windows

Start your work in Word with these quick instructions on how to create, share, and edit documents on your own or with your team.

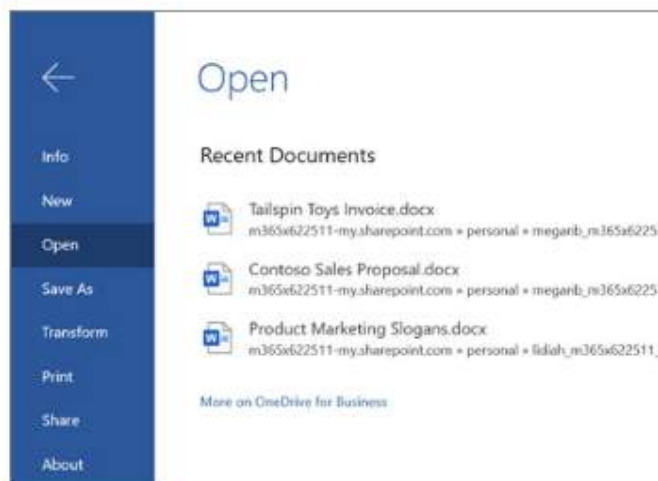
Create a document: Select File > Home.

Select Blank document, select one of the templates, or open a Recommended or Recent file.



Open a document: Select File > Open, and select the document you want.

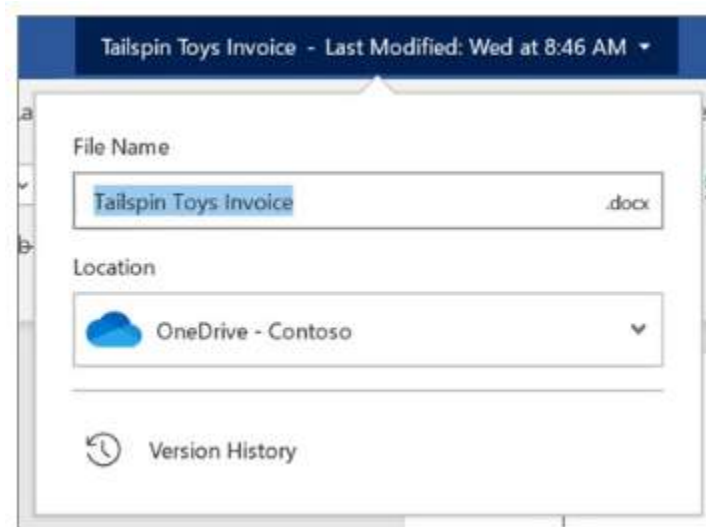
If your files are saved to OneDrive and you're working offline, your revisions will be saved the next time you connect online.



Rename a document: To rename your document:

If your files are saved to OneDrive, select the file name at the top, and type in what you want.

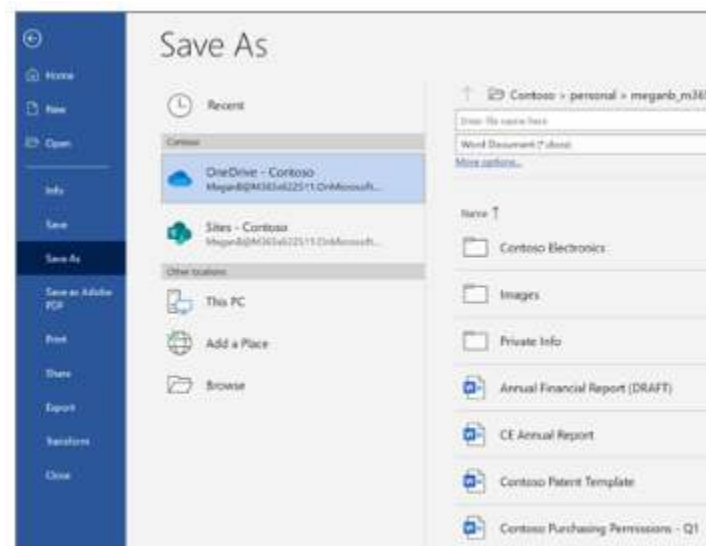
Select File > Save As, choose a location to save to, and rename your file.



Choose a save option

Save your document to OneDrive If your file's been saved to OneDrive, your changes are automatically saved.

Select File > Save As. Select OneDrive. Type in a name and select Save. Other save options



There are more ways to save your document. First, select File:

To save your file, select Save, or select File > Save.

To save your file as a PDF, select File > Save as Adobe PDF.

Keyboard Review

You should now have a basic understanding of how to use a keyboard. Keep the following in mind:

1. Alphanumeric keys are A-Z and 0-9.
2. Modifier keys are Alt, Shift, Ctrl, and Cmd (Mac only).
3. Action keys are the Spacebar, Enter, Delete, Backspace, the arrow keys, Tab, and Caps Lock.

Keyboard shortcuts

In addition to entering information by using single keystrokes and simple key combinations, you can use advanced keyboard shortcuts and key combinations, which speed up your productivity.

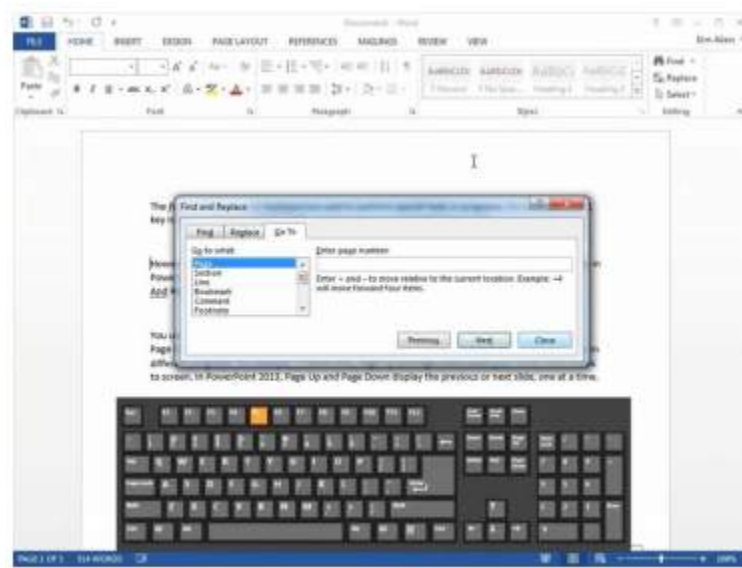
The function keys on your keyboard are used to perform special tasks in programs.



For example, the F1 key is most commonly used to display a Help page associated with the program.



For example, the F5 key in PowerPoint 2013 is used to start a slide show, whereas in Word 2013, the same key displays the Find and Replace dialog box.



You use the navigation keys to move around in a document. These keys include Home, End, Page Up, Page Down, and the four arrow keys. What the navigation keys do is similar, but not exactly the same, in different programs.

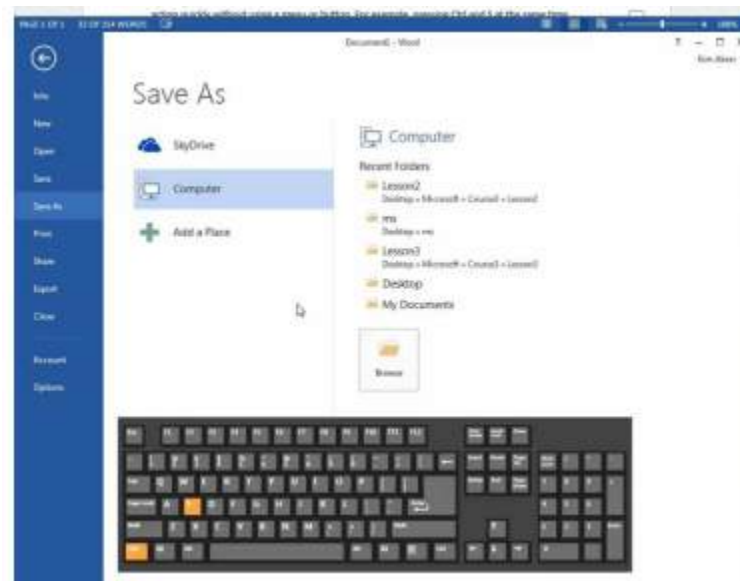


You can even hold down Shift while pressing Page Up or Page Down to select large chunks of text.



By using keyboard shortcuts, in which two or more keys are pressed simultaneously, you can perform an action quickly without using a menu or button.

For example, pressing Ctrl and S at the same time. Ctrl+S is the shortcut to save an open document.



You can press the Alt key to display letters and numbers on the File menu and on the ribbon. You can press Alt and one of the letters displayed to open that menu without using your mouse.



For example, pressing Alt and F together displays the File menu.



Keyboard shortcuts review

You should now have a basic understanding of how to use keyboard shortcuts. Keep the following in mind:

- Modifier keys are Alt, Shift, Ctrl, and Cmd (Mac only).
- A key combination consists of pressing two or more keys.
- Shortcut keys will perform an action quickly

LESSON 2: WORD PROCESSING

OBJECTIVES:

- 2.1. PERFORM BASIC TASKS BY USING A WORD PROCESSOR
- 2.2. EDIT AND FORMAT TEXT.
- 2.3. WORK WITH TABLES AND PICTURES.
- 2.4. WORK WITH LANGUAGE TOOLS.
- 2.5. IDENTIFY THE VARIOUS BENEFITS OF USING DESKTOP PUBLISHING (DTP).

Introduction to Word Processors

Microsoft Word 2013 is a word processor that can help you create certain items; which are listed below:

- Professional resumes
- Agendas
- Brochures
- Certificates
- Birthday cards
- And more

In this lesson, you will learn about the basic tasks that you can perform by using a word processor. You will learn to work with text, tables, and pictures.

Basic tasks in Word 2013

A word processor is a software program that you use to create and work with text on a computer. It performs the same functions as paper, a pen, a typewriter or keyboard, an eraser, and a dictionary. Most computers come with basic word processors such as Notepad and WordPad. You can use advanced word processors, such as Word 2013, to perform more complex tasks that include checking for spelling errors, finding synonyms, inserting graphics, and printing documents.

Creating a document by using a template

Office 2013 provides various types of templates. These templates contain predefined formats and layouts that you can use to create a document quickly. Standard templates are available locally on your computer. However, advanced templates are also available on the Office.com website.

Edit and formatting text

Word 2013 helps you to easily cut, copy, and paste text, undo and redo actions, justify text, change font parameters and paragraph spacing, and perform various other editing and formatting functions. In this demonstration, you will see how to perform basic tasks that are related to editing and formatting of text in Word 2013.

With Word 2013, you can easily change the text of your document and make it look more interesting. One way to move text in your document is to copy and paste it.

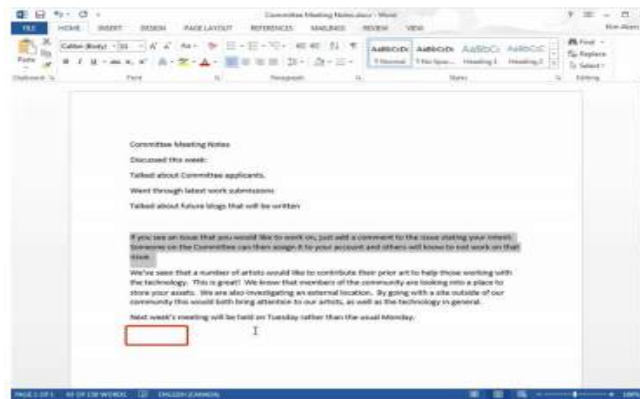
The copy option in Word 2013 helps you duplicate information. When you select and then copy text no changes are made to the original copied text.



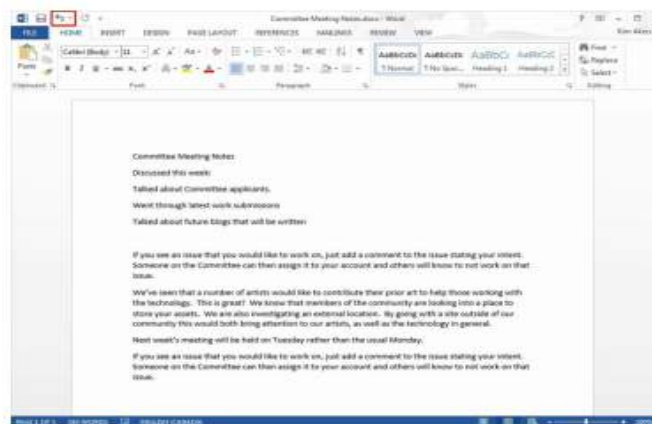
Start by selecting text to copy with your mouse or the keyboard and then click the Copy button in the toolbar or press control and C.



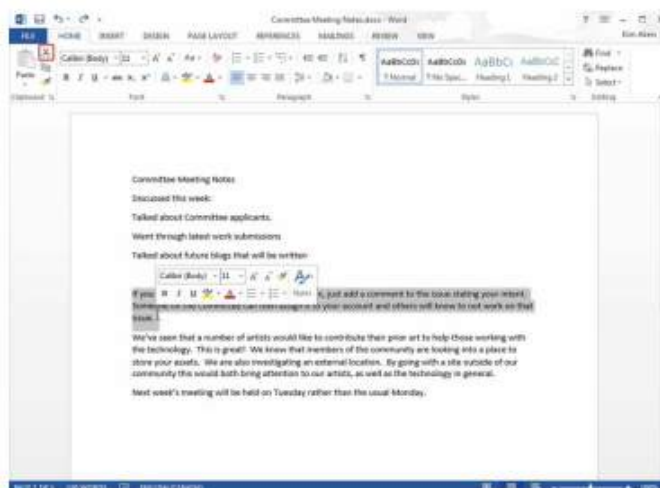
The Paste option inserts text that has been cut or copied from a document into a different location. That can be within the same document or even into a different document.



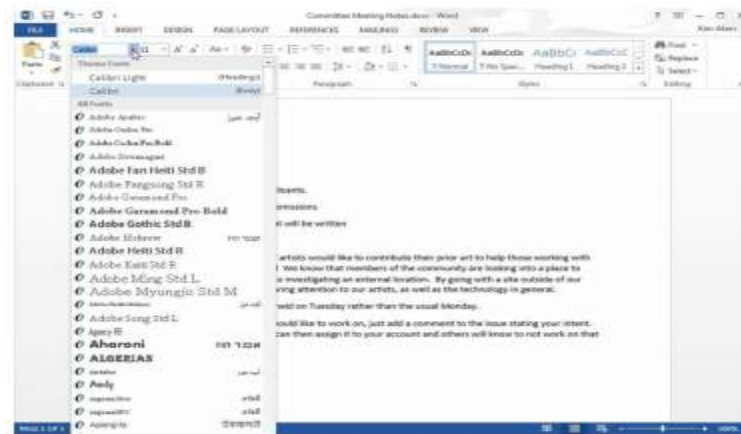
You can reverse specific and repeat actions that you performed. Click the undo button or press control Z to reverse your last action. Pressing it again will reverse the action that happened before that and so on.



You can use the Cut option to remove selected text from its original location by clicking the cut command or pressing control X.



You can change your selected text by clicking the Font selection box and picking an entry from the list.



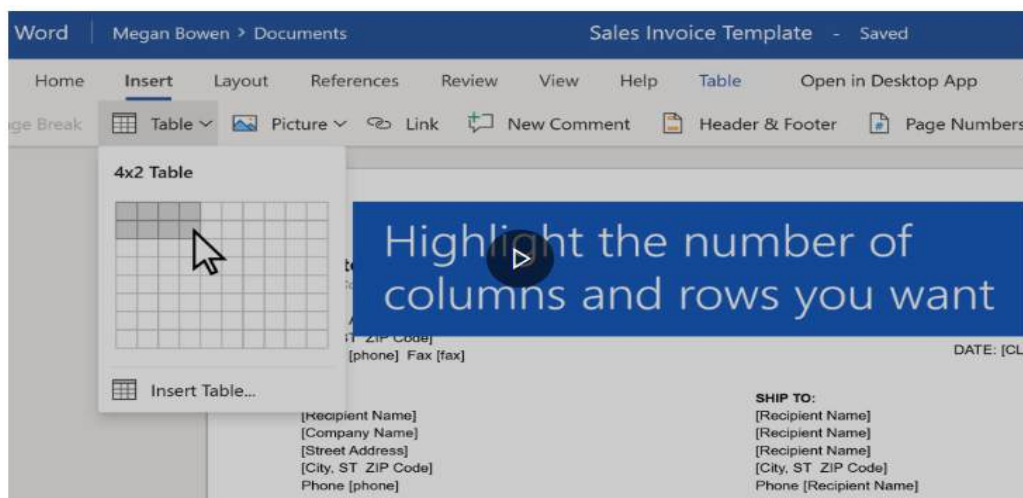
Working with tables and pictures

Suppose you want to create a report that contains information about the weekly sale of museum tickets during a month. Instead of presenting this information as plain text, you can present the information in an easy-to-understand format using a table. You can divide the information into logical sections within a table to group different types of data, making it easier to read the entire document. You can also include pictures in a document to make it visually appealing.

Insert a table or picture into your document to give it structure or a visual touch.

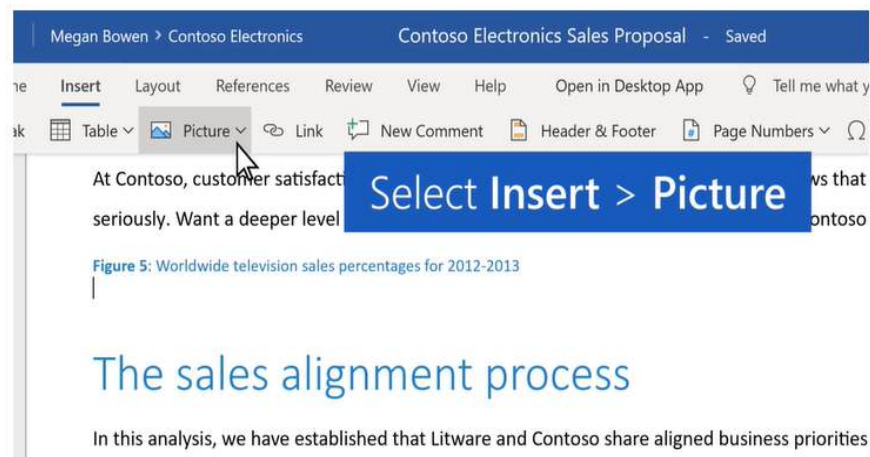
Insert a table

1. Select Insert > Table.
2. Highlight the number of columns and rows you want, and then select them.
3. To create a larger table, select Insert > Table > Insert Table.
4. Choose the number of columns and rows you want.



Insert a picture

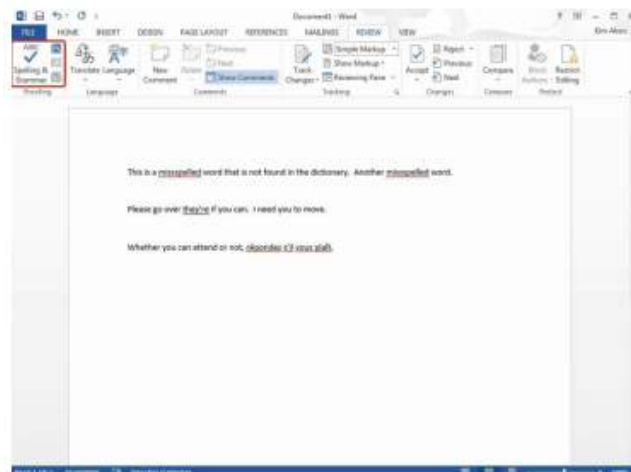
1. Select Insert > Picture.
2. There are two options to choose from:
3. Select This Device to insert a file from your PC.
4. Select Bing to use a photo from the web.
5. If you choose Bing, type what you're looking for, choose a photo, and select Insert.

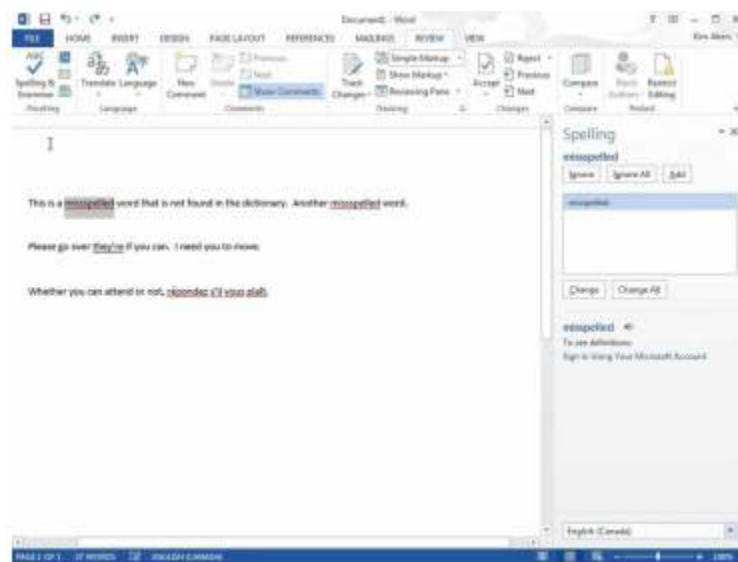
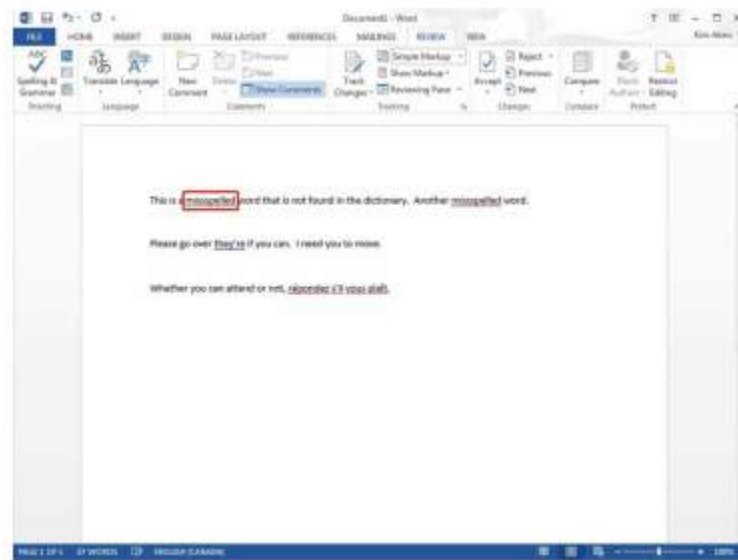


Language Tools

Advanced word processors, such as Word 2013, can automatically check the spelling, grammar, and punctuation of a document based on the language of the typed text. You may also search for synonyms using the built-in thesaurus. Word 2013 also features an AutoCorrect function that will automatically correct misspellings and grammatical errors as soon as you start typing the text. For example, typing “teh” will automatically be corrected to “the”.

Any time you are writing a report or an essay you may check your document for spelling and grammar mistakes. This is done using the proofreading tools that Word 2013 provides as found under the Review tab on the Ribbon bar.





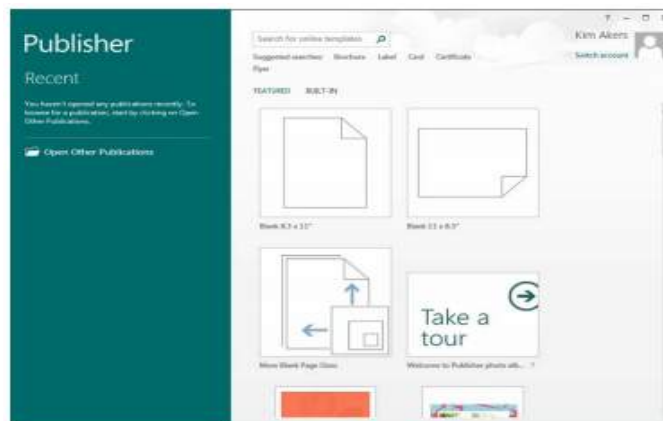
Benefits of desktop publishing

Up to this point you have learned all about Word Processors, such as Word 2013. You will now be learning about the benefits of publishing the work that has been done on a word processor. There is a specific program that will allow you to do some final touches and move stuff around much easier for a very long document.

A word processor such as Word 2013 provides a lot of flexibility in creating your document. But if you are looking to create documents with a specialized text and image layout a desktop publishing (DTP) program such as Publisher 2013 can help out. DTP makes it easier to arrange and modify information for documents that are very long, or have special formatting needs. And DTP programs also offer built-in templates for documents such as newsletters, brochures, flyers, signs, resumes, business forms, letter formats, and catalogs.

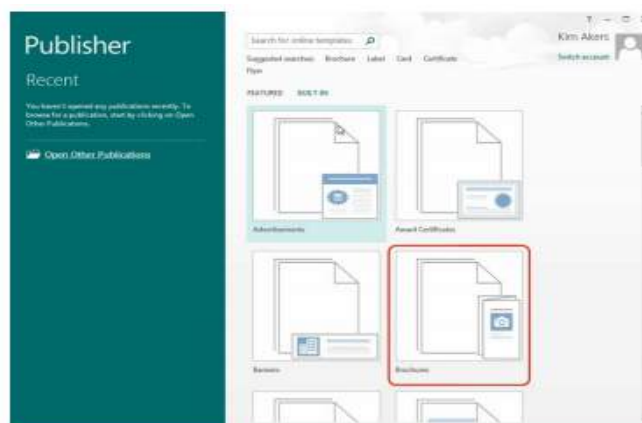
A word processor such as Word 2013 provides a lot of flexibility when creating your documents. But if you are looking to create documents with a specialized layout such as a brochure or a photo calendar, a desktop publishing application like Publisher 2013 gives you the power to combine text and images in a variety of ways.

Let's step through creating a brochure using Publisher 2013. When we start up Publisher 2013 we are provided with a list of templates to choose from.

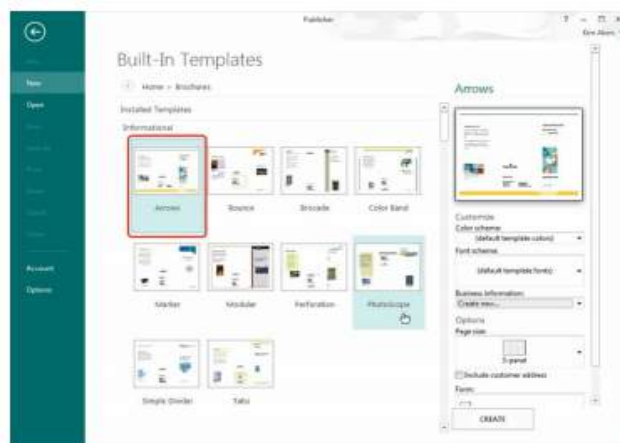


We will start by selecting BUILT-IN. We are now provided with a list of template categories.

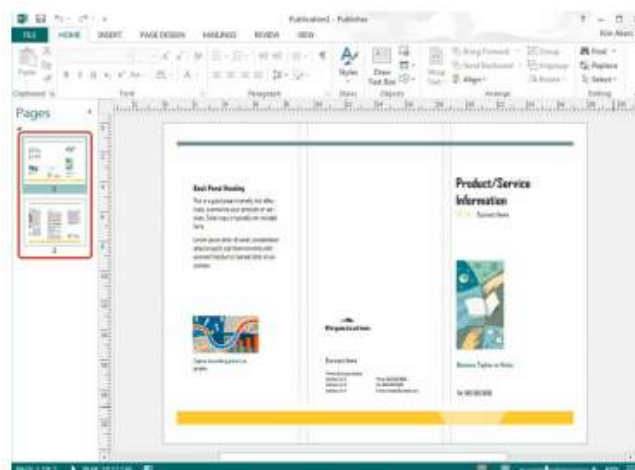
We will select the Brochures category.



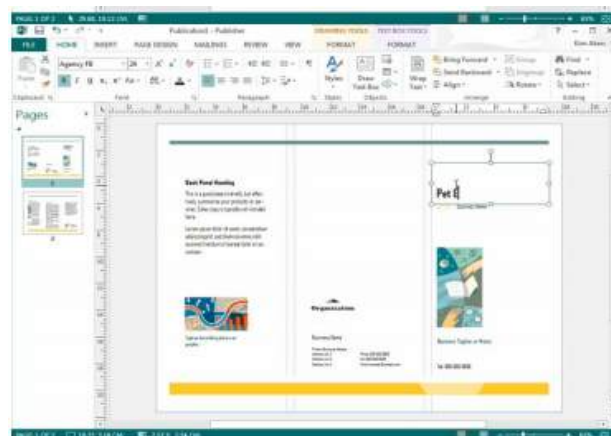
Publisher 2013 now provides us a list of brochure templates to choose from. will select the first brochure template called Arrows. And click on the Create button to create a new document based on this template.



Once Publisher 2013 has created our document, we are provided with a list of the pages on the left.



The work area of Publisher 2013 is where we may change the provided example text and images according to our project's needs. Let's start with the brochure's title.



Benefits of desktop publishing review

This demonstration introduced you to the following tasks you can perform with Publisher 2013:

- Create a DTP document based on a template
- Modify text to suit your particular project
- Change the provided default images
- Browse the Office 2013 royalty-free image catalog

HOOR TWO (2)

LESSON 3: SPREADSHEETS

OBJECTIVES:

3.1. IDENTIFY THE DIFFERENT COMPONENTS OF A SPREADSHEET.

3.2. ENTER DATA INTO A SPREADSHEET.

3.3. PERFORM BASIC MATHEMATICAL OPERATIONS IN A SPREADSHEET.

3.4. INSERT CHARTS INTO A SPREADSHEET.

3.5. EXPLAIN THE PURPOSE OF OPTIONS AVAILABLE FOR PRINTING A SPREADSHEET.

Introduction to Spreadsheet Programs

Excel 2013 is Microsoft's Spreadsheet Program. This specialized program allows you to create spreadsheets, such as:

- Schedules
- Budgets
- Calendars
- Invoices
- Plans
- And more

In this lesson, you will learn about the components of a spreadsheet and how to store data in it. You will also learn to perform basic mathematical tasks and create charts and graphs in a spreadsheet. In addition, you will learn about the various options available for printing a spreadsheet.

Microsoft Office 2013 includes Excel, the leading spreadsheet program. A spreadsheet is a table consisting of rows and columns designed to organize your data. Data usually consists of words or numbers used to describe an item. A number might represent the cost of groceries, or even the date they were purchased. A word might describe where you purchased the groceries. Organizing data into a table has two

advantages. First off it makes it easy to find a piece of data, answering the question “How much did I pay for the groceries last time?” Secondly, it allows you to do calculations on the data, “How much have I spent on groceries this month?” In addition you can take that data and use Excel to build graphs and charts, but let’s not get ahead of ourselves, time to learn the basics and gain some terminology!

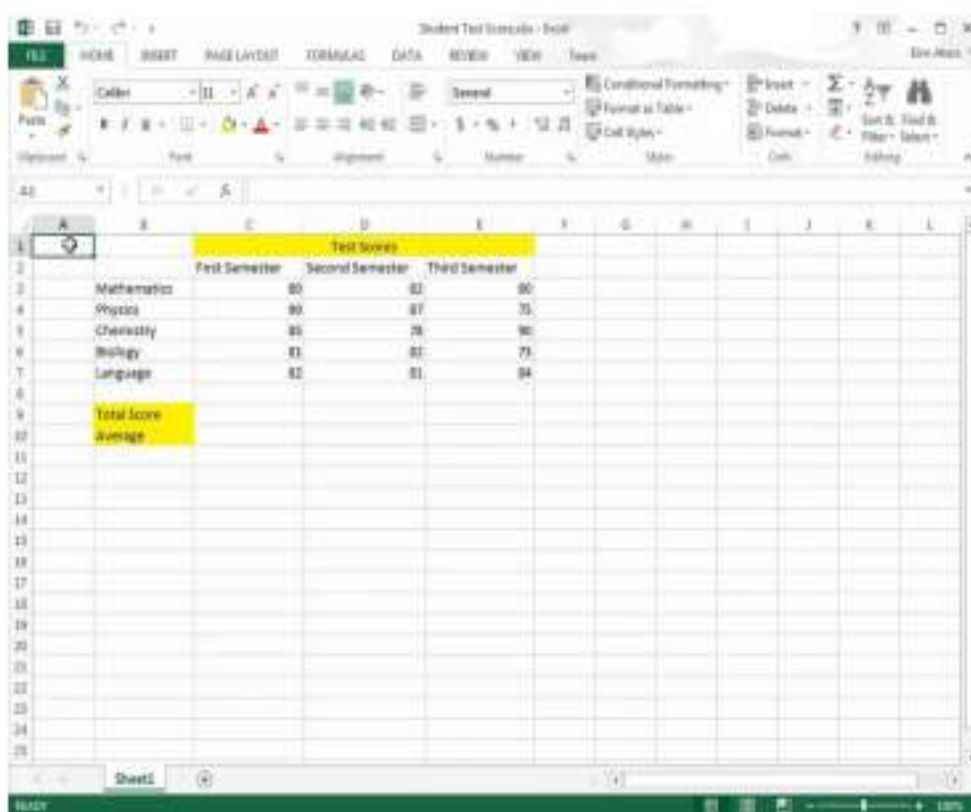
Inserting data

The time will come when you need to make a budget and keep track of how much your bills are each month. In order to do this, you will need to know the details about each bill such as bill name, due date, bill amount, date paid, etc. Then you will need to calculate the overall amount of your bills for the month so you can see how much of your paycheck goes out each month. All of this can be easily tracked, and tabulated, with Excel. You do this by entering the information into the cells of an Excel spreadsheet. If the bill ever changes, you can even edit the cell with the new value and it will automatically replace the old value with the new. In the following demonstration you will learn how to insert data into a worksheet in Excel 2013.

Performing basic mathematical tasks in a spreadsheet

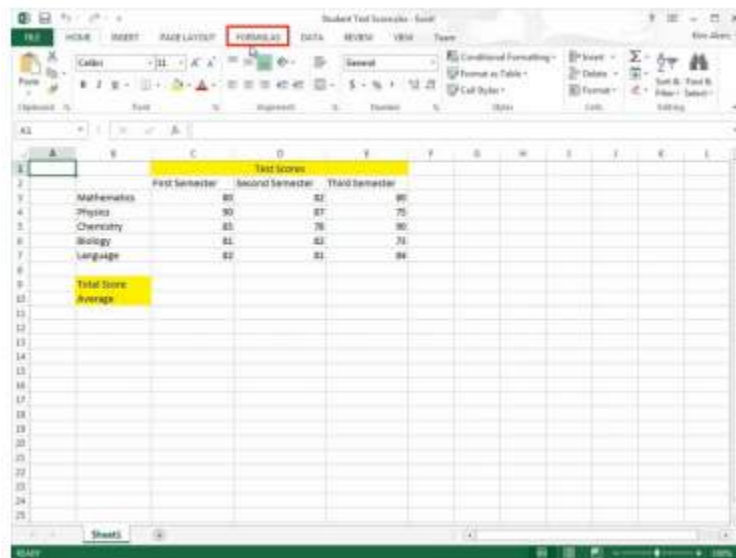
In addition to storing and manipulating data, a spreadsheet program such as Excel 2013 allows us to perform various mathematical tasks on our data quickly and accurately. The spreadsheet program provides functions that perform calculations on the cells of the spreadsheet. For example, you could add together all of your household expenses to reach a total, or use the Average function to calculate the average of your grades.

A spreadsheet program such as Excel 2013 allows us to perform various mathematical tasks on our data quickly and accurately. To perform these tasks, we use functions provided by Excel 2013.

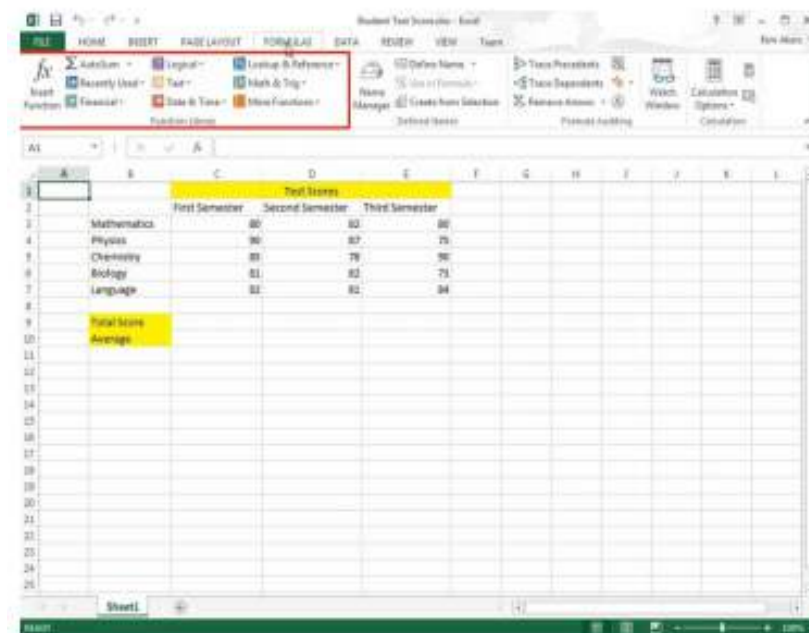


Student Test Scores - Book1				
		First Semester	Second Semester	Third Semester
1				
2				
3	Mathematics	80	82	80
4	Physics	80	87	75
5	Chemistry	85	78	90
6	Biology	81	82	79
7	Language	82	81	84
8				
9	Total Score			
10	Average			
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				
21				
22				
23				
24				
25				

We begin by clicking on the Formulas tab in the Ribbon Bar.



The Function Library group provides access to all of the functions provided by Excel 2013.

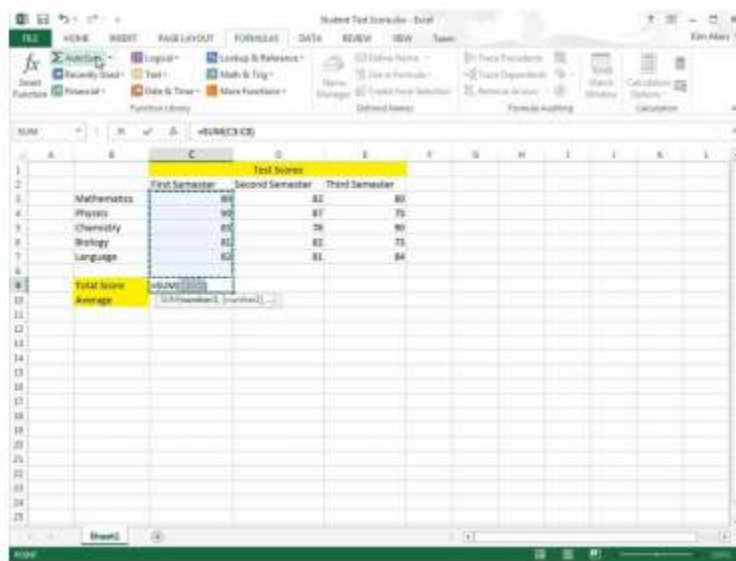


Let's begin by adding together the students' first semester test scores in our example.

Start by clicking on the cell we wish to place the total. In our example it is cell C9.

We then click on the AutoSum command. This will automatically add all the numbers in the cells directly above our selected cell.

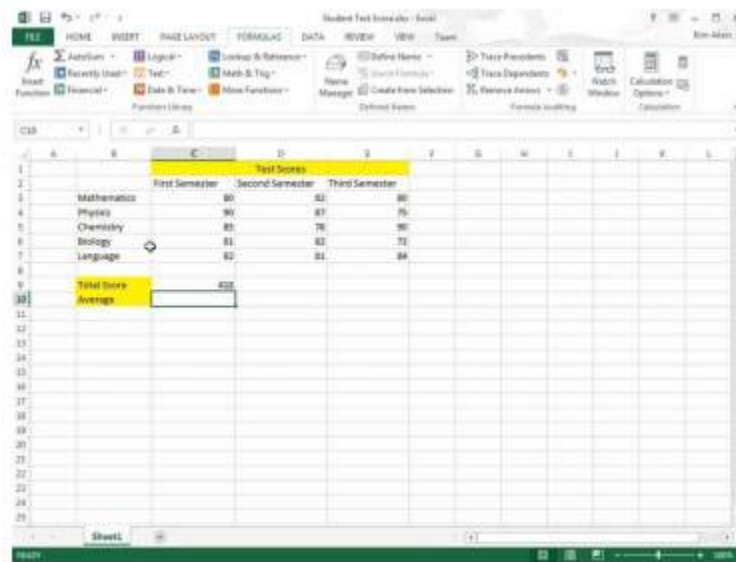
Notice that Excel 2013 shows us which cells and the numbers that will be added together



If we wanted to use a different range of cells we could click and drag on the different cells.

For now, let's limit our range to just add the numbers in the cells C3 through C7. To accept the cell range, press the Enter key.

Excel 2013 calculates the total and displays it in a selected cell.



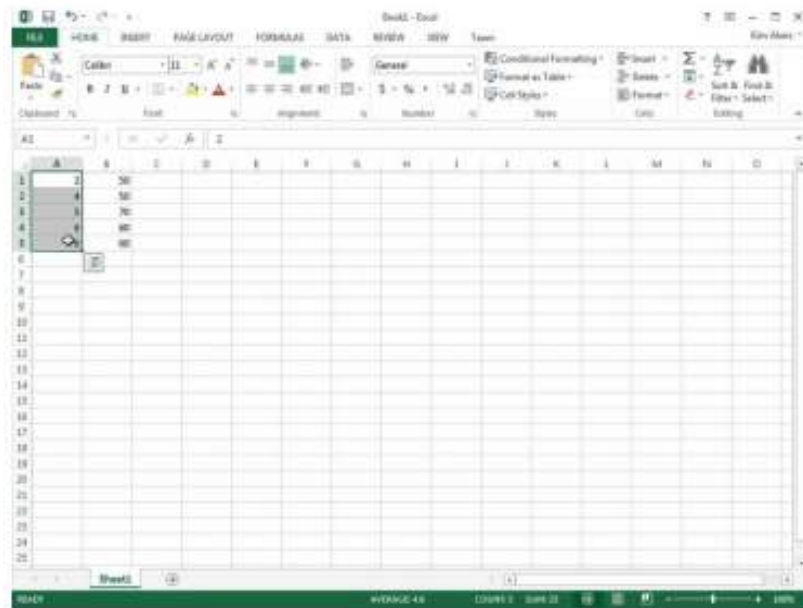
We can see the Sum Function used for the C9 cell by clicking on it again and looking in the formula bar.

Inserting charts

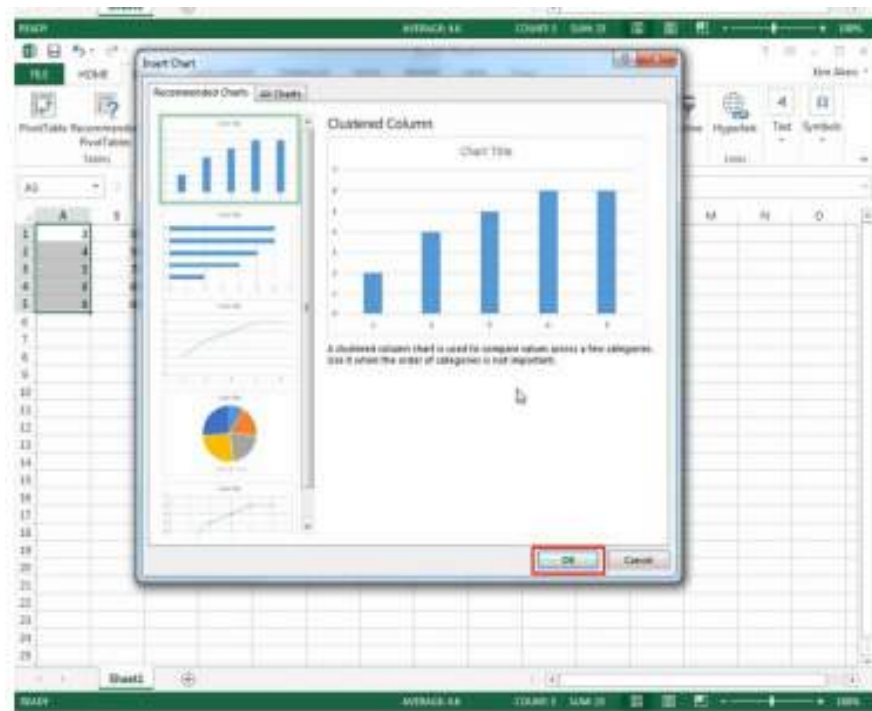
You have learned how to insert data and operate data now we will show you how to display data with a chart. Excel provides a wide variety of charts and built-in functions for the chart. Perhaps you would like to find a linear relationship to a chart. All you have to do is plot the data on a scatter chart and insert a trend line. Excel will give you the $y = mx + b$ relationship of the data! Let me show you.

Often you can learn from data more easily when it is displayed graphically, such as in a chart.

Let's start with a simple data set consisting of one column.



Click on the insert tab and then select Recommended Charts in the Charts group and click Ok to create the chart excel recommends.



When you select the chart, additional tabs for Design and Format appear. You can customize a chart to include a title and labels



To edit the title, first click to select it and then click on it again to edit the text.

You can remove the chart title by going to the Chart Layouts group on the Design tab.

Select Add Chart Element and Chart Title then select None in the options.



LESSON 4: PRESENTATION PROGRAMS

OBJECTIVES:

- 4.1. IDENTIFY THE BASIC FUNCTIONALITIES OFFERED BY PRESENTATION PROGRAMS.
- 4.2. CREATE A NEW PRESENTATION.
- 4.3. ADD GRAPHICS AND MULTIMEDIA TO A PRESENTATION.
- 4.4. IDENTIFY THE OPTIONS AVAILABLE TO PRINT PRESENTATIONS IN DIFFERENT FORMATS.

Introduction to Presentation Programs

PowerPoint 2013 is Microsoft's Presentation Program. This specialized program allows you to create a presentation for business, school, or for personal use. A presentation can include the following:

- Text
- Graphics
- Charts
- Video
- Audio
- And more

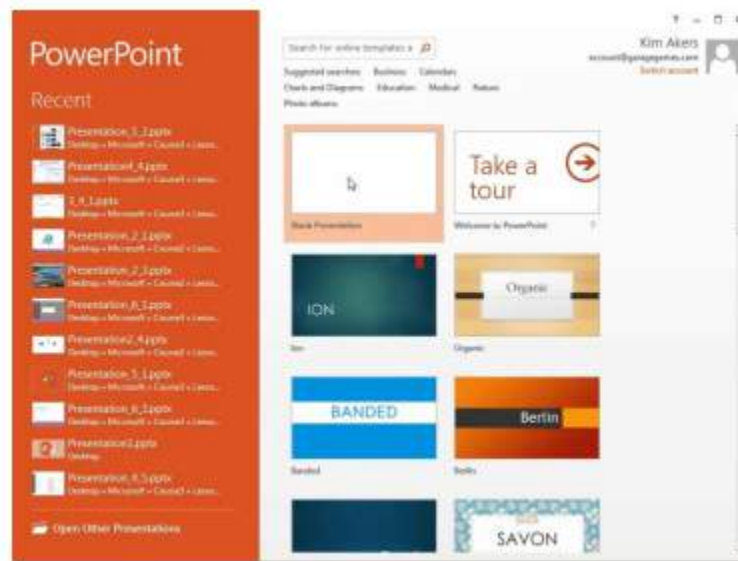
In this lesson, you will learn about the basic elements of a presentation program, how to create a presentation, and how to add graphics and multimedia to a presentation. You will also learn to identify the various options available to print your presentation in different formats and create a new online presentation.

PowerPoint 2013 allows you to develop presentations. Presentations consist of slides containing text and images. Presentations can be displayed visually as a slide show or printed for distribution. PowerPoint uses the Office 2013 interface to quickly add text and images to slides.

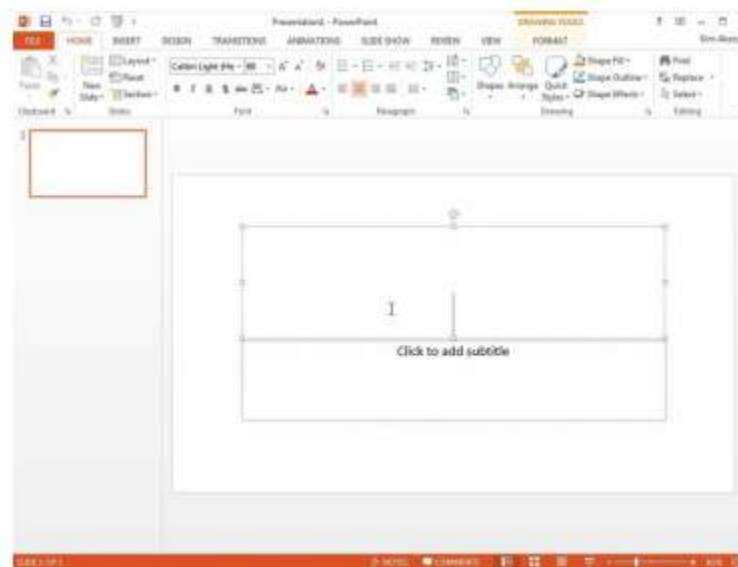
Creating a presentation

Most presentations consist of several slides of content. The appearance of the slide can be themed to improve the look of your presentations. Many times, transitions are added to the content to add higher quality of professionalism. After you create a presentation, it is important to review the presentation to ensure it behaves as intended.

Text is the most direct way to add content to a slide.

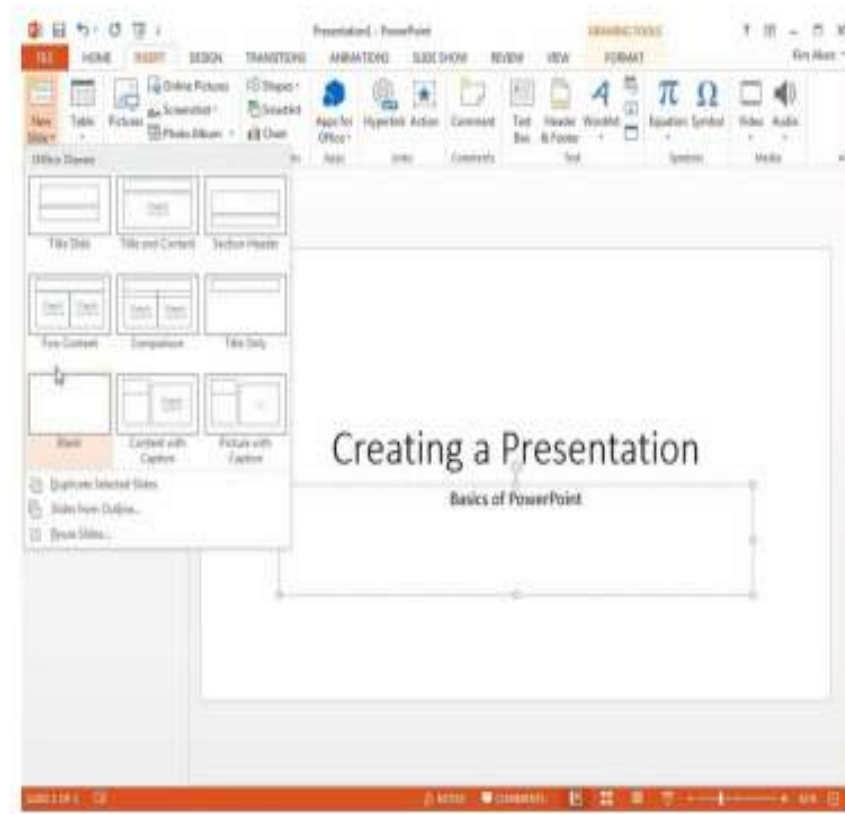


The default slide has two text areas, one for a title and another for a subtitle. To add text simply click on the text area and start typing.



To add a new slide, select the 'New Slide' Command from the Insert Ribbon.

There are several layouts to choose from, we will select the blank slide layout and add our own elements.



Adding graphics and multimedia

Most presentations are designed to show the audience some content. Many times the best way to express the content is with images, tables and charts. Powerpoint 2013 supports content from other Office 2013 products. The content can be pasted from other Office 2013 products or built directly within PowerPoint 2013. Let's look at some of the ways we can improve the content of a presentation.

Multimedia exists to help reinforce our key speaking points. Here are a few ideas for working with images, audio, and video to make your presentation more interesting:

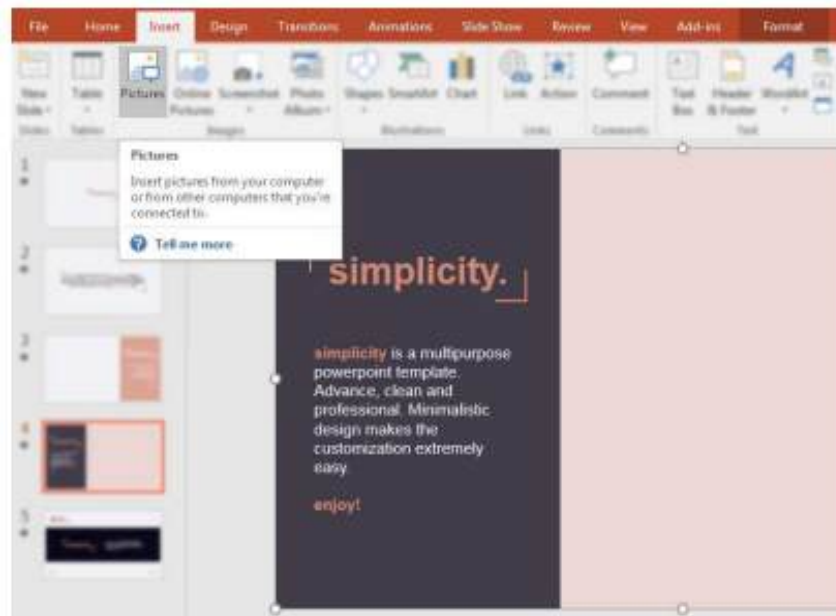
Images: These could be used for everything from simple stock photos that add slide variety, to logos and icons that showcase your brand.

Audio: Audio is a great way to add narrations to your slides if you can't present directly to your audience or to add music and sound effects.

Video: You can add video clips from your computer or sites like YouTube. Videos could be used for showing instructional walkthroughs from YouTube, TED talks, or a recorded screen cast of your own.

How to Insert Images into PowerPoint

The simplest way to add an image to PowerPoint is to simply copy and paste. You can copy an image to your clipboard in practically any app like a web browser or image editor, and then right click to paste it into PowerPoint. A better way to insert images is to save them to your computer, then insert them using the Insert menu. Go to the Insert > Pictures menu to browse to an image to insert.



Insert an image from the Insert > Pictures menu on the PowerPoint ribbon. On the pop-up menu, browse to where your images are stored and click on them to add them to the presentation. By Insert menu you can add multiple images at the same time. Hold the Control button on your keyboard and click on multiple images before you press Insert to add them all to a slide.



Select multiple images by holding Control on your keyboard and clicking on them before pressing Insert to add them all to a slide.

Add an Audio File to PowerPoint

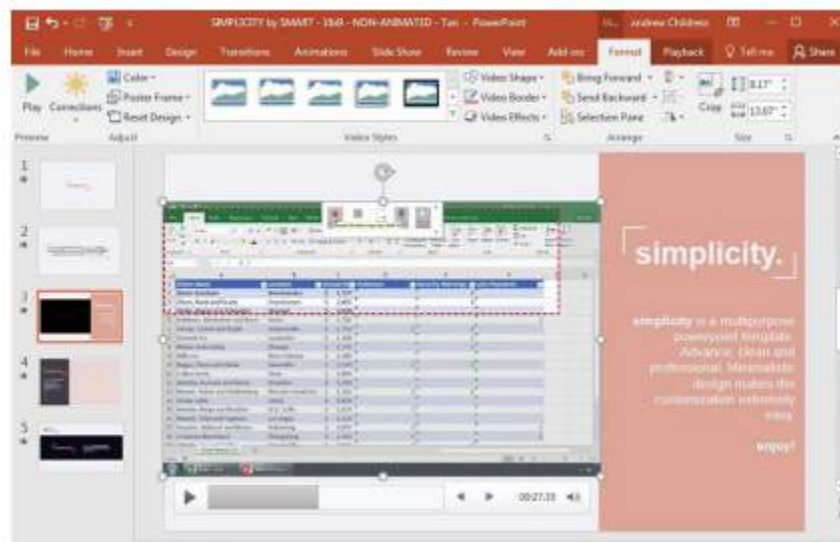
Audio is another multimedia type you can work with in PowerPoint. If you have an audio track on your computer like an MP3 or WAV file, you can easily add them to your presentation. Choose the Insert > Audio > Audio on My PC option to browse for and choose an audio file.



Go to Insert > Audio > Audio on my PC option to browse for and choose an audio file to add. After you choose the audio file, you'll see a small speaker icon on your PowerPoint slide. You can hover over it and press the play button to play it while presenting.

Add a video Clip from Your Computer

If you have a video clip on your computer, you can quickly drop it into your presentation by using the Insert menu. Go to Insert > Video > Video on My PC to browse to your video file.



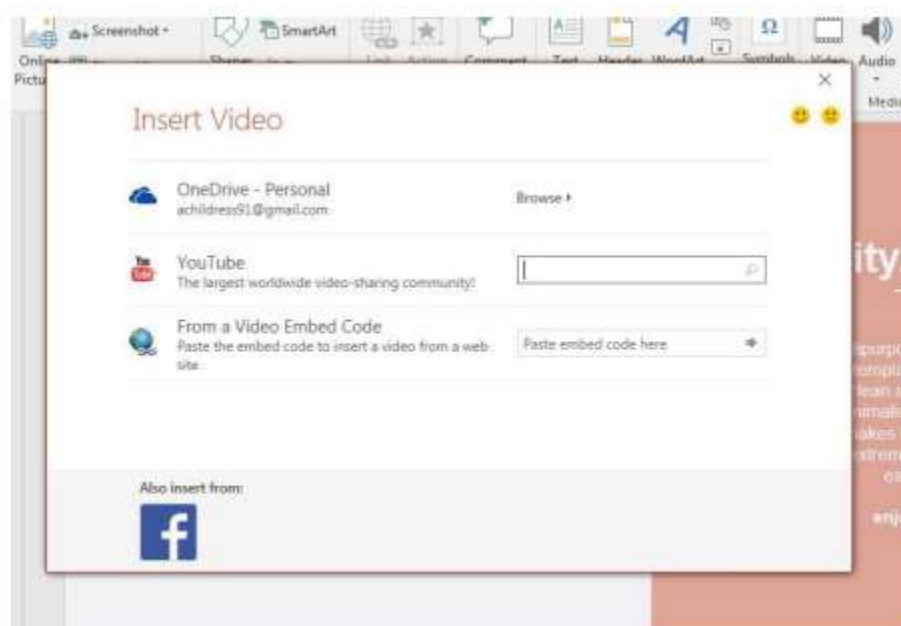
After browsing to the video and inserting it, you'll see it placed on your slide with the playback controls. Once the video is on your slide, you can drag the corners to resize it and control how much of the slide it covers. PowerPoint has good support for most of the major video file types, like MP4, QuickTime, and Windows Media files.

Add a Video Clip from Online

Finding an inspiring or insightful YouTube video can help drive your point to reach an audience. Instead of finding a workaround to download and convert a YouTube video, PowerPoint can link to and insert them directly. To add a YouTube (or other online) video, go to Insert > Video > Online Video to launch the menu.



To add an online video, choose Insert > Video > Online Video to get started. Now, PowerPoint will show a new window to search for or add an online video. If you want to add a YouTube video, you can drop in the URL for a video, or search directly inside this box for a relevant clip. Other sites might give you embed codes on their site. On sites like Vimeo or TED, try out the Embed code and drop the HTML into the bottom box.



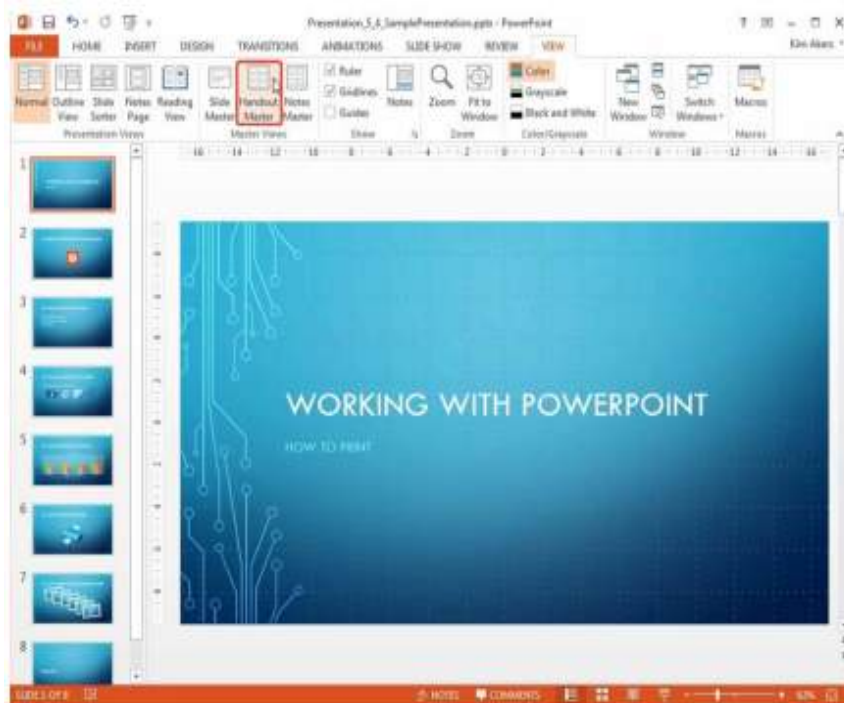
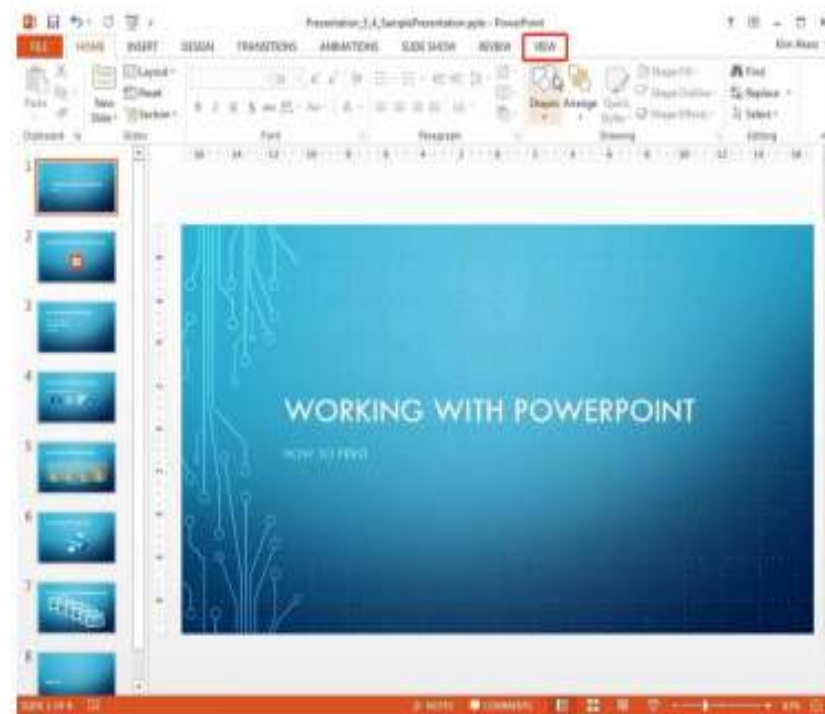
Search for videos using the box or drop in a URL for a YouTube video of your choice.

Adding a YouTube video is a great way to include a video without increasing your PowerPoint file size. Instead of embedding the video inside the file, the link will play a YouTube video, right inside the presentation.

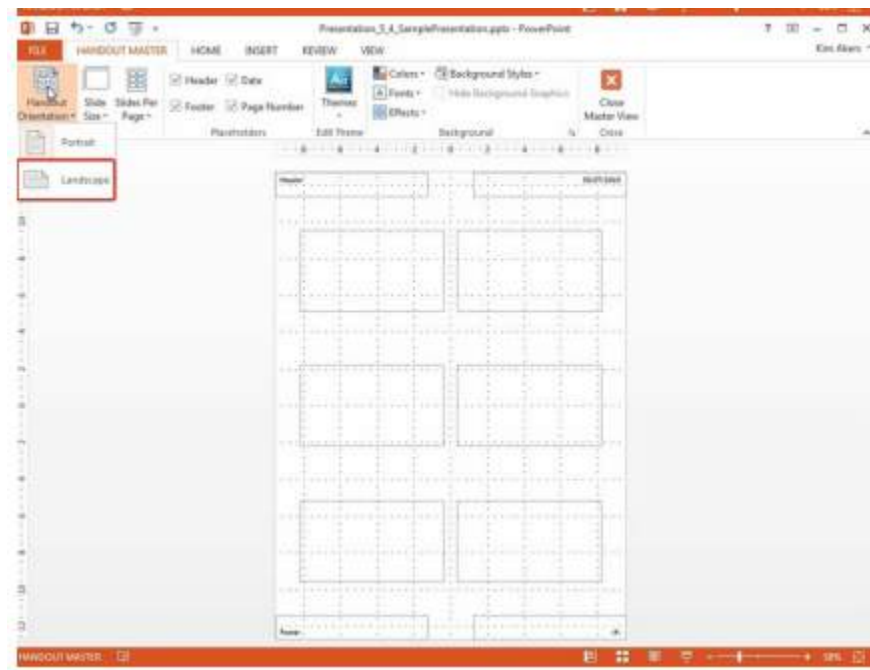
How to print a presentation

PowerPoint 2013 provides many options for printing spreadsheets, for instance you can adjust how many slides appear on a page. You may want to include your notes when printing or show just the text. Let's look at some of the ways you can print spreadsheets in PowerPoint 2013.

PowerPoint 2013 lets you adjust the layout of the handouts of your Presentation and select Handout Master from the View Ribbon.

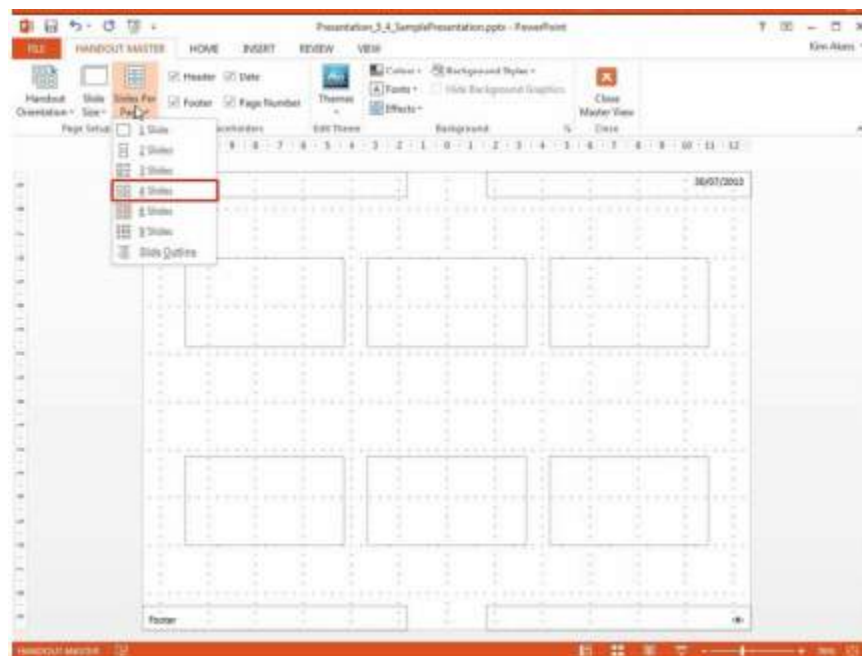


The Master View offers many options, here we change the handout's layout to be Landscape.

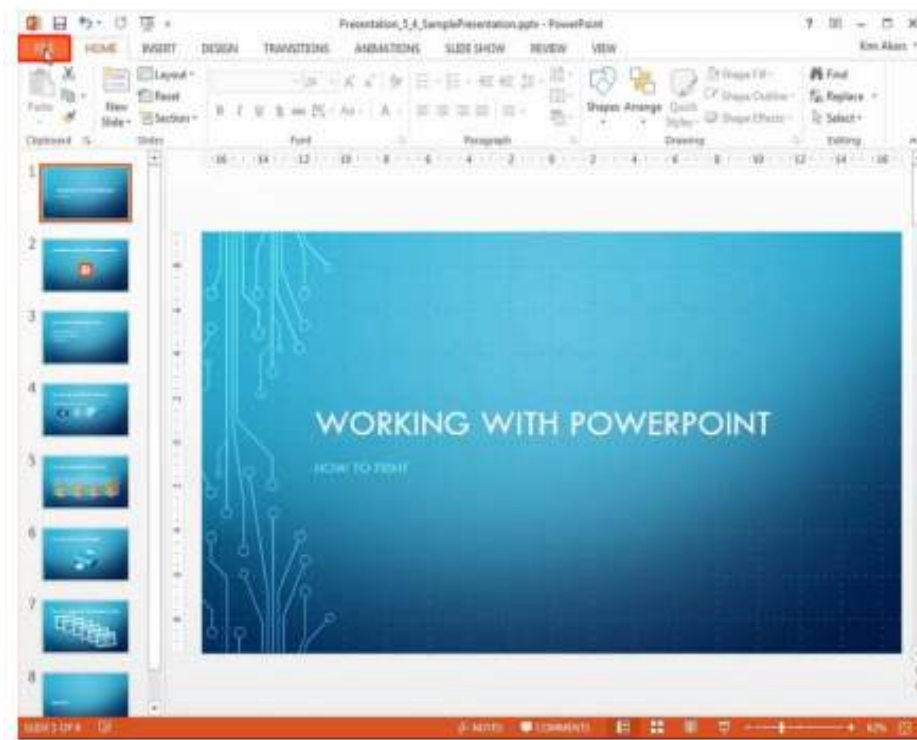


You can also have more than 1 slide to be printed out on each page.

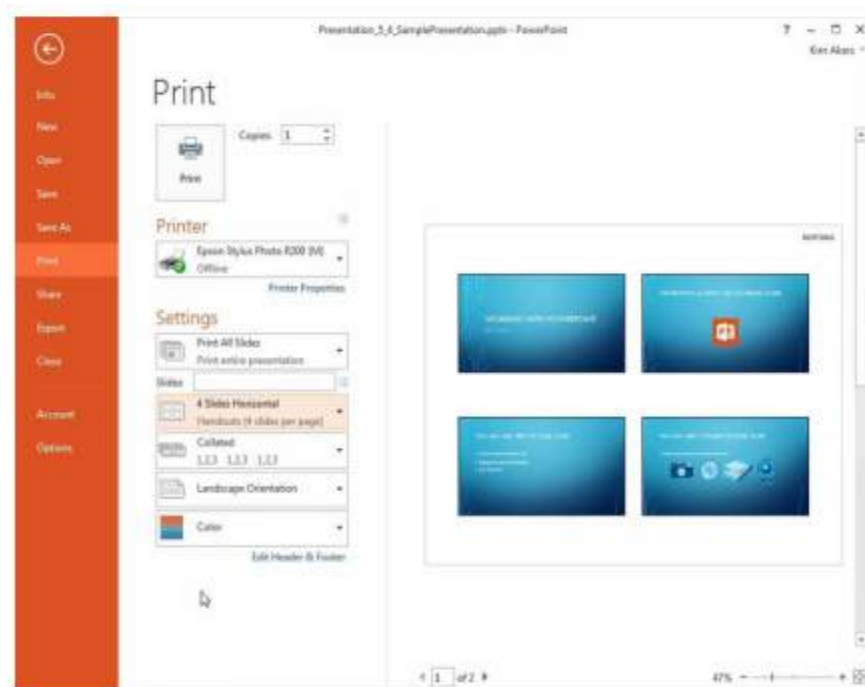
Here we will set PowerPoint to put 4 slides on each printed page.



When you are ready to print, select the Print button in the backstage view.



An example of the Layout is displayed in the preview window.



Now you have the basics of printing for PowerPoint 2013 presentations.

LESSON 5: DATABASES

OBJECTIVES:

- 5.1. EXPLAIN BASIC DATABASE CONCEPTS.
- 5.2. CREATE A DATABASE.
- 5.3. WORK WITH RECORDS IN A DATABASE.
- 5.4. EXPLAIN WHAT DATABASE QUERIES ARE AND HOW THEY WORK.
- 5.5 EXPLAIN WHAT REPORTS ARE AND THEIR USES.

Introduction to Database Programs

Access 2013 is Microsoft's database program. This specialized program allows you to create a database for:

- Providing information
- Keep certain items organized
- Keep track of money
- classify certain things
- And more

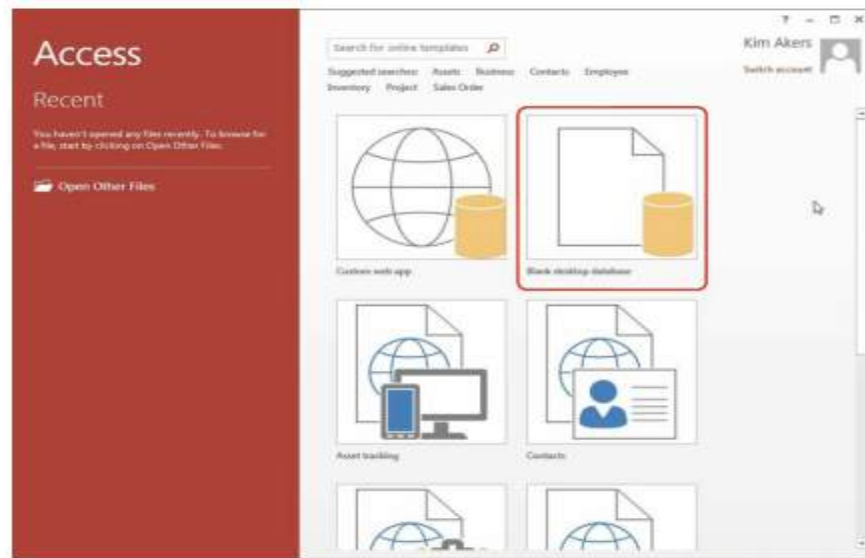
In this lesson, you will learn about the basic tasks that you can perform by using a database program. You will learn how to create and use a database. In addition, you will learn how to generate various types of reports by using queries.

Businesses generate a lot of data, considering how many times you have been asked for your name and email just to sign up for a new account on a website. Your name and address are examples of data. Databases are designed to store and access data. This data is usually stored in several different tables, when someone wants to verify your account they check the submitted data to that stored in the database. Access 2013 allows you to build these tables, fill them with data and then look at the data. Let's look at an overview of the process.

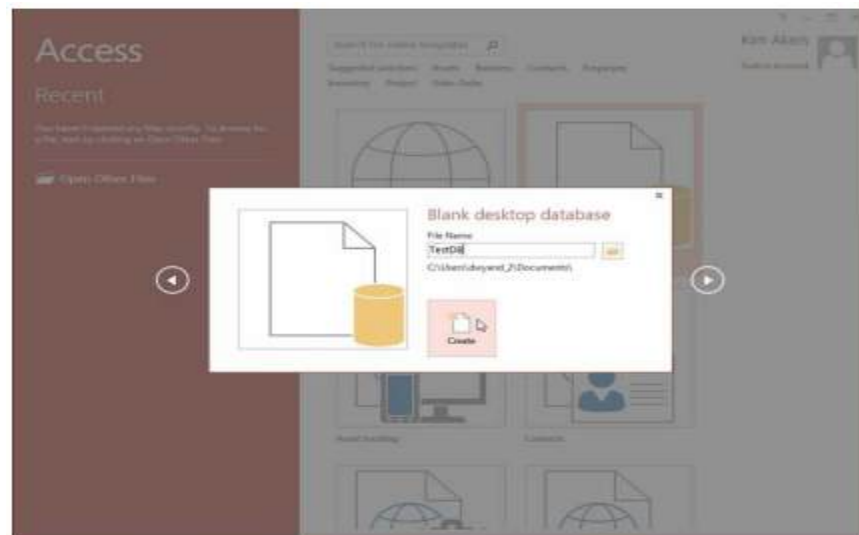
Creating a database

Databases contain several tables. A table needs a description of the data before data can be entered. For any column we must specify if the data will be text, a number, a date or something more sophisticated. By specifying the data type, we can ensure the data is valid when it is added to the table. Let's look at how we added a table to a database.

As you can see there are several types of Databases, we will create a Blank Desktop Database and name the file TestDB.

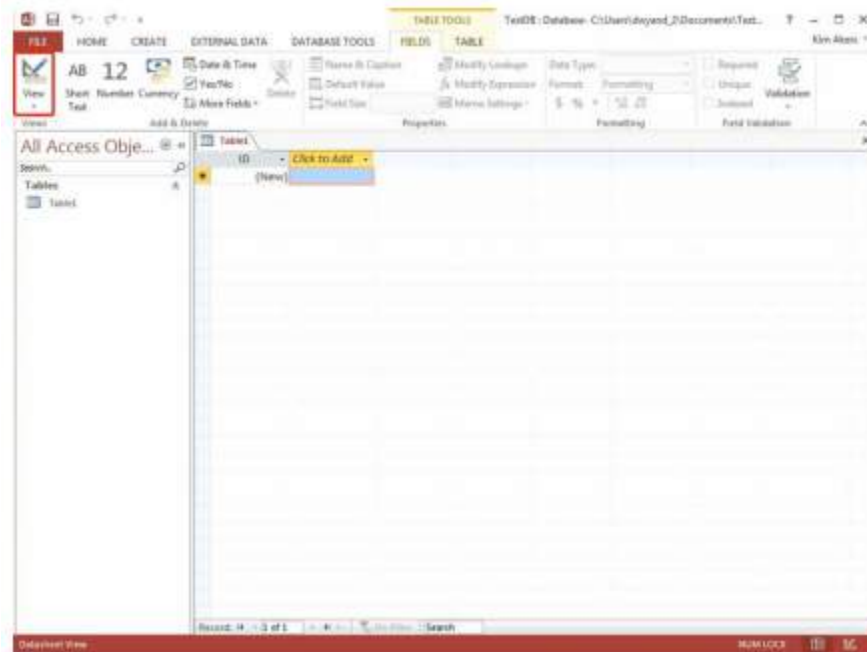


Click the Create button to open the database file.

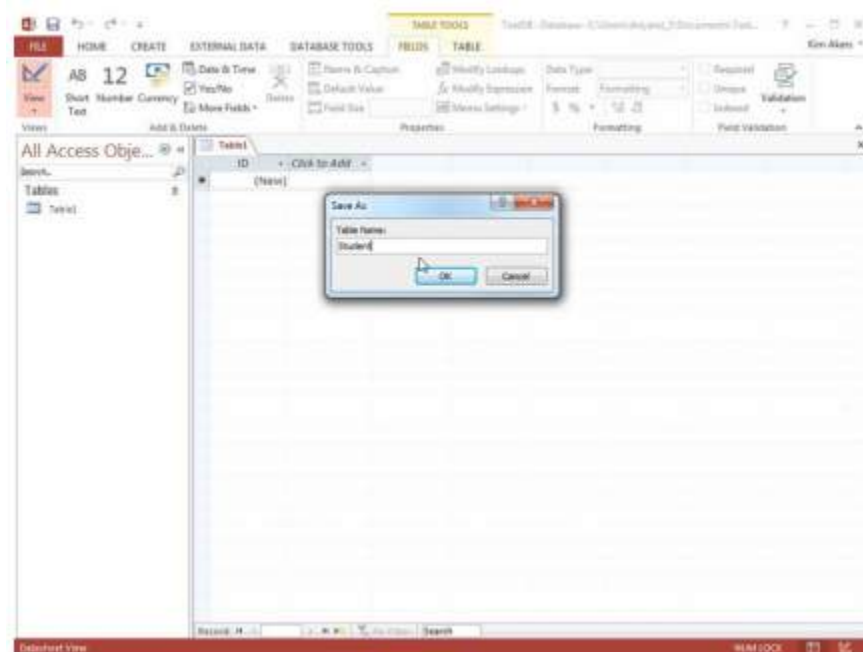


By default, Access 2013 shows you a table in DataSheet View.

You can change views, with the View button on the Home Tab.

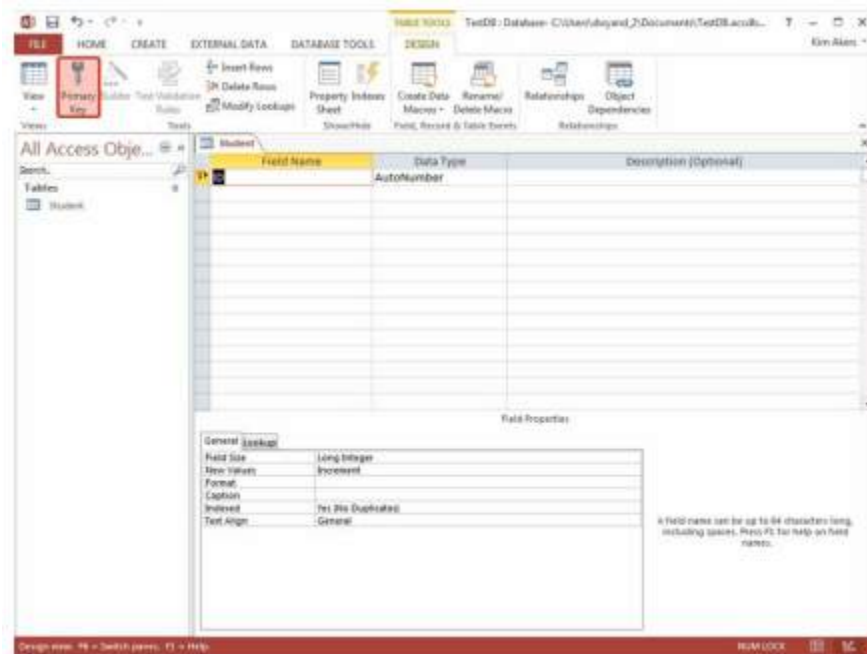


Notice how Access 2013 will ask you to provide a name for your table. Let's call our table "Student" and click OK.

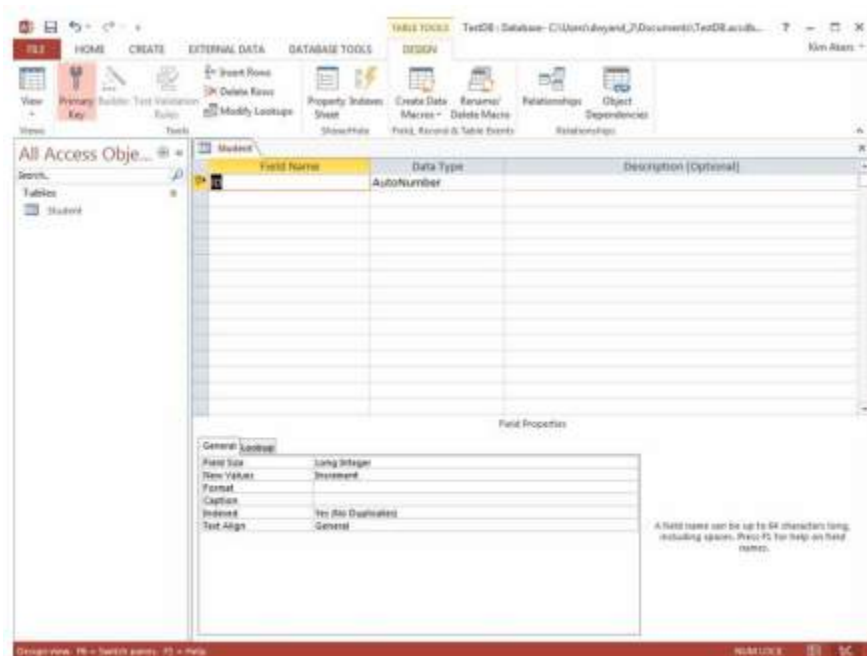


Each table starts with a piece of data called ID whose data type is Autonumber.

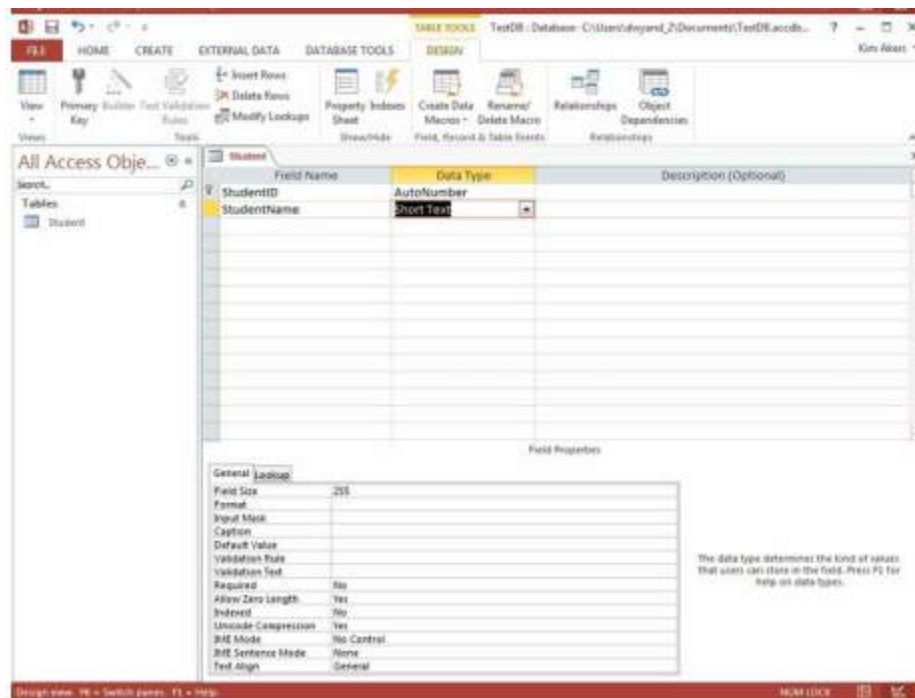
Notice how the Primary Key button is highlighted in the Ribbon when the ID row is selected.



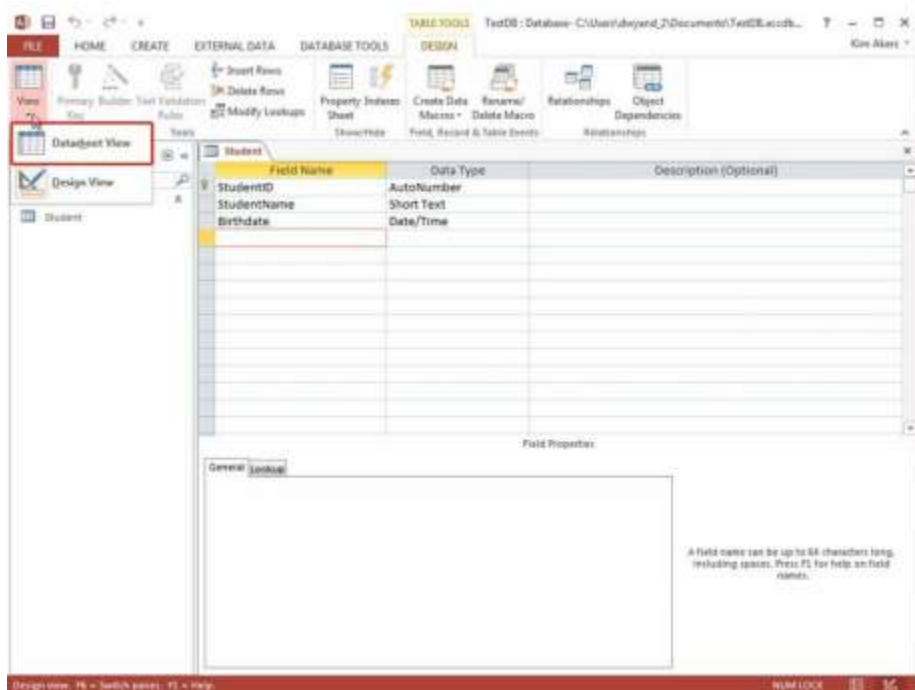
You can toggle which row will be the table's Primary key



Lets add a couple more rows of entries on our table one for StudentName as a Short Text Data Type and one of Birthdate as a Date and Time Data type.

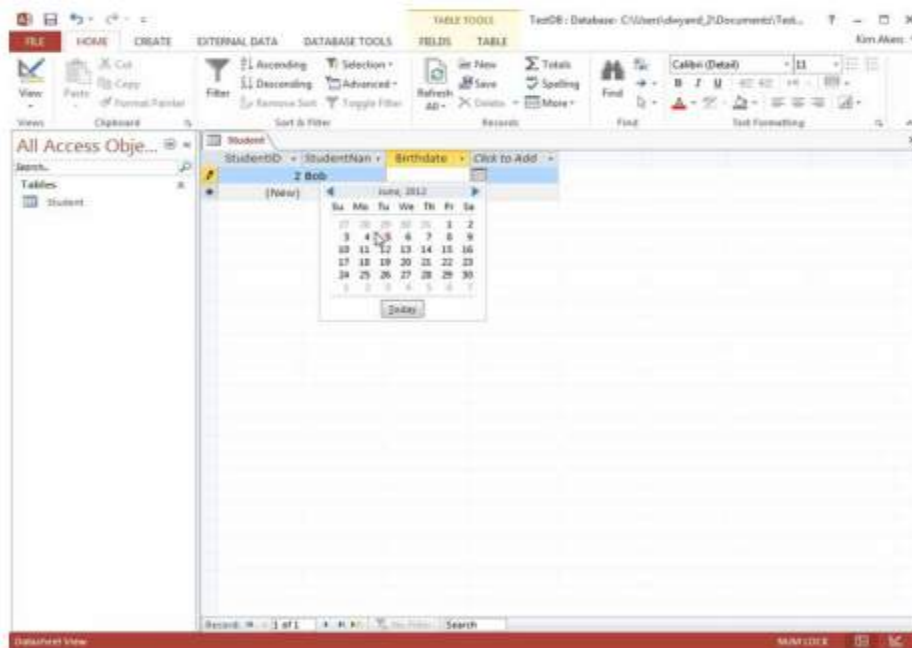


In DataSheet View notice that the field names we created are now the column headers.



In DataSheet View notice that the field names we created are now the column headers

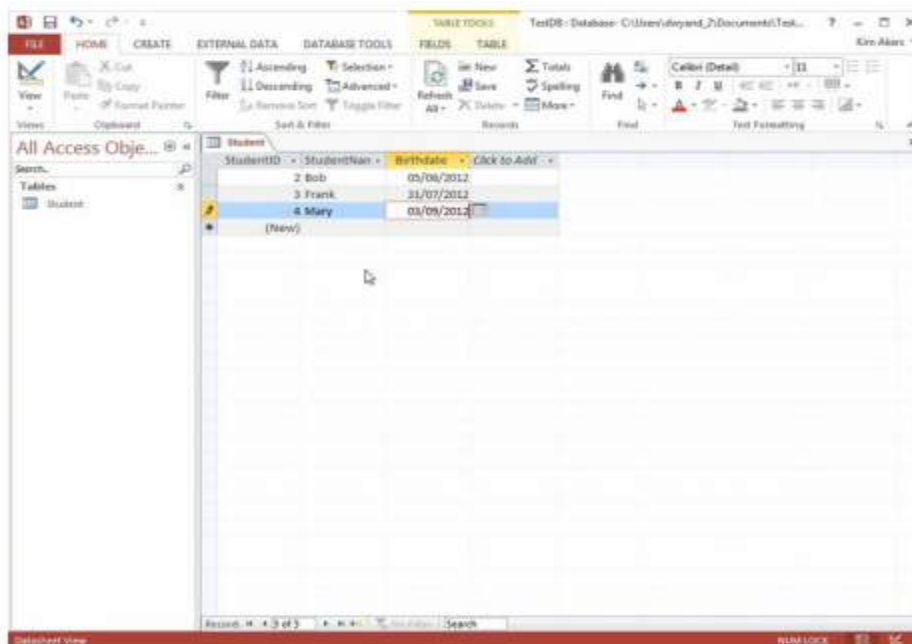
You can type directly into the cell or in certain cases, like a date, we can choose a value with the Calendar widget.



Notice that with each new entry in the table Access automatically updates the StudentID number.

Access 2013 allows database designers an interface to create their tables quickly.

Take some time to learn which data type will best suit your information and records.



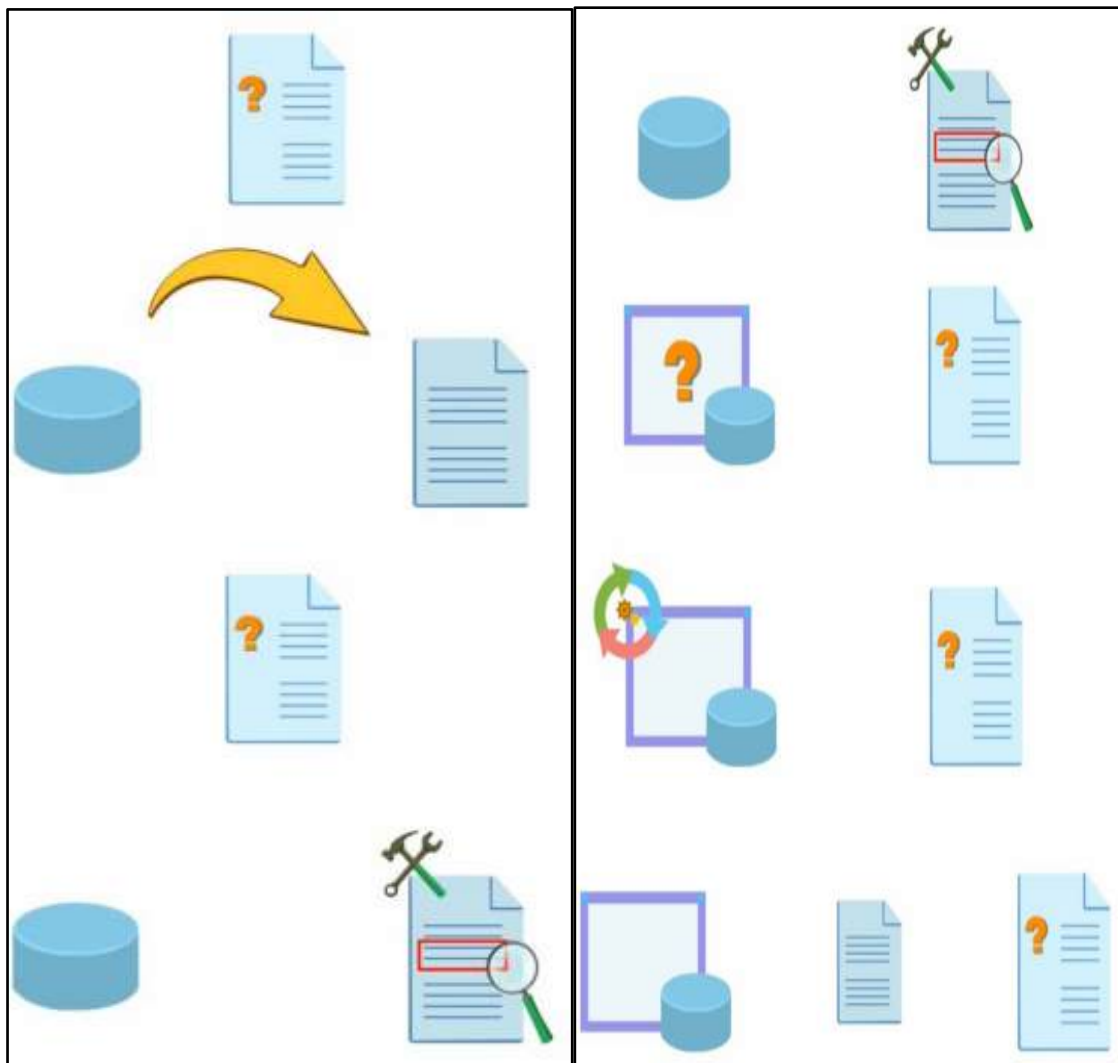
Working with records in a database

Access 2013 databases hold a collection of tables and Excel 2013 creates tables. It is natural to wonder 'How does one import an Excel table?' Access 2013 supports a wide range of formats for importing and exporting. In this lesson let's focus on the Excel Importer Wizard.

Database queries

Queries allow database designers to change or retrieve data from a database. Access 2013 allows queries to be built with a graphical interface. There is also support for SQL queries. Let's look at how we can create a query.

You use a query to retrieve specific information from a database.



You use a database program to enter a query in the form of a question. The database program then performs the required operations to present the answer in the form of a report.

Suppose you own a coffee shop and you store detailed information about your customers in a customer table and data about the sales of all the products in a Product table.

These two tables form part of a database.



Queries work on search conditions called filters to retrieve specific information from a database.

After you execute the query, the result appears as a report.

Access 2013 allows you to develop reports on the content of the database. Reports are useful for meetings and presentations when you want to distribute part of your database. The Report Wizard also offers options to show statistical analysis on the data.

Uses of Report

The objective of business reports is to give the information in an organized manner which helps to make major decisions and plan for the future. Reports help businesses plan budgets and decide on promotions and advertisements based on the drafted information.

Break: 1 Hour

HOUR THREE (3)

COURSE 4: COMPUTER SECURITY AND PRIVACY (FIREWALLS AND ANTI-VIRUS SOFTWARE)

DESCRIPTION

This course helps you to identify the various threats to your computer and the data stored on it. You will explore how to protect your computer from these threats by taking some preventive measures. Finally, this course explains the ethical and legal issues related to Internet usage.

BENEFITS

Like any other electronic device, your computer is at risk from damages caused by accident or by intention. Some of these damages can be permanent. You can prevent your computer hardware, software, and the data stored on it from a number of damages by taking certain preventive measures.

LESSON 1: OVERVIEW OF COMPUTER SECURITY AND PRIVACY

OBJECTIVES:

1.1. EXPLAIN WHAT THE TERMS SECURITY AND PRIVACY MEAN AS THEY APPLY TO COMPUTING.

1.2. IDENTIFY VARIOUS THREATS IN THE WORLD OF COMPUTERS AND EXPLAIN THEIR CORRESPONDING SOLUTIONS.

Computer Security and Privacy

If you use computers regularly, you may have a lot of information stored on the computer. You need to ensure that this information is not viewed by other people without your permission. You also need to protect this information from getting damaged. Any information can be damaged or stolen; so, protecting it is important, to protect information like the following:

- Tax Details
- Personal Letters
- Business Correspondence
- Projects
- And more

In this lesson, you will explore the need for protecting your computer hardware, software, and electronic data from damage, loss, and theft. You will also learn about the various solutions and devices that you can use to protect the data on your computer.

Computer security and privacy are two significant issues today. Computer security pertains to intrusion with personal accounts. Privacy pertains to how companies handle information you have given them. As we grow with more technology the social perception will evolve. In this lesson we will look at various aspects of computer security and privacy.

Natural threats

Computer security and privacy can be threatened by natural dangers such as earthquakes, flooding or lightning storms. These natural threats can lead to fires, extreme temperatures and even electric shocks to your computer, causing potential physical damage and loss of data.

Fire: It can be devastating to your computer. Beyond direct damage from flames, the heat generated can melt sensitive components within the computer. Smoke can damage the CPU fan which can also cause overheating. In case of a fire, always get yourself to a safe place before trying to rescue your computer.

Extreme temperature: Generally, computers are designed to function within a moderate temperature range. Excessive heat or cold can cause important components to malfunction or break. In cases of drastic temperature changes, it is best to allow the computer to return to room temperature before using it.

Lightning strikes are a huge threat to your computer, even if it never actually touches your hardware. Sometimes a lightning strike will create an electric surge (or spike) that amplifies the voltage to your

computer through the power supply. This sudden surge of electricity can destroy crucial components of your computer including your motherboard.

Measure for protection from natural threats

Natural threats can cause considerable damage to your computer. The following table explains the measures that you can take to protect your data and computer from natural threats.

1. **Back up data:** Backing up data involves creating multiple copies of your data. Events like floods and earthquakes can strike without warning. Making a backup helps you recover your data in case of any data loss using an external hard drive or using an online data backup site such as Cloud. This way, you can view and save your data online on any computer. To provide better recoverability, keep a copy of your important data in a physically separate location, such as in another building or city.



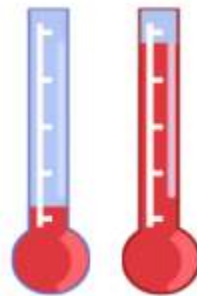
2. **Install computers in secure locations:** Install your computer in a place where it is not likely to get damaged due to natural factors. For example, avoid installing computers in rooms that are exposed to excessive dust or moisture.



3. **Insulate computers from fire:** Insulate the computers from fire by housing them in fire retardant surroundings. In addition, you can install adequate fire safety equipment and procedures for quick damage control.



4. **Maintain appropriate temperatures and humidity:** You should maintain an optimum temperature and humidity level to ensure the smooth functioning of your computer. You can do this by installing devices such as air conditioners and humidity controllers. If you are using a desktop computer (a computer designed to fit conveniently on top of a typical office desk), then you can install or upgrade a fan that attaches to either the front or back of the case, install a water-cooling kit, etc.



5. **Install protective electrical devices:** Install devices such as an Uninterruptible Power Supply (UPS) that can provide battery backup in case of a power outage. A UPS prevents software damage caused by abrupt shutting down of your computer. A UPS also provides surge protection and line conditioning features, which help protect your computer against spikes and surges on the power line. You can also install separate surge protectors and line conditioners. However, in case of a strong surge, caused by events such as a major storm, you should turn off the computer and unplug it from the power to avoid damage.



Threats from human actions

1. **Theft:** Anyone can steal your computer or its components, if they have access to it. With the popularity of portable computers, such as laptops, physical theft of computers has become very common.

You can also become a victim of virtual theft, when your computer is connected to the Internet. An example of virtual theft is identity theft. Identity theft happens when someone, usually a hacker, steals your personal information and uses it without your permission to assume your identity. The hacker will use your identity to gain access to your finances or perform illegal actions. The hacker can also wreck your reputation, which is more reason to prevent identity theft.

Another example of virtual theft is software piracy, which is theft of a computer design or program. It may also mean unauthorized distribution and use of a computer program or confidential documents.



2. **Viruses, worms, and trojan horses:** Viruses, worms, and trojan horses are computer programs that can damage the data or software on your computer or steal the information stored on your computer. These viruses can reach your computer, without your knowledge, through the Internet or through storage devices, such as floppy disks and CD-ROMs.

Worms are malicious programs that self replicate, and then penetrate an operating system with the spread of malicious code.

The worms are designed to spread by computer networks, sending copies of the original code to other computers. This can cause harm by consuming bandwidth or possibly the deletion of files or sent documents.



A Trojan horse is a type of malware designed to provide authorized access to a user's computer. Unlike worms, trojan horses do not have the ability to replicate. Trojan horses can deal critical damage to your computer. Trojan horses can lead to viruses being installed on a machine because it is being controlled by the trojan's creator. This can allow data to be stolen, installing more malware, key logging, and various illegal actions.

3. **Spyware** **Spyware:** are programs that can be installed on your computer without your knowledge. Spyware collects the owner's private information such as your browsing habits, downloads, or your personal data like your name or address. There are many different strains of spyware. Some spyware can redirect you to a different site, install additional software that causes to slow down your computer's performance, change your computers' settings, change the homepage, or serve you a personal advertisement known as adware.



4. **Internet Scams:** While using the Internet, you might come across some attractive offers through email messages or chat room communication. They may appear real at first, although they are usually fake. They can deal damage by stealing your personal information and can steal money from you.



5. **Online predators:** Online predators are individuals who lure anybody online into inappropriate and unethical relationships and habits. It is common that an online predator

can attack a victim by manipulating them causing them to be unaware of the dangers. Normally, an online predator can be found in instant messaging rooms, chat rooms, and social networking sites.



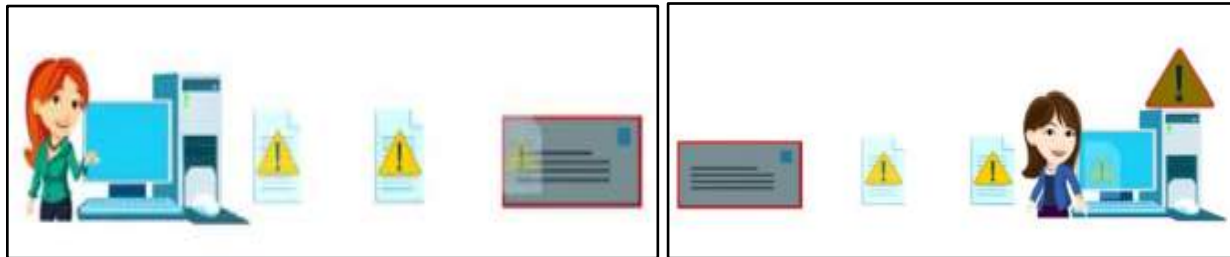
6. **Accidental deletion of data:** Many times, damage to a computer is due to unintentional human errors. Accidental deletion of an important file can disrupt the integrity of data or prevent other files or programs from working. For example, you may accidentally delete an important file, causing the computer to malfunction.



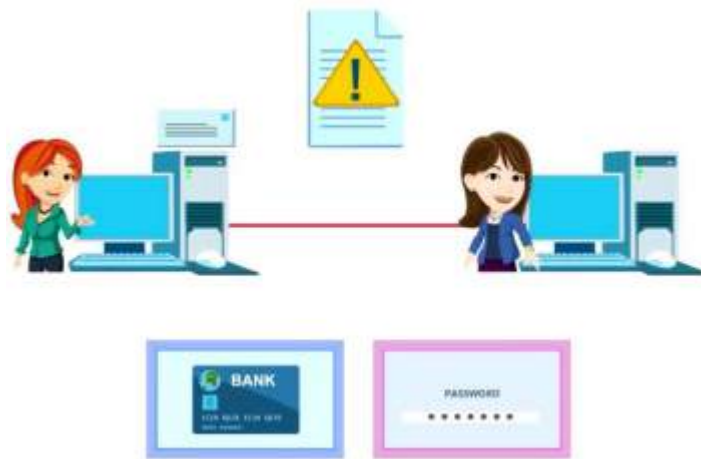
7. **Accidental damage to hardware:** Computer components are extremely delicate; being delicate makes them run the risk of getting damaged due to carelessness. For example; if you accidentally drop your laptop computer, this might result in damage to the hardware components, such as the motherboard or CD-ROM. As a result you could lose the data stored on the computer. In addition, physical damage to data due to spilling of food and beverages on storage devices can affect your computer.



8. **Phishing** is a common activity used to extract personal information from computer users and then use the information for malicious purposes



For example, someone may send an email message to you pretending to be from a bank or another trustworthy organization.



The email asks you for sensitive information such as your credit card number or password. This information might be sold to someone else or immediately used to cause you financial loss.



It's important that you verify the authenticity of any email message that asks you to respond with personal information. There are various phishing websites on the internet that will use fake email messages to try and collect your personal information.

Internet Explorer 11 helps protect you from phishing websites with privacy settings and the smart screen filter.

Measures for protection against threats from human actions

You can take some simple measures to minimize the risks associated with malicious human threats and human errors.

1. **Store data safely:** To minimize the possibility of theft or tampering of data, keep your data in safe and secure locations that people have limited access to. A safe and secure location could be your private home office or a place where people cannot have easy access to your files. If you feel as if you do not have a safe place to store your data, then put a strong password on your account. Make sure your password is not simple, like the name of your pet or your birthday. Your password should be longer than eight characters, have numbers and special characters, and have both upper and lowercase letters.



2. **Install antivirus and antispyware programs:** When using the internet, you might encounter a spyware or virus. Viruses and spyware can do great damage to your computer. It is important to download an antivirus and antispyware program that can prevent and remove the viruses. Antivirus and spyware programs present in the computer's memory can alert and prevent viruses from entering. You must regularly update these programs because they update their database about new viruses and spyware, which further protect your computer. Most programs offer an automatic update feature that automatically installs the updated version of the program.

Because many viruses, spyware, and Trojan Horses do come from email sources, and email software. Windows Live Mail, allows you to block junk email messages, filter messages from viruses, Trojan horses, and spyware, and allows you to scan your attachments.



In Windows 8, there is a built-in feature called Windows Defender which is a free program that helps protect your computer against pop-ups, slow performance, and security threats caused by viruses, spyware, and other malicious software. Windows Defender runs in the background. It can warn you if there is any suspicious program or activity, although you can run your own scan. Another built-in feature is Windows Smart screen which is a new safety feature in Windows 8 and Windows RT (Windows 8 operating system designed for mobile devices and tablets). Windows Smart screen helps protect you from new malware that antimalware software hasn't detected yet. Smart screen features anti-phishing protection, download manager protection, application reputation, and antimalware protection.

3. **Back up data:** You may lose your data by accidental deletion or the destruction of data. You can back up important computer data to help restore lost or deleted items. Your windows computer comes with a backup system. You can locate the backup data option by clicking the Star button, clicking Control Panel, click System and Security, and clicking Backup and Restore. Click Backup now, if you are not an administrator, you may need the administrator password or confirmation. If you follow the steps in the Wizard, you'll be able to choose where you would like to back up the information and what you want to back up. If you have backed up your information before, you can wait until the next scheduled backup maintenance or continue to back up your information. It is good to back up your information regularly because you may have new programs or files that cannot be found when restoring your old information. It is best to back up in multiple places such as an external hard drive or a cd. This way you always have a backup if anything such as digital or physical damage and theft is done.



4. **Encrypt data:** To prevent lost, stolen, or hacked documents and files from being seen, you can encrypt data by using the BitLocker or BitLocker To Go features available for you on Windows 8. BitLocker is compatible with Windows 7, Vista Enterprise, Windows 7 and Vista Ultimate or Windows Server 2008 and R2. BitLocker will not encrypt any hard drives from Windows XP, Windows 2000, or Windows 2003. BitLocker is a feature that protects your computer's files from being readable. If your information ever becomes lost, stolen, hacked, or tampered. Your files will be encrypted which means that the contents can be read only by someone who has the right encryption key to unscramble it. BitLocker does not allow anybody to have access to your internal hard drive, and even external hard drives like Universal Serial Bus (USB) drives. BitLocker helps protect attacks offline as well as online. Although it is not likely, an offline attack comes from a malicious user taking your hard drive from your computer and connecting it to another computer, so they can harvest your data.



5. **Install firewall:** Another effective step you can take to protect against malicious threats is to install a firewall. A firewall system prevents unauthorized access to a private network from network traffic. A firewall system forms a barrier between trusted and untrusted websites from threats like hackers and worms. You can configure your firewall management into two different ways, default-deny policy and default-allow policy. A default-deny policy allows only the listed network services. A default-allow policy blocks the listed network services, and accepts everything else. Windows 8 comes with a Windows Firewall, which blocks unwanted access to your computer.



A firewall program does not protect you from all viruses, so it is necessary to install an antivirus or anti-malware app. Window Defender can help you with protection against viruses that the firewall does not protect from.

6. **Keep computer in a safe environment:** Keep the computer in an area that is room temperature, dust-free, free from vibrations, and out of the way of possible impact. A table or shelf housing the computer should be steady and stable to keep the computer from falling, even if anybody bumps into the computer.



Recommended room temperature should range from 65° to 70°F (18° to 21°C) for optimal system reliability. If the room temperature is too cold, it can create humidity levels to be too high which can cause water condensation to occur resulting in to hardware corrosion and component failure. When the room is too hot it can cause humidity levels to become low. If the humidity is too low, it can allow electrostatic discharge that can affect sensitive components. Hot rooms can also cause overheating, which can greatly reduce the life of electronics. You should maintain your computer space to the relative humidity. Relative humidity is the amount of moisture in the air at a given temperature in relation to the maximum amount of moisture that can be held at the same temperature. To keep computers at a safe temperature, you can install a fan into your computer. Specific fans can cool systems using liquid circulation systems.

LESSON 2: PROTECTING YOUR COMPUTER AND YOUR DATA

OBJECTIVES:

- 2.1. IDENTIFY VARIOUS METHODS OF PROTECTING THE OPERATING SYSTEM, SOFTWARE, AND DATA ON YOUR COMPUTER.
- 2.2. IDENTIFY VARIOUS WAYS OF SECURING ONLINE AND NETWORK TRANSACTIONS.
- 2.3. IDENTIFY COMMON MEASURES FOR SECURING E-MAIL AND INSTANT MESSAGING TRANSACTIONS.

Protecting Device

A malicious human source could be a threat to your computer. A hacker is a person who tries to illegally access your computer when you connect it to the Internet. Hackers are normally programmers who can gain unauthorized access to other computers. They can sabotage your personal information, finances, your computer, other computers, and programs. Hackers send out personal information to advertisement companies. In addition to malicious human threats, human errors such as accidental deletion of data and physical damage are also a threat to your computer.



Protecting your computer

You need to provide your identification to access your bank locker or your safe deposit box. This identification is to ensure that no one else is able to access your items. Similarly, you can implement various security measures to minimize the threat to your computer and the data on it.

This lesson introduces you to some common best practices that will help you protect your operating system, software, and data on your computer. You need to secure your computer to protect the data and files on it and to protect your identity. The simulation on this page will introduce you to ways you can protect your computer.



Data Protection: the legal procedures that must be adhered to in relation to access, collection and use of data, including data stored on a computer

1. personal data shall be obtained only for a specified and lawful purposes.
2. Personal data should not be processed in any manner incompatible with those purposes.
3. If personal information has been stored, it should be destroyed once it is no longer needed
4. Data subjects should be notified if data about them has been distributed in an unauthorised way.
5. data protection laws may differ by country or between regions

Privacy and Security

Always take reasonable measures to protect your privacy and security

How can you stay safe and keep sensitive information secure online?

1. Restrict available Personal Information
2. Private Messaging
3. Disable Location information
4. Block / Report Unknown users

What guidelines should one follow while communicating online, sharing information?

Accurate and Brief: Avoid long and complex messages and accurately express the message you are passing

Clear Subject Headings: Clear message headings help to avoid filtering the messages into junk or spam mails.

No circulation of inappropriate content: Consider the recipient and use appropriate tone and content

Avoid inappropriate disclosure of personal details: Be careful on the recipient of your personal details to avoid misuse

Spell check the content: Avoid grammar and spelling mistakes always.

Best practices for securing online and network transactions

Having your computer connected to the Internet gives you access to all kinds of activities and services. But it also introduces risks. Unauthorized people may gain access to your computer or you may download files and programs that can harm your computer. Read through the text on the right to learn some ways to secure your computer.

1. **Strong password:** A strong password is one that is complex enough that it cannot be guessed or accidentally discovered. A strong password combines different password elements such as letters in both uppercase and lowercase, numbers, and unique keyboard characters such as '!', '#', or other symbols. The best passwords are not complete words and could not be guessed by someone who knows you or by a stranger.

A strong password is most crucial to protect your data on websites but is also used to limit access to networks, sites with personal or sensitive information on file and even to protect data on your own computer.

2. **Spyware and hacking protection:** Spyware programs can be picked up from the Internet without your knowledge and transmit personal information about you to a hacker. This illegal attack on your computer could come from anywhere in the world, regardless of where you are. Hackers try to put spyware on your computer so they can steal your confidential data.

You should utilize all the protection you can get including Windows built-in antispyware and online security support from your Internet service provider (ISP). Often this support will be in the form of anti-spyware software recommendations, firewall protections and email screening and spam protection.

3. **Periodically clear your browsing history:** For your convenience, websites you visit on the Internet while browsing are saved and can be easily found again in your browser history. Your computer uses temporary memory called a cache memory to store your Internet browsing history. Often, some of this temporarily stored data is in the form of personal information you have given to a site, such as credit card information you have given to purchase an item. Hackers can attempt to recover this information but you can limit this risk by regularly deleting your browser history and cache memory.

Cookies are another type of file created to make Internet browsing and shopping easier. They are used to save preferences and often needed information to sites that you visit often. They allow things like your name and password to be remembered and offer people a more personal, less hassle experience.

Cookies can also become a threat to your computer privacy because, once again, hackers can steal this personal data. As with deleting your browser history, making it a practice to regularly delete cookies is a smart routine.

4. **Avoid sharing personal information:** Before you give a website, any personal information makes sure you are dealing with a reputable company and that the information requested is appropriate for the services offered. Making online purchases will often require credit card information and leave you vulnerable to hackers and fraud. To minimize your risks be sure any Internet commerce is on a secured website and read carefully how your information will be shared with others. Best practice is to avoid giving any personal information on the Internet unless you know exactly who you are giving it to and what they will do with it.
5. **Make online transactions only on secured sites:** Eventually you are going to use the Internet to make a purchase. This will require you sharing your credit card number or some other form of

personal data. When you are ready to make a transaction, it is crucial you make sure the website is a secure website. To identify a secure website be sure its site name has the prefix https. This prefix tells you the website implements the Secure Sockets Layer (SSL) protocol and uses encrypted communications and certifies the website is what it says it is.

Make sure to always check the security certificate of any website before offering your personal information or performing an online transaction.

6. **Disable active content:** Active content are small programs that install into your computer to enhance your browsing experience on the Internet. Often, they provide toolbars or videos to make interacting on the Internet easier or more immersive. In some cases, however, these programs are used by malicious persons to attack your computer and steal your data without your knowledge. You can disable active content on your computer through your browser settings.

Measures for securing email and instant messaging

You can send email attachments to share files with your friends and you might receive a photograph or a music file as an attachment in an email message, however, you need to be cautious while opening any mail containing an attachment because it is the most common pathway for the spread of viruses.

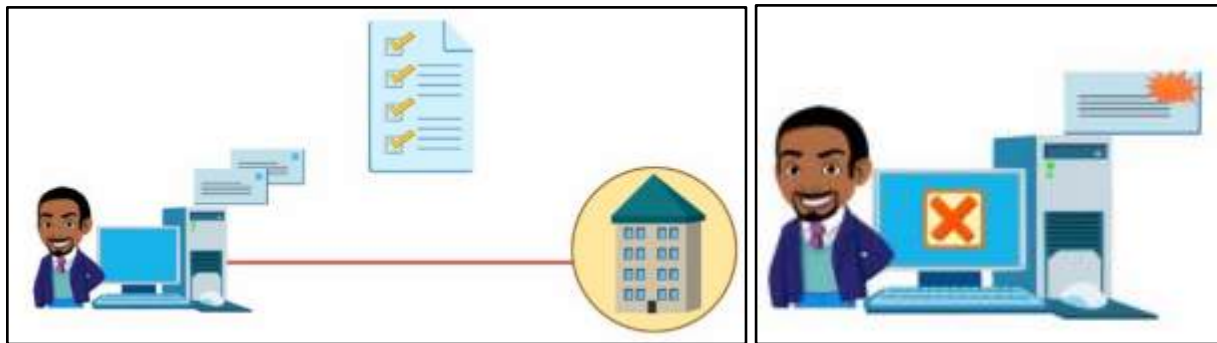


Spam and junk mail

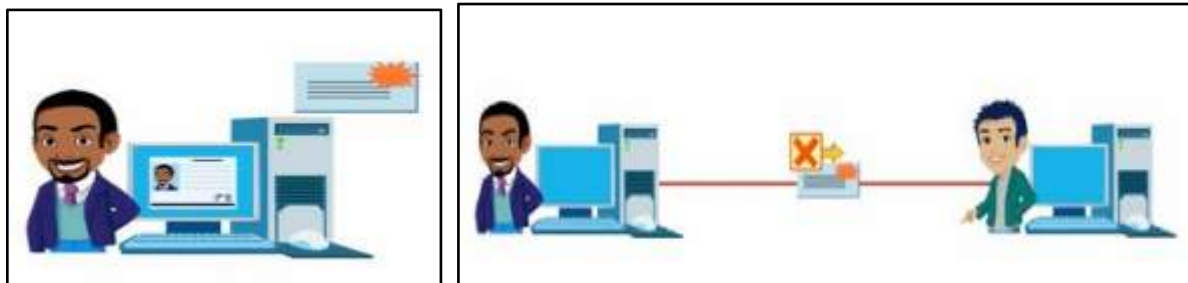
You may receive a number of unwanted email messages from unknown senders; including individuals and companies that are advertising their products or services.



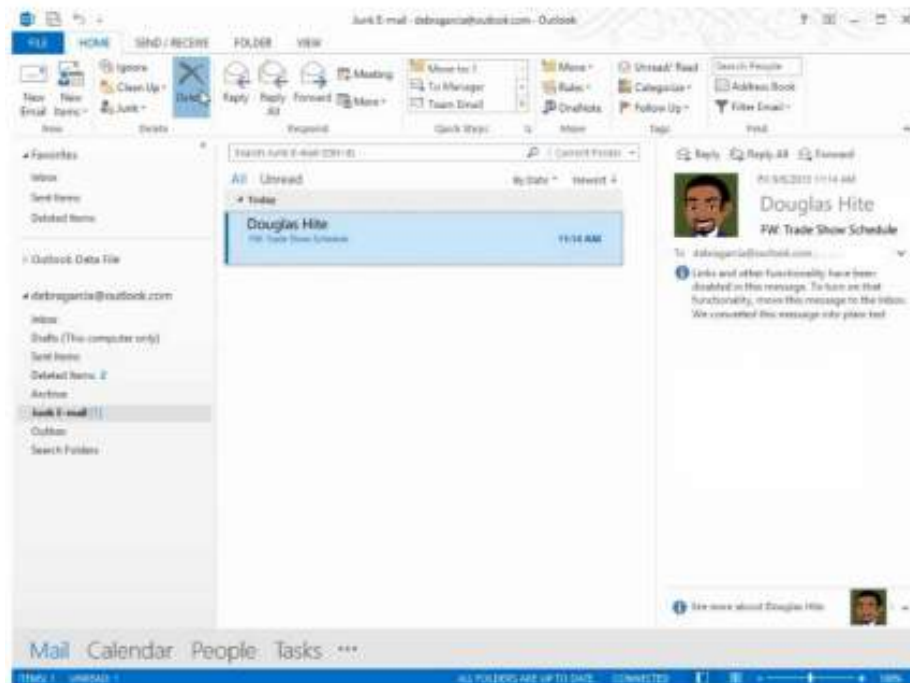
These messages may also be in the form of online surveys that require you to fill out personal information. These unsolicited messages are known as spam or junk mail.



Junk mail or spam can often include content that can be harmful to your computer, In addition junk mail is often used for stealing identities and you might accidentally share sensitive information by responding to certain messages.



You should always avoid replying to any spam or junk mail message. You should delete any spam or junk mail message when you receive it. Email programs such as Microsoft Outlook include a spam or junk mail folder where spam and junk mail messages can be directed and automatically deleted.



DAY THREE: HOURS ALLOCATED: 3 HOURS

HOUR ONE (1)

LESSON 3: PROTECTING YOURSELF AND YOUR FAMILY FROM SECURITY THREATS

OBJECTIVES:

3.1. IDENTIFY COMMON MEASURES USED TO PROTECT PRIVACY.

3.2. DESCRIBE HOW ONLINE PREDATORS WORK.

3.3. IDENTIFY THE GUIDELINES TO PROTECT CHILDREN FROM ONLINE PREDATORS.

Protecting your family from security threats

Computers are not only used at schools, colleges, and offices, but are also commonly used in homes. Every member of your family can find some use for the computer. With the increase in the use of computers at home and at work, it is important that you and your family understand the various threats associated with the use of computers and the Internet. Since every member can use the computer for various purposes, such as:

- Keeping household accounts
- Writing email
- Browsing the Internet
- Playing games
- Music
- And more

In this lesson, you will learn about the various measures that can help protect your computer from these threats.

Measures to protect your privacy

Protecting your computer ranges includes preventing malicious software from infecting your computer to hide private information on websites. Antivirus software, email filters and privacy settings are methods to protect your privacy.

How To Protect Your Online Privacy

- Commit to sharing less online.
- Use strong, unique passwords and two-factor authentication.
- Tighten privacy settings for your online accounts.
- Purge unused mobile apps and browser extensions.

- Block search engines from tracking you.
- Browse online with a secure VPN.

Online predators

Online predators target children and the elderly for inappropriate relationships. In this lesson you will learn some strategies to reduce the risk of being a victim.

Guidelines for protection from online predators

Do not let you or your family members become victims of online predators. Use the following guidelines to prevent online predators from harming you and your family.

1. **Identify signs of predator behaviour:** Online predators can display certain behaviours that should seem odd to the careful observer. For one, they have a goal in mind to get to know you as well and as soon as possible. This often comes out being overly friendly or familiar early in the acquaintance. They are impatient for the natural progression of a normal online relationship and will push unearned affection, quickly getting frustrated with even slight rejection. If an online contact wants to meet a minor at any time in the real world, it should be a warning sign.
2. **Be cautious of offers by strangers:** Online predators often try to gain trust by offering gifts or tempting offers of kindness. Always be aware of gifts offered to minors. Even adults need to be careful of offers that seem unearned or inappropriate. Even if the gift is genuine, the intentions behind the gift may not be.
3. **Educate friends on online safety:** There are many precautions you can take to create greater online safety for your friends and family. Instruct loved ones to never use their real name in a chat room. In addition, they should never divulge any personal information about themselves including age or any contact information such as where they live or even a phone number. Above all else, never under any circumstances reveal any of your private usernames or passwords to anyone, stranger or not.
4. **Monitor your children on the internet:** Parents or guardians should always keep an eye on what their children are doing on the Internet. They must be protected from inappropriate websites and potential online predators. It is always recommended that a parent or guardian be present when a child is visiting online sites.
5. **Know what sites your child visits:** It is important to know what sites your child has visited. If you cannot be there in person to always monitor their online activities, it is a good idea to track what sites they have been to through their browser history. There is also special software you can buy to monitor your children's computer habits and website history.
6. **Block access to inappropriate websites:** Be sure to block any website you find unsuitable for your child, including adult content sites, message boards where minors unknowingly interact with adults and any other site that you do not want your child visiting.
7. **Monitor chat history:** There are also programs parents can buy that will track and monitor your child's chat room activities. These programs can instantly flag and call your attention to common issues in chat rooms such as inappropriate language or images.

LESSON 4: KEEPING YOUR COMPUTER SECURE AND UPDATED

OBJECTIVES:

- 4.1. EXPLAIN THE PURPOSE OF DIFFERENT SECURITY SETTINGS ON YOUR COMPUTER.
- 4.2. IDENTIFY THE OPTIONS AVAILABLE FOR KEEPING YOUR COMPUTER UP TO DATE.

Securing a computer

When you are on the Internet, the biggest security risk to your computer comes from viruses and hackers. Often, threats such as these can be greatly reduced simply by making sure your computer security settings are correct and that your security software is up to date.

Windows 8 can help keep your computer secure and free of viruses by alerting you to security and maintenance issues and by offering virus protection, a network firewall and system backups. Each of these security settings can further be adjusted depending on your security needs.

The following can all be accessed via Windows 8 to help protect your computer:

1. Firewall settings to restrict unauthorized access to your computer while online.
2. Settings to allow your computer to automatically download and install security updates to protect against the latest viruses.
3. Privacy and security settings you can set on individual Internet sites that you visit.
4. Malware protection settings to detect and eliminate dangerous software that could harm your computer.
5. Maintenance tools that can perform tasks such as troubleshooting Windows and offering the latest updates.



Keeping your computer secure and updated

When you connect your computer to the Internet, your computer software and data are accessible to the rest of the world. Connecting to the Internet increases the threat to your computer from viruses, spyware, and hackers. However, you can minimize these security threats by configuring the security settings on your computer and keeping security-related software up-to-date.

In this lesson, you will learn the importance of securing a computer and how to maximize the security of your computer by configuring the security settings on your operating system. The lesson also explains how to configure your computer to automatically update its security software.

Keeping the computer up-to-date

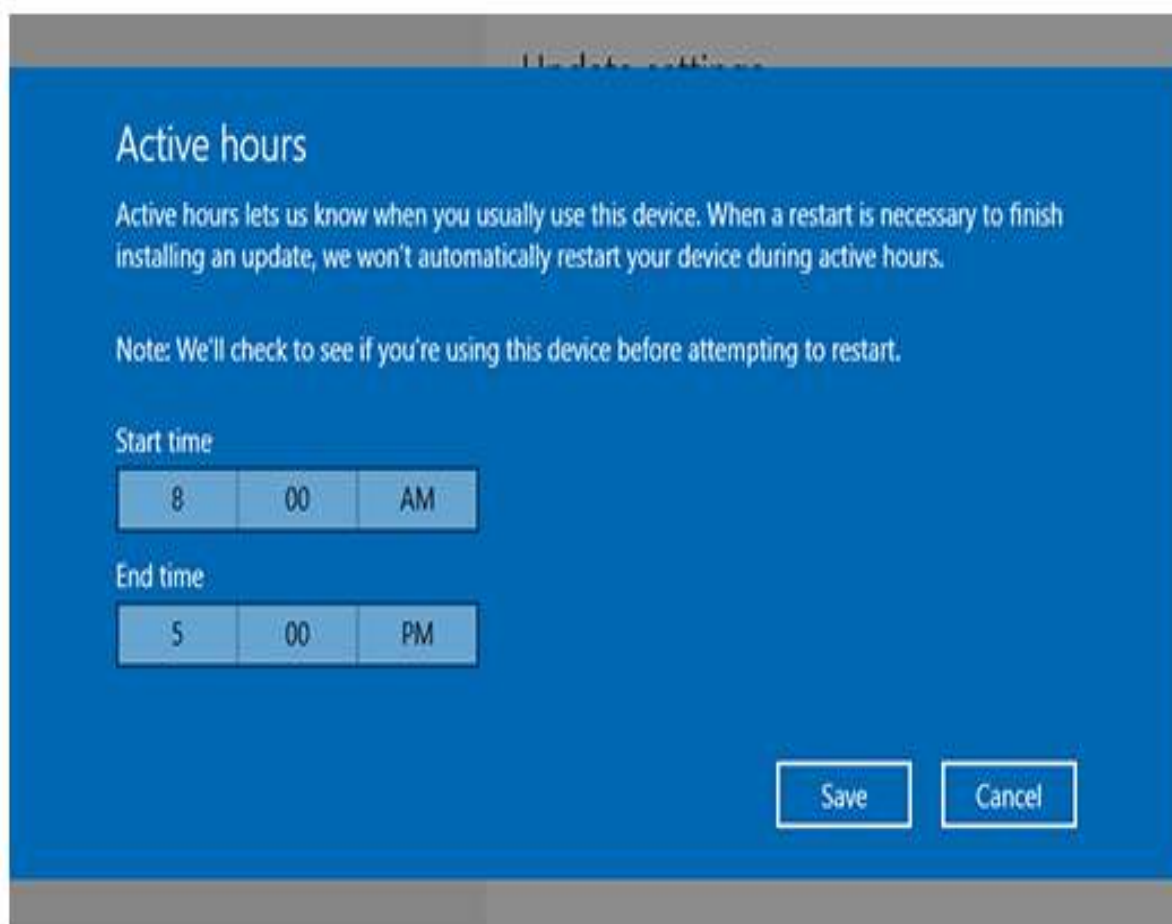
Making sure your computer is updated with the most current security and system updates will not only keep your computer secure but will also help with speed and performance.

We've made it easier to keep Windows current with the latest features and improvements—and with fewer interruptions and restarts when you're using your PC the most.

The great news is that you don't have to do anything to get the latest updates, as they'll automatically download and install when they're available. (Unless you're on a metered connection, then updates won't download until you get them.)

Occasionally, your PC may need to restart to finish installing the updates. You can set active hours to let us know when you're usually using your PC to help prevent inconvenient restarts.

Select Start, then select Settings > Update & Security > Windows Update, and then select Change active hours.



LESSON 5: COMPUTER ETHICS

OBJECTIVES:

5.1. EXPLAIN WHAT THE TERM INTELLECTUAL PROPERTY MEANS AS IT APPLIES TO COMPUTING.

5.2 IDENTIFY THE VARIOUS COPYRIGHT VIOLATION ACTS AND THEIR PREVENTIVE MEASURES.

5.3. IDENTIFY THE VARIOUS LEGAL CONCERNS ASSOCIATED WITH INFORMATION EXCHANGE.

Computer ethics

Searching the Internet is often the fastest and easiest way to gather information, at any time. For example, you can use the Internet to search for information for your school assignments, or ideas to include in an office presentation. You can also download songs and movies from various Websites. You do not have to pay for downloading information from most of the Web sites. But these free downloads may not actually be free. The information in a Web site is legally owned by the author who created it or by the Website that published it. Therefore, you may need the permission of the author or the owner of the Web site to use the contents. You need to be aware of the rights or permissions you have on the contents available in a website before downloading them. That includes the downloads of the following:

- Songs
- Movies
- Software
- And more

This lesson explains the meaning of intellectual property in the field of computing, and how unauthorized use of intellectual property can lead to copyright violation. In this lesson, you will also learn about the various legal concerns associated with information exchange.

Copyright: legal right that gives the creator of an original work, such as text or an image, exclusive rights for its use and distribution.

1. An Example of copyright is the protection of Software.
2. Copyright applies too to all you may find online e.g. research paper work always cite or quote the sources
3. Copyright infringement or plagiarism could be punishable by law

Intellectual Property: work that has been created and can be protected by laws such as copyright

Copyright violation and prevention

Tips that could help you to avoid copyright infringement

1. Do not copy anything.

2. Avoid non-virgin development.
3. Avoid access to prior design work.
4. Document right to use.
5. Negotiate for enhanced warranty and indemnity clauses.
6. Document your own work.

Legal concerns with information exchange

With the extensive use of the Internet, there is potential for you to get involved in illegal and unethical activities, such as gambling and defamation. You need to be aware and cautious of these illegal and unethical issues. Also, remember that these issues vary from country to country, and even within parts of a country.

1. **Defaming someone's reputation:** When you communicate with others by using email, chat, or online public forums, be careful not to make any statements that may result in defamation of someone. Defamation means making false statements about a person that can negatively affect the person's reputation. For example, suppose you post a message on an online forum falsely stating that your neighbour, who is a celebrity, owns illegal property. This can be considered as defamation because you are spreading false information that may harm your neighbour's credibility.



Libel and slander are two forms of defamation. Libel means written defamation which has been published, whereas slander means verbal defamation. The legal system of most countries considers both libel and slander as punishable offenses. You may face criminal or civil penalties according to the legal system of your country. The punishment may range from monetary penalty to severe punishment such as jail terms. Sometimes, the severity of the offense depends on the situation. For example, in some countries, any insult to the President is a criminal offense. However, in some other countries, public officials have less protection than the average citizen. Therefore, be aware of the local laws before making any defamatory statements.

Note: A false statement can be considered defamatory, even if it is not derogatory. Sometimes, even true statements can be considered as defamatory if the statements harm someone's reputation.

2. **Visiting inappropriate websites** Some websites offer services that carry out activities that are restricted by the legal system of your state or country. It is possible to access these sites over the Internet because there are no boundaries and policing over the Internet. For example, you can

access a gambling site even if the law of your country bans gambling. But this may cause legal troubles for you.



You also need to be aware that the legal system is different in different states and countries. For example, the products that you can legally buy or sell in one country may be an illegal purchase or sale in another country. So, though a website may not restrict you from buying an item that is illegal in your country, you may face legal charges for buying the item.

HOUR TWO (2)

COURSE 5: MULTIMEDIA SERVICES (DEDUCING INFORMATION FROM MULTIMEDIA)

COURSE BENEFIT

Computers can be used as TVs. You can watch your digital videos that you have made, view your photos, and listen to recordings or songs. Not only that, but you can buy TV movies, subscriptions to shows, and different sources of media, to view on your own computer. In this lesson, you will learn

LESSON 1: THE MODERN DIGITAL EXPERIENCE

OBJECTIVES:

- 1.1. IDENTIFY THE BENEFITS OF THE EXPANDING SCOPE OF DIGITAL TECHNOLOGY.
- 1.2. EXPLAIN HOW MERGING TECHNOLOGIES EXPAND THE FEATURES OF DIGITAL DEVICES.

Digital technology

Technology is expanding with different devices to enhance our digital outlook and the features of media devices.

In this lesson, you will learn the different benefits of digital technology. Computers, laptops, tablets, and phones have made difficult tasks simple. You will also learn how digital technology expands the features of

media devices such as cameras, digital audio devices, and camcorders. Digital technology has allowed sharing, organizing, and editing different media fast and easy.

Digital technology has changed the way you manage your work, communicate with others, and entertain yourself.

Benefit of Digital Technology

Digital technology is used in all areas of life today. It has changed the way people work and entertain themselves. This module explains the scope of digital technology at home, at work, and at play. It also identifies the different career opportunities made available because of the extensive use of computers in various fields

1. **Compactness:** Digital technology has enabled devices to be more compact, faster, lighter, and versatile. Large amounts of data and information can be stored or accessed locally, remotely and virtually instantly.
2. **Connectivity:** Digital technology has enhanced social connectivity by manifolds. Now it's so much easier to stay in touch and connected with friends, family, and your workplace, being present in any part of the world.

You can communicate by text, visually via a video or video chat and through audio messages and calls.

3. **Keeps you Updated:** Through digital mediums like websites, apps, and other digital platforms nobody is isolated in the digital world. News and local events update users regularly about what's happening around them and across the globe.
4. **Speedy Technology:** Internet quality and speed have improved exponentially with digital transformation strategy leading to ongoing advancements in the field of communications.

Faster broadband connections facilitate the transfer of large amounts of data and information across the World Wide Web almost instantaneously, making it possible to stream videos and audio files faster and almost in real-time.

Sending large data files or accessing information from virtually anywhere in the world now is a matter of seconds. For example, Apps and integrations like Google Assistant and WhatsApp have made our life so much easier and resourceful.

5. **Versatile:** The nature of work has been transformed in the digital age. Increased connectivity has led people to work online or from home. Online selling, blogging, and websites are some great examples of how digital technology has transformed the way we used to look at things.
6. **Global Access Advantage:** Anybody with access to the internet now has access to a huge proportion of the world's knowledge over the web. Lessons and courses can now be delivered virtually online.

7. Digital Device

8. Today, the functions of telephones, cameras, video recorders, media players, and Internet devices are being merged into smaller and portable devices.
9. From cell phones to MP3 players to digital cameras, computers are changing our everyday lives. Which includes digital audio, digital video, and digital photography. It explores how these and other computing technologies are creating new career opportunities and shaping the world we live in.

10. **features of digital devices** (portability, performance, storage, user interface, connectivity, media support, energy consumption, expansion capability, security features, applications);

LESSON 2: DIGITAL AUDIO

OBJECTIVES:

- 2.1. IDENTIFY THE CHARACTERISTICS OF DIGITAL AUDIO.
- 2.2. EXPLAIN THE CONCEPTS OF RECORDING, COPYING, AND CONVERTING DIGITAL AUDIO.
- 2.3. IDENTIFY THE FEATURES OF ADVANCED SPEECH TECHNOLOGIES.

Digital Audio

Organizing CDs and creating playlists is easier with the use of technology. In this lesson, we will learn how to identify the characteristics of digital audio. There are different types of audio files and we will learn the concepts of recording, copying, and converting digital audio. You will be able to identify the features of speech technology to be able to work with your computer by speaking.

Characteristics of digital audio

Digital audio has many characteristics that allow you to edit, organize, and share your music. With audio editing software you can add sound effects to your audio files. By compressing audio files, you can save space on your computer and more easily transfer your audio files to the Internet.



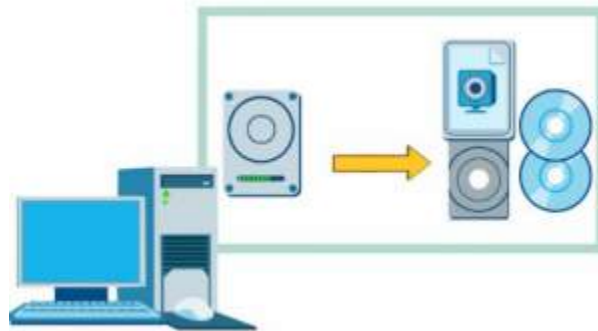
You can stream digital audio files to listen to audio without having to download a large audio file. It is important to know the different audio file formats and how they are used.

WAV: A wave (WAV) file is a universal audio file format developed by Microsoft. This format retains decent audio quality but the files are uncompressed and are larger than other formats.

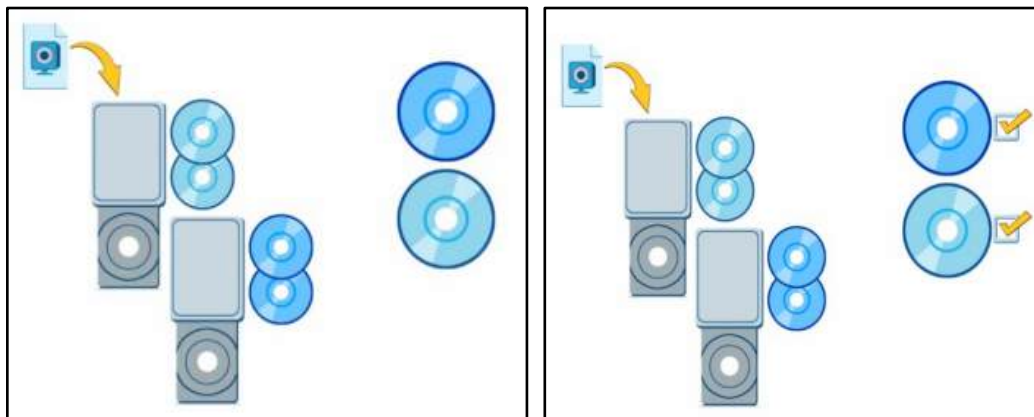
MP3: MP3 is an audio file format designed by the Moving Picture Experts Group (MPEG). This is a common audio file format for streaming music on the Internet because the files are compressed and smaller but still have a good quality sound.

WMA: WMA or Windows Music Audio is a compressed audio file format that works with the Windows Media Player and other Windows software.

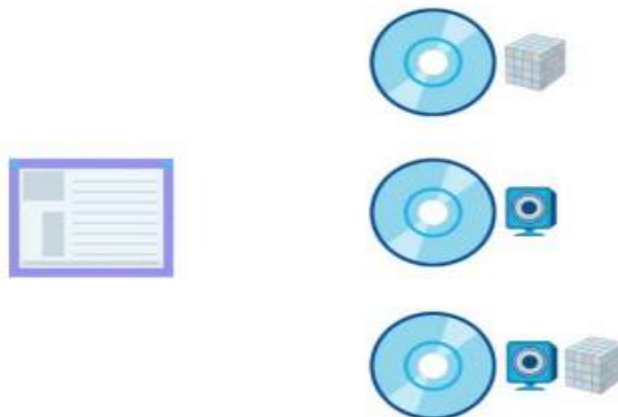
Copying digital audio: You can copy audio from storage devices such as a computer hard drive and store it on a recordable CD or DVD.



This process of copying audio and storing it on a recordable CD or DVD is referred to as burning. You need special hardware such as a CD writer or a DVD writer to burn audio files.

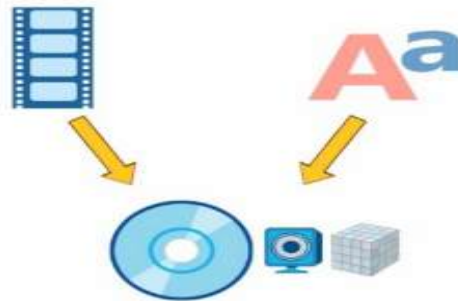


A CD writer allows you to burn only recordable CDs. Most DVD writers allow you to burn both recordable CDs and DVDs. Along with hardware you will also need software to copy audio to a recordable CD or DVD.

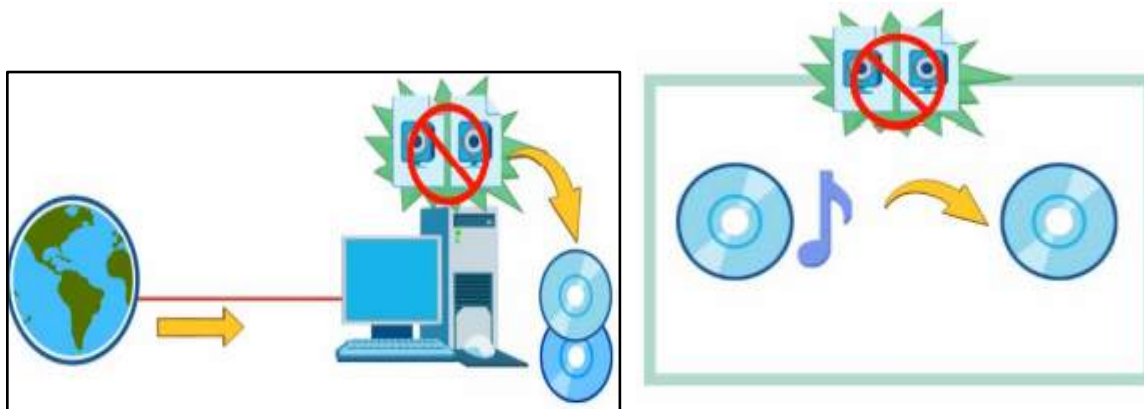


You can use the software to create different types of CDs such as data CDs, audio CDs, and mixed mode CDs.

For example, you can create an audio CD and then add video files and text files to it to create a mixed mode CD.

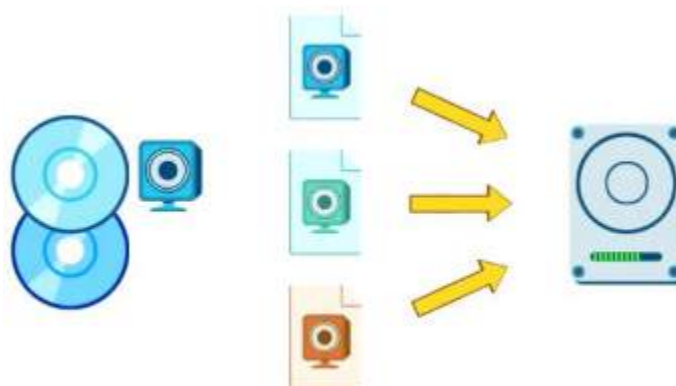


Copying audio directly from the Internet and storing it on recordable CDs can be illegal.

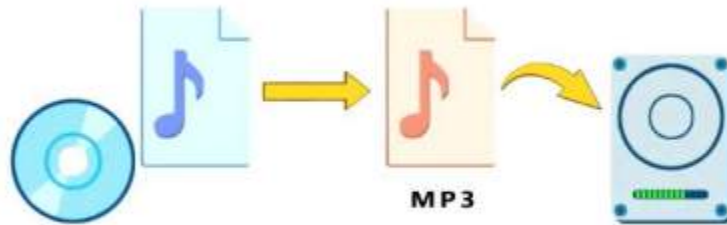


It can also be illegal to copy music from CDs and DVDs. Ensure that you have permission before you copy any material.

Converting digital audio: You can convert audio from a CD or a DVD to a different format to be able to store and play it from a hard drive.

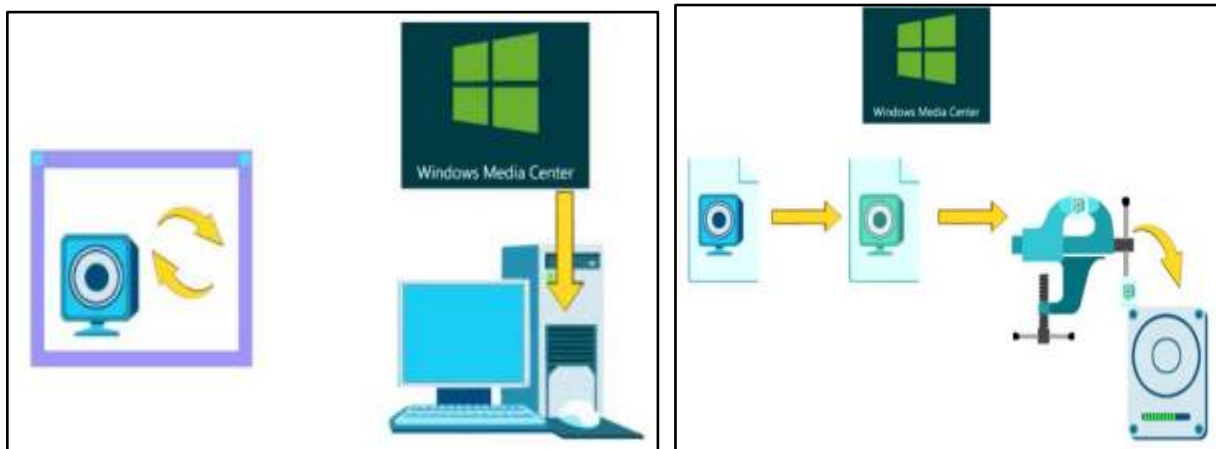


For example, you can convert your favourite song from a CD to an MP3 file that can be stored and played on your computer.

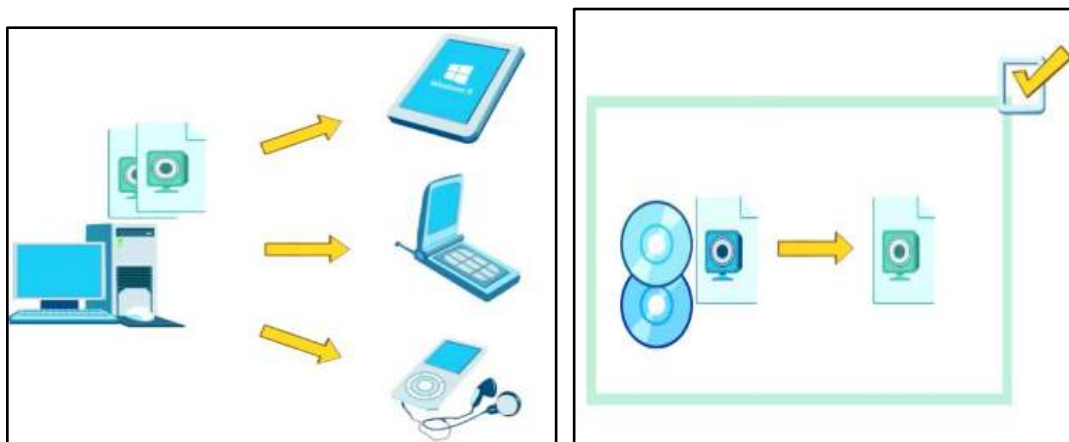


You must have audio conversion software such as Microsoft Windows Media Center installed on your computer to convert audio files.

The software changes the format of the audio and may also compress the audio so the files take up less space on the hard drive.



You can then transfer these audio files from your computer to a digital device such as a tablet computer a mobile phone or an MP3 player



Converting audio from CDs and DVDs to a different format can be illegal so be sure that you have permission to convert audio to a different format before doing any conversions.

Break: 1 Hour

HOUR THREE (3)

LESSON 3: DIGITAL VIDEO

OBJECTIVES:

- 3.1. IDENTIFY THE CHARACTERISTICS OF DIGITAL VIDEO.
- 3.2. DESCRIBE WHAT DIGITAL VIDEO EDITING IS AND THE VARIOUS OUTPUT FORMATS FOR DIGITAL VIDEO.
- 3.3. IDENTIFY THE FEATURES OF DIFFERENT WEB VIDEO TECHNOLOGIES.

Digital Videos

Digital videos are not only used by professional movie producers. Anyone can make digital videos with the use of a camcorder or webcam. In this lesson, we will identify the characteristics of digital video. Then we will identify what digital video editing is and the different output formats for digital video. You will also learn about the features of video editing software

Characteristics of digital video files

One characteristic of digital video is that you can edit the video files on a computer with digital video editing software. You can edit videos to include titles, add a soundtrack, or special effects.

Digital video editing

With digital video editing software, you can improve your videos with special effects, audio, and enhance the video quality.

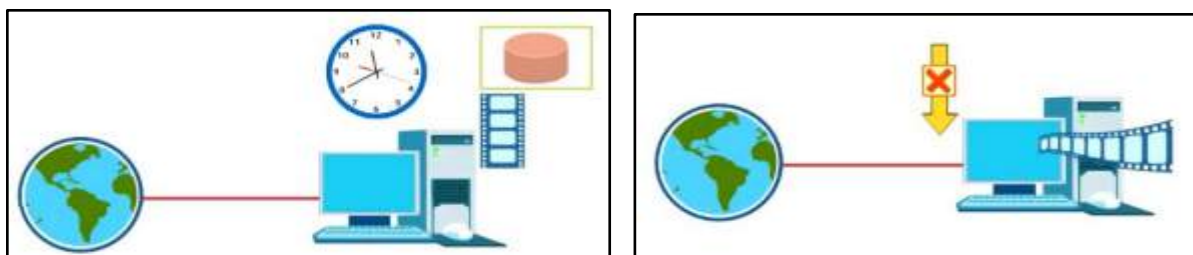


Output Format

These are the most common digital video formats and their most frequent uses.

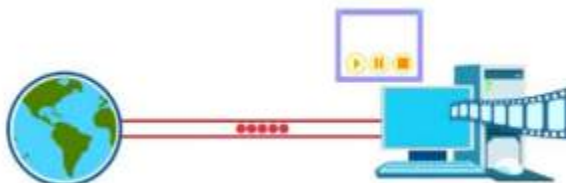
1. **MP4** (MPEG-4 Part 14) is the most common type of video file format. Apple's preferred format, MP4 can play on most other devices as well.
2. **MOV** (QuickTime Movie) stores high-quality video, audio, and effects, but these files tend to be quite large.
3. **WMV** (Windows Media Viewer) files offer good video quality and large file size like MOV. Microsoft developed WMV for Windows Media Player.
4. **AVI** (Audio Video Interleave) works with nearly every web browser on Windows, Mac, and Linux machines. Developed by Microsoft, AVI offers the highest quality but also large file sizes. It is supported by YouTube and works well for TV viewing.
5. **AVCHD**: Advanced Video Coding High Definition is specifically for high-definition video. Built for Panasonic and Sony digital camcorders, these files compress for easy storage without losing definition.
6. **FLV, F4V, and SWF**: Flash video formats FLV, F4V, and SWF (Shockwave Flash) are designed for Flash Player, but they're commonly used to stream video on YouTube. Flash is not supported by iOS devices.

Digital streaming: Generally digital video files are large and can take a long time to download.



Video streaming allows you to view video files over the Internet without downloading them.

In this case the video file is sent to your computer in a continuous stream. You need a streaming video player to view the video file over the internet.

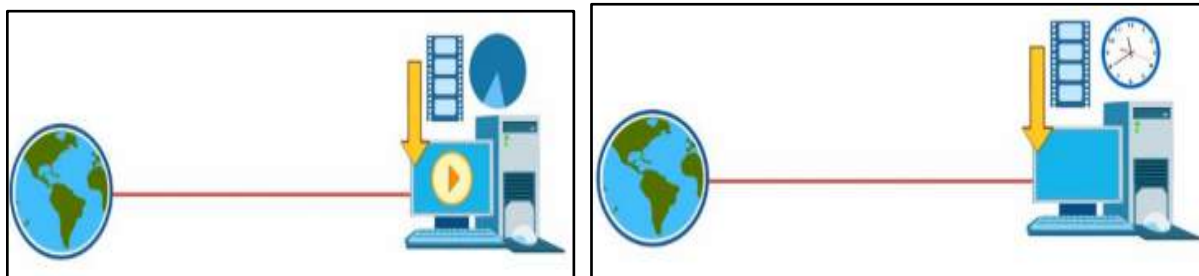


You might need to download a streaming video player and install it on your computer before you can play streaming video files.

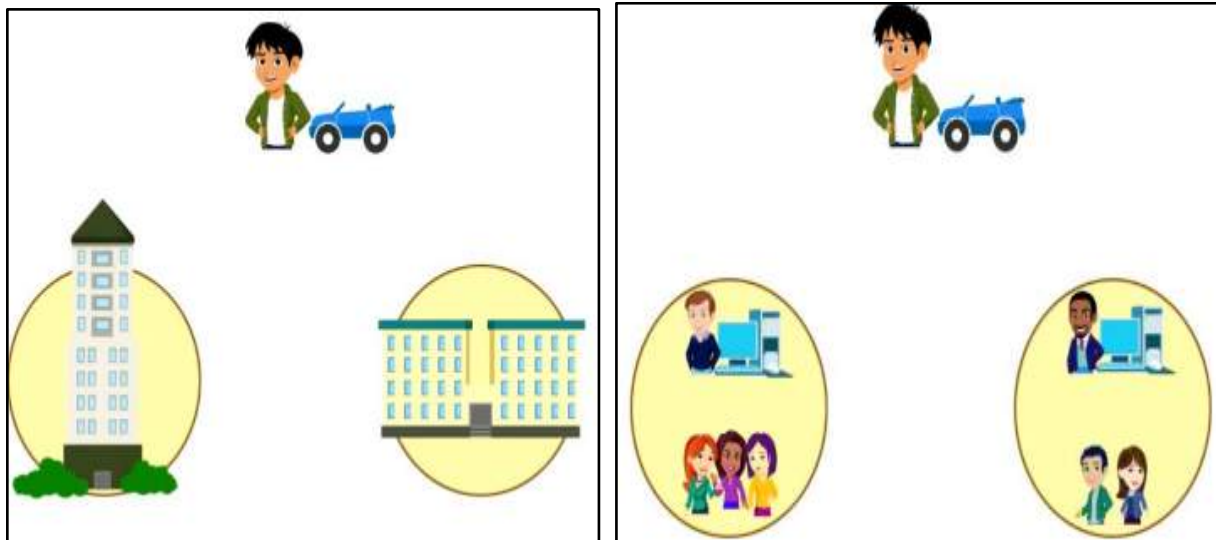
Digital downloading: You can also download video files such as video presentations, movie trailers, or music videos from the Internet. In this case a copy of the video file is downloaded onto your computer.



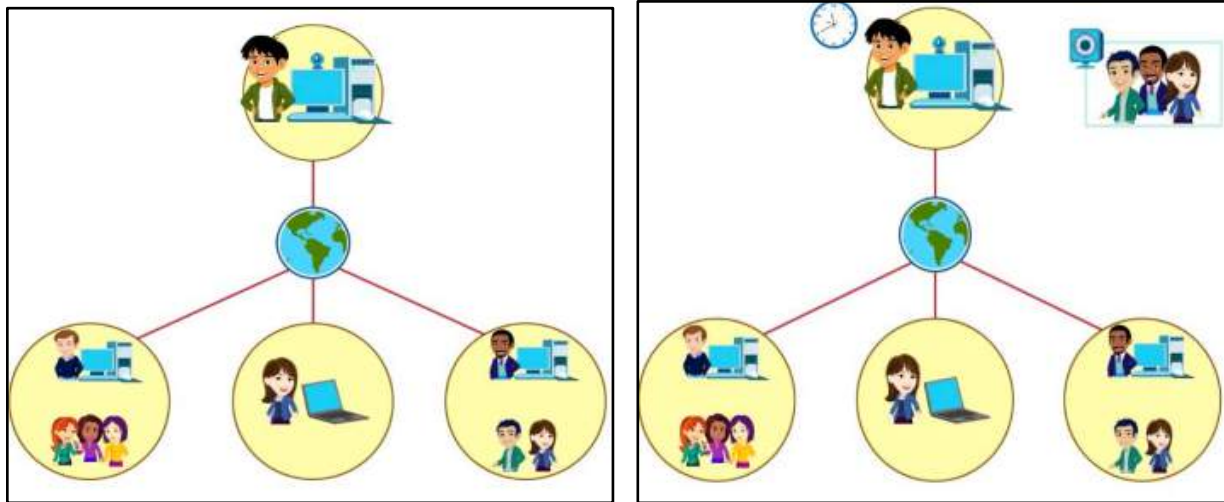
Sometimes when you are downloading a file you can play the file after a major portion of the file is downloaded. But usually, you need to wait until the entire file is saved on your computer.



Web conferencing: Today you do not need to travel from one place to another to attend conferences and seminars.



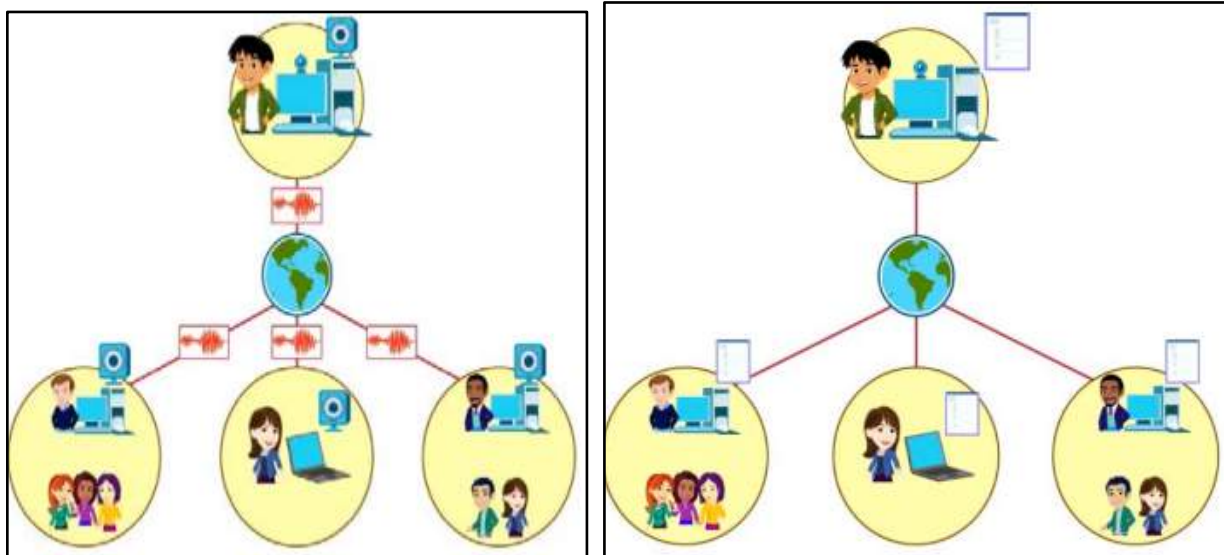
Instead, you can use web conferencing so that all the participants can communicate with each other over the Internet. One of the features of web conferencing is that it allows you to hold a virtual conference where you can see and hear others in real time.



You can also use tools such as virtual white boards and slide presentations during the web conference. Another feature of web conferencing is screen sharing where the participants can see what is on the presenter's desktop.

Web conferencing also allows voice communication between the participants so they can share their views and discuss their opinions.

Voice communication could be voice over internet protocol VOIP or voice communication by using telephone conferencing. Web conferencing also allows you to use text chatting if voice communication is not possible.



Microsoft Team is one application that enables web conferencing.

LESSON 4: DIGITAL PHOTOGRAPHY

OBJECTIVES:

- 4.1. EXPLAIN THE BENEFITS, FEATURES, AND WORKING OF A DIGITAL CAMERA.
- 4.2. DESCRIBE HOW DIGITAL IMAGES CAN BE MANAGED AND EDITED.
- 4.3. IDENTIFY THE FEATURES OF DIFFERENT TYPES OF PRINTERS THAT ARE AVAILABLE FOR PRINTING PHOTOS.

Digital Photography

Instead of using a film camera, you can use an electronic camera that can take pictures in an instant without the use of film. Cameras are easy to use and have become inexpensive over the years.

Digital camera: A digital camera, unlike a traditional camera, does not use film to store pictures.

A digital camera stores each picture in a digital format on a storage device such as a flash memory card or a hard drive inside the camera.



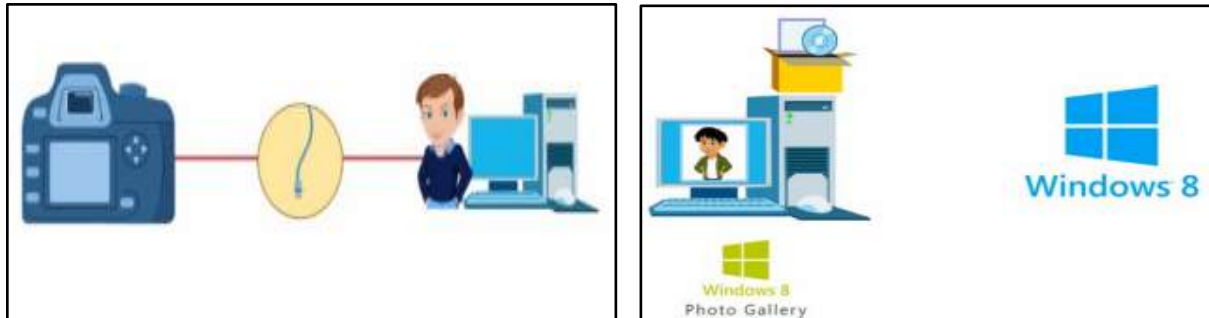
Like a traditional camera, a digital camera has a dark box with a lens that focuses light to create an image.

However, unlike traditional cameras, which focus the light onto a piece of film, a digital camera focuses light on a sensor. The sensor creates the image which is then stored on the camera's storage device.



You can transfer pictures taken with a digital camera to a computer. You can use a USB cable to transfer pictures.

You can then enhance the pictures on a computer by using image editing software including software available with Windows 8.



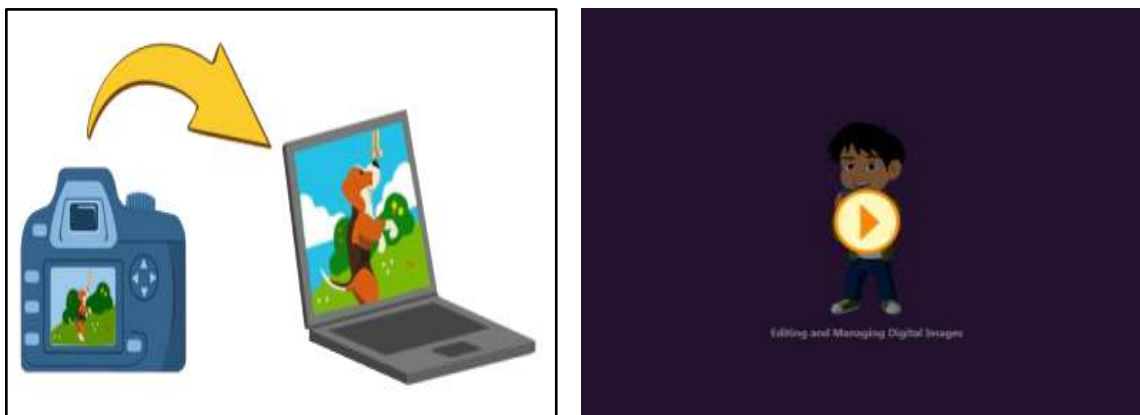
Editing and managing digital images

Image editing is done for removing unwanted elements such as dust specks and scratches, adjusting the geometry of the image like rotating and cropping, correcting for lens aberrations, sharpening or softening the image, making colour changes or adding special effects to the image.

Editing and managing digital images with digital cameras, you can transfer pictures to a computer where you can edit the digital images to enhance their quality

Managing of Digital Images

1. Transfer the Images to computer
2. Edit and organise the images in a folder
3. Identify (Rename) the images and the folder
4. Create a backup in the cloud in case of lost
5. Keep the storage system up-to-date
6. Ensure future access to your digital photo collection.



Organizing digital media on your computer

You can view and play digital media, such as music, photographs, and videos, on your computer. As your media library grows, it is important to keep it organized.

Printing digital images

To share pictures and digital images, you can print the pictures using a personal printer or a professional photo printer.



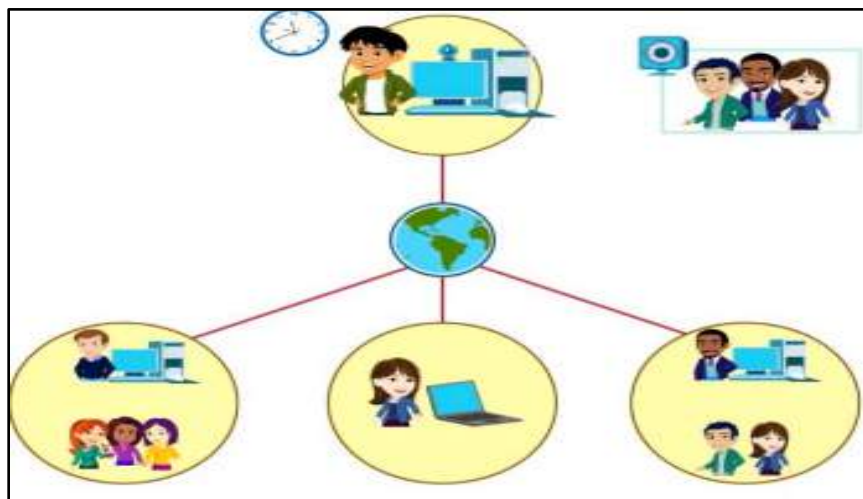
LESSON 5: DIGITAL TECHNOLOGY AND CAREER OPPORTUNITIES

OBJECTIVES:

- 5.1. EXPLAIN HOW DIGITAL TECHNOLOGY ENABLES YOU TO WORK REMOTELY.
- 5.2. IDENTIFY DIFFERENT CAREER OPPORTUNITIES AVAILABLE IN THE DIGITAL TECHNOLOGY WORLD.
- 5.3. IDENTIFY DIFFERENT CAREER OPPORTUNITIES AVAILABLE IN THE DIGITAL TECHNOLOGY WORLD.
- 5.4 DISCUSS THE BENEFIT DIGITAL MEDIA ON YOUR COMPUTER

Digital Technology

Digital technology has changed the way you manage your work, communicate with others, and entertain yourself. With the development of technology, a student can attend lectures from his/her comfort zone, an employee can work from home and a businessman can buy and sell product online as a result of digital technology



Digital Technology and Career Opportunities

With the expansion of technology, many jobs have required the use of technology allowing the way people work. It has allowed people to become more efficient in their careers. Advanced technology has allowed people to be able to work from remote locations whether you are working from home, in your office, or even in a different country. Technology has allowed different career opportunities to open such as IT professionals, developers, and information workers.

Digital technology and the workplace

Digital technology has changed the way people work. Today, work is no longer a place where you have to go. Advanced digital technology allows people to work from remote locations.

Career opportunities as information workers

1. **Information worker careers:** With computer skills and knowledge you can find a career as an information worker. There are many occupations that fall under this category of gathering and recording information. The following list is just a start.



2. **Graphic artist:** Computer graphics are in high demand and require the latest in graphic editing software. Graphic artists often work to create graphics and animation for video games and movies.



3. **Technical writer:** As a technical writer you will create documents to help explain and troubleshoot technical issues. You may be working for a tech company writing manuals for employees or as journalist, writing about the latest devices and digital advances.



4. **Receptionist:** Modern receptionists' positions constantly utilize digital technology and most jobs require excellent computer skills these days. Organization, employee and customer files, and everything else that maintains office efficiency is greatly facilitated by computers

Career opportunities as IT professionals

Information Technology is a growing field for those with technical expertise, experience and qualifications. The following describes some opportunities that are in demand right now.

1. **Network administrator**

Companies depend on their computer networks operating correctly and accurately. Network administrators make sure the network is working at full capacity. They are also in charge of maintaining the list of authorized users and generally keeping track of user access.

2. **Database administrator**

Databases are maintained by database administrators, who are responsible for keeping company databases organized and useful for storing information. They are also in charge of allowing or restricting access to the database.

3. Web administrator

Web administrators maintain and manage websites on the Internet. They can facilitate a website's initial creation or manage its functionality. Often a web administrator runs the site of a client who owns it and ensures it is secure, legal and updated.



How to enjoy digital media on your computer

Do not let you or your family members become victims of online predators. Use the following guidelines to prevent online predators from harming you and your family

1. **Online entertainment services:** With an Internet connection, your computer can function like a TV. There are many website services that you can download or stream thousands of choices of movies and entertainment either for free or for a fee. Purchases are automatically accessed by your computer and then can be viewed on any of your smart devices you have linked to your account. You usually have options of buying your own copy or just rent for a single viewing. Movies you purchase by download can also be accessed via the Cloud on your smart devices.
2. **DVD:** Your computer can also play a DVD of a movie or show that you purchased from the store. Place the DVD into the DVD player and it will automatically give you the option of starting your movie. Be sure to check if your computer is capable of playing a DVD or a Blu-ray disk before you make your purchase
3. **Digital copies:** newer movies purchased on DVD often come with a digital download option that allows you to view the movie on your computer or smart device via a code that comes with the DVD. This option allows you to watch the movie anytime without having the disk through your media player



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